Machine Learning versus Economic Restrictions: Evidence from Stock Return Predictability*

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Abstract

This paper shows that investments based on deep learning signals extract profitability from difficult-to-arbitrage stocks and during high limits-to-arbitrage market states. In particular, excluding microcaps, distressed stocks, or episodes of high market volatility considerably attenuates profitability. Machine learning-based performance further deteriorates in the presence of reasonable trading costs due to high turnover and extreme positions in the tangency portfolio implied by the pricing kernel. Despite their opaque nature, machine learning methods successfully identify mispriced stocks consistent with most anomalies. Beyond economic restrictions, deep learning signals are profitable in long positions and recent years and command low downside risk.

Keywords: Machine Learning, Return Prediction, Neural Networks, Financial Distress, Fintech

JEL Classification: G10, G11, G12, G14

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I. Introduction

Financial economists have uncovered a plethora of firm characteristics that predict stock returns in the cross-section. However, recent work has challenged the credibility of the evidence on stock return predictability. Harvey, Liu, and Zhu (2016) examine 296 published significant factors and conclude that 80 to 158 of them are likely to be false discoveries. Hou, Xue, and Zhang (2020) further show that 82% of 452 anomalies become statistically insignificant after excluding microcap stocks and when employing value-weighted returns. There is also mounting evidence that anomalies tend to concentrate in distressed stocks and extract most of their profitability from short positions.[1] Notably, it is increasingly difficult to exploit anomalies in recent years due to increased market liquidity and arbitrage activity.

Counter to this "anomaly challenging" strand of literature, there has been an emerging body of work that reports phenomenal investment profitability based on signals generated by machine learning methods.[2] Applying machine learning routines to financial data has been implicitly motivated by the AFA presidential address of Cochrane (2011), who suggests that in the presence of a vast collection of noisy and highly correlated return predictors, there is a need for other methods beyond cross-sectional regressions and portfolio sorts. Indeed, machine learning offers a natural way to accommodate a high-dimensional predictor set and flexible functional forms, and it employs "regularization" methods to select models, mitigate overfitting biases, and uncover complex patterns and hidden relationships.

A close look at these two strands of literature, namely, diminishing *individual* anomalies in contrast to the outstanding profitability of machine learning signals that aggregate *multiple* anomalies, suggests that our understanding of the economic significance of machine learning signals is inconclusive. While vast evidence shows that individual anomalies concentrate in difficult-to-arbitrage stocks and high limits-to-arbitrage market episodes, machine learning methods could indicate a promising direction to

[1] See, e.g., Stambaugh, Yu, and Yuan (2012) and Avramov, Chordia, Jostova, and Philipov (2013, 2018).
[2] See, e.g., Rapach, Strauss, and Zhou (2013), Heaton, Polson, and Witte (2017), Feng, He, and Polson (2018), Feng, Polson, and Xu (2019), Rapach, Strauss, Tu, and Zhou (2019), Choi, Jiang, and Zhang (2020), Freyberger, Neuhierl, and Weber (2020), Gu, Kelly, and Xiu (2020, 2021), Han, He, Rapach, and Zhou (2020), Rapach and Zhou (2020), Bianchi, Büchner, and Tamoni (2021), Chinco, Neuhierl, and Weber (2021), Cong, Tang, Wang, and Zhang (2021), and Kim, Korajczyk, and Neuhierl (2021). Others focus on high-dimensionality cross-sectional asset pricing models, e.g., Kelly, Pruitt, and Su (2019), Chen, Pelger, and Zhu (2020), Kozak, Nagel, and Santosh (2020), and Lettau and Pelger (2020a, 2020b).

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identify mispricing when trading frictions attenuate due to their advanced mechanism of aggregating individual anomalies. Thus, a priori, it is impossible to make clear inferences about the economic significance of machine learning signals based on the knowledge we have gained from individual anomalies.

This paper aims to fill this gap by comprehensively examining whether investors can harvest extra profits generated by various machine learning signals that are detailed in the following paragraph. To do so, we first impose several economic restrictions. In the cross-section, we limit the universe of stocks to those that are relatively cheap to trade by excluding microcaps or distressed firms. In the time series, we examine whether investment profitability is more pronounced during high limits-to-arbitrage market states, such as high volatility and low liquidity. We then assess the turnover and the corresponding transaction costs associated with implementing machine learning-based investments and finally explore the economic foundations of trading strategies advocated by seemingly opaque machine learning methods.

We consider a comprehensive set of linear and nonlinear models, study both spread (long-minus-short) and tangency portfolios, and account for both the stochastic discount factor (SDF) and beta pricing formulations.[3] We first implement a neural network with three hidden layers (NN3) as in Gu, Kelly, and Xiu (2020) (GKX) and then follow Chen, Pelger, and Zhu (2020) (CPZ) to incorporate a no-arbitrage condition into multiple connected neural networks, including feed forward networks (FFNs), recurrent neural networks (RNNs) with long short-term memory (LSTM) cells, and a generative adversarial network (GAN). We then analyze two conditional beta pricing models in the machine learning universe, namely, instrumented principal component analysis (IPCA) per Kelly, Pruitt, and Su (2019) (KPS) and the conditional autoencoder (CA) extension by Gu, Kelly, and Xiu (2021).

Note that all these machine learning methods may differ in sample coverage, input variables, and optimization objectives; hence, we do not aim to identify the best method for trading purposes or conduct a rigorous comparison across the models. Instead, we examine whether the machine learning-based investment payoff could still extend to economic significance in the presence of plausible

[3] While both formulations are equivalent theoretically, their empirical performance could be different. See, e.g., Kan and Zhou (1999), Cochrane (2001), and Jagannathan and Wang (2002).

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restrictions on the investment universe. The empirical experiments are based on a large sample of U.S. stocks from 1987 to 2017. The analysis proceeds as follows. To set the stage, we replicate the results reported in the original papers. The value-weighted long-short portfolio return across all stocks is 1.56% (2.18%, 0.95%, 1.16%) per month based on the GKX (CPZ, IPCA, CA) signal, and the corresponding Fama-French 6-factor-adjusted (FF6-adjusted) return is 0.92% (1.87%, 0.62%, 0.75%). Such large and significant figures reflect the impressive success of machine learning techniques in generating outstanding performance relative to traditional methods such as nonregulated regressions and portfolio sorts based on individual anomalies.

Imposing economic restrictions reveals that the predictability of deep learning methods (i.e., GKX, CPZ, and CA) weakens considerably. Relative to the full sample evidence across all stocks, the value-weighted FF6-adjusted return based on the GKX (CPZ, CA) signal is 66% (71%, 48%) lower after excluding microcaps, 53% (77%, 75%) lower after excluding firms that do not have credit rating coverage, and 78% (69%, 94%) lower after excluding financially distressed firms that face further deteriorating credit conditions. None of the deep learning methods generates significant value-weighted FF6-adjusted returns at the 5% level after excluding distressed firms. For perspective, we apply the same economic restrictions to traditional methods and find a similar proportional magnitude of performance deterioration. While IPCA underperforms deep learning models for the full sample, its performance only deteriorates modestly for subsamples that consist of cheap-to-trade stocks. Unlike deep learning models that facilitate nonlinearities, IPCA draws on the linear dependence between stock returns and firm characteristics. The evidence is thus consistent with the concept that accounting for nonlinearities is especially useful for difficult-to-value and difficult-to-arbitrage stocks.

Our findings are robust to alternative ways to implement neural networks, such as imposing economic restrictions on the training and validation samples (rather than using the entire universe of stocks), considering an alternative loss function that value weights (rather than equal weights) forecast errors, and predicting risk-adjusted (rather than raw) returns.

We then show that all four machine learning signals generate portfolio turnover that is considerably higher than most individual anomalies. The monthly turnover in the long-short portfolio is at least 87% (163%, 113%, 148%) for the GKX (CPZ, IPCA, CA) method. Altogether, in the presence of reasonable

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trading costs, the machine learning-based investments that we analyze would struggle to achieve a statistically and economically meaningful risk-adjusted payoff.

The above-described findings are further confirmed in an investment universe consisting of equity portfolios rather than individual stocks. In particular, we employ the Kozak, Nagel, and Santosh (2020) (KNS) approach to estimate the SDF. Since the SDF slope coefficients correspond to weights of the mean-variance efficient (MVE) portfolio (Hansen and Jagannathan 1991) and the MVE portfolio tends to take extreme stock positions associated with poor estimates of mean returns and the covariance matrix (e.g., Merton 1980; Green and Hollifield 1992), we examine the implications of economic restrictions on the SDF-implied MVE portfolio in terms of performance and portfolio weights. We find that the performance of the SDF-implied portfolio also deteriorates in subsamples with economic restrictions. Moreover, conditional on a predetermined level of market volatility, the SDF is estimated based on rather extreme portfolio positions. For instance, the SDF implies a -234% (-91%) short position in an individual anomaly portfolio at the 5th (25th) percentile and a 190% (96%) long position at the 95th (75th) percentile. Thus, the pricing kernel might be inadmissible from a real-time investment perspective. Restricting the investment universe to relatively cheap-to-trade stocks considerably mitigates equity positions.

We next examine whether and how machine learning-based performance varies with market conditions. Economic theory implies that fewer trading frictions and more arbitrage activity should improve price efficiency.[4] Consistent with the economic notion of limits to arbitrage, we demonstrate that the investment strategy based on GKX and CPZ signals (across all stocks) is considerably more profitable during periods of high investor sentiment, high market volatility, and low market liquidity. For instance, the monthly value-weighted FF6-adjusted return based on the GKX signal is statistically insignificant at 0.22% at times of low *VIX* and increases dramatically to 1.66% at times of high *VIX*, while the full sample average is 0.92%. In contrast, IPCA and CA models display relatively modest

[4] Prior research documents that the profitability of anomaly-based trading strategies is higher during periods of high investor sentiment (Stambaugh, Yu, and Yuan 2012; Avramov, Chordia, Jostova, and Philipov 2018), high market volatility (Nagel 2012), and low market liquidity (Chordia, Subrahmanyam, and Tong 2014).

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time series variation in trading profits and could even outperform in low limits-to-arbitrage market states.

We also examine the most recent years and find that unlike traditional anomaly-based trading strategies, all four machine learning signals continue to predict cross-sectional stock returns in the post-2001 period across all stocks. This finding supports the concept that machine learning techniques combine multiple weak sources of information into a meaningful composite signal. Consistent with our main results, anomalous return patterns in recent years are also confined to difficult-to-arbitrage stocks.

As emphasized by Karolyi and Van Nieuwerburgh (2020), it is imperative to examine the economic foundations of seemingly opaque machine learning methods. We propose two experiments to gauge the economic foundations. We first examine whether stocks with similar machine learning signals share common characteristics that are known to predict returns. The evidence shows that all four machine learning methods successfully identify mispriced stocks consistent with most anomaly-based trading strategies. Stocks in the long positions of machine learning-based investments are typically small, value, illiquid, and old stocks with a low price, low beta, high 11-month return (medium-term winners), low asset growth, low equity issuance, low credit rating coverage, and low analyst coverage. Therefore, despite their opaque nature, machine learning signals successfully identify mispriced stocks consistent with well-established empirical facts, without preselection of truly useful characteristics and models. Our findings highlight the merits of employing machine learning methods to avoid the data snooping problem in the anomaly literature and suggest that black-box-like machine learning models are reasonably interpretable, which is essential for a robust and credible assessment of out-of-sample predictability.[5]

Second, we control for the industry benchmark in machine learning signals and decompose the unconditional payoff into two components: intra-industry and inter-industry payoffs. Taking the GKX signal as an example, we construct three trading strategies based on NN3-predicted returns for all investable stocks (beyond the extreme long and short portfolios), including (1) an unconditional strategy

[5] Such interpretability does not imply that we attempt to explain *why* firm characteristics predict future returns. Instead, we confirm that machine learning generates economically interpretable trading strategies that are in line with the most common return predictors.

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that takes long (short) positions on market winners (market losers); (2) an intra-industry strategy that takes long (short) positions on industry winners (industry losers); and (3) an inter-industry strategy that takes long (short) positions on winner industries (loser industries). We show that the intra-industry strategy outperforms the unconditional strategy and the inter-industry strategy, suggesting that GKX signal is more informative for stock selection than for industry rotation. Our findings are robust to similar return decompositions based on IPCA and CA signals. Consistent with the finding that machine learning signals identify mispricing in difficult-to-arbitrage stocks, adjusting for industry averages further controls for similar firm fundamentals within the same industry and thus better predicts subsequent corrections due to market frictions.

With the recent development of financial technology (Fintech), using machine learning tools to identify new signals on price movements and to develop investment systems that can outperform human fund managers is gaining popularity (FSB 2017).[6] Our findings further support the concept that machine learning-based investments could hold considerable promise for asset management. First, we find that they are able to mitigate the downside risk and provide a good hedge during market crises. For instance, for major episodes of market downturns (e.g., the 1987 market crash, the Russian default, the bursting of the tech bubble, and the recent financial crisis), the GKX (CPZ, IPCA, CA) method generates, on average, a monthly value-weighted return of 3.56% (0.68%, 1.49%, -0.53%) after excluding microcaps. For perspective, the contemporaneous market excess return is -6.91%. Second, although the profitability of individual anomalies is driven primarily by short positions and often disappears in recent years, machine learning signals yield considerable profit in the long positions and remain viable in the post-2001 period. The performance of machine learning signals could further improve upon industry adjustment.

To conclude, our paper is the first to provide large-scale evidence on the economic significance of machine learning methods. The machine learning (especially deep learning) techniques that we analyze face many of the challenges regarding cross-sectional return predictability. In particular, anomalous return patterns characterize difficult-to-value and difficult-to-arbitrage stocks. In addition, to the extent

[6] See, e.g., Wigglesworth (2016), "Money Managers Seek AI's 'Deep Learning,'" *Financial Times*, and Zuckerman and Hope (2017), "The Quants Run Wall Street Now," *The Wall Street Journal*.

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that deep learning signals predict cross-sectional stock returns for the full sample, the trading strategy is more profitable during periods of high market volatility and low market liquidity. Machine learning signals also involve remarkably high turnover and often require taking extreme long-short positions in the tangency portfolio implied by the pricing kernel. Beyond economic restrictions, machine learning-based trading strategies nonetheless display less downside risk, yield considerable profit in long positions, and remain viable in the post-2001 period and the crisis period. Finally, black-box-like machine learning methods still generate economically interpretable trading strategies and are more meaningful for stock selection than for industry rotation.

Our analysis initiates a protocol for future work to demonstrate the feasibility of trading profits, such as (1) constructing value-weighted portfolios after excluding difficult-to-arbitrage stocks, (2) considering portfolio turnover and the corresponding transaction costs, (3) ensuring that the proposed tangency portfolio mitigates extreme long-short positions, (4) excluding high limits-to-arbitrage market states, (5) emphasizing performance over recent years, and (6) focusing on long positions.

Some recent studies have already responded to our findings and considered these economic hurdles when advocating new machine learning tools. These studies indicate the potential promise of understanding the cross-section of equity returns through the lens of deep learning and artificial intelligence (AI). For instance, CPZ continue to find a high annual Sharpe ratio of the SDF portfolio at 1.73 after removing 40% of the smallest stocks. Cong, Tang, Wang, and Zhang (2021) employ reinforcement learning to directly construct an AlphaPortfolio that maximizes the out-of-sample Sharpe ratio. The AlphaPortfolio is robust to imposing various economic restrictions in both the cross-section and time series as we propose; hence, it could be a competent and implementable approach for practitioners and advance our understanding of cross-sectional return predictability.

Taken together, our paper enriches the academic and policy discussions surrounding the adoption of machine learning techniques in asset management, including the effectiveness and sustainability of new trading signals, the lack of transparency and economic interpretability in complex machine learning algorithms, and the potential regulatory and supervisory implications related to financial stability.

The rest of the paper is organized as follows. Section II describes the methodology and data. Section III presents evidence on return predictability and other characteristics of machine learning portfolios

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for the full sample as well as subsamples with economic restrictions. Section IV studies the time-varying return predictability of machine learning methods. Section V investigates the economic foundation for machine learning methods. Section VI concludes.

II. Methodology and Data

A. Methodology and Data Sources

Our empirical analysis starts with the use of two deep learning methods that have been empirically successful in predicting future stock returns. We first implement a feed forward neural network with three hidden layers having 32, 16, and 8 neurons per layer (NN3), using batch normalization and the Lasso penalty for training.[7] According to the comparative analysis of GKX, the NN3 model displays superior out-of-sample performance relative to traditional and more advanced return forecasting benchmarks. The second approach, advocated by CPZ, combines four neural networks, including two feed forward networks (FFNs) and two recurrent neural networks (RNNs), with long short-term memory (LSTM) cells. Each of the LSTMs is connected to one FFN. The two FFN outcomes interact in the loss function to formulate a minimax optimization problem, termed the generative adversarial network (GAN).

As GKX and CPZ rely on multilayer neural networks, their works are considered deep learning routines. While GKX studies a reduced-form setup in that they do not explicitly impose economic restrictions on data, CPZ incorporates a no-arbitrage condition to estimate the SDF and its stock loadings. Specifically, CPZ employs a minimax loss minimization problem formulated as a zero-sum game. One player, the asset pricing modeler, aims to choose the best-performing model, while the other player, the adversary, attempts to choose conditions under which the model delivers the worst performance. Therefore, CPZ employs an adversarial approach to select moment conditions that lead to the largest mispricing, consistent with the findings in the seminal paper by Hansen and Richard (1987).

[7] We adopt the adaptive moment estimation algorithm (Adam) for the stochastic gradient descent used in the optimization.

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We next consider two conditional beta pricing setups in the machine learning universe, namely, instrumented principal component analysis (IPCA) and the conditional autoencoder (CA). As proposed by KPS, IPCA formulates stock returns as a linear function of latent factors and allows factor loadings to vary with observable firm characteristics. In our main analysis, IPCA is estimated in a setup that imposes a zero-alpha restriction and considers six latent factors.[8] For robustness, we also report results for unrestricted IPCA. Gu, Kelly, and Xiu (2021) further relax the linearity assumption of KPS and employ conditional autoencoder neural networks to model latent factor loadings as a flexible nonlinear function of firm characteristics. We focus on the autoencoder for which loadings on latent factors are modeled through neural networks with two hidden layers having 32 and 16 neurons per layer. The factors' neural network output layer has five neurons, indicating five latent factors (CA2).[9]

In addition to the above described routines at the stock level, we also employ the KNS approach to estimate the SDF using equity portfolios. The KNS approach aims to minimize the Hansen-Jagannathan distance (Hansen and Jagannathan 1991) using ridge regression with three-fold cross-validation.

To summarize, our battery of tests consists of **(1)** nonlinear deep learning methods (GKX, CPZ, and CA) along with linear models (IPCA and KNS); **(2)** pricing kernel formulations (CPZ and KNS), beta pricing representations (IPCA and CA), and a reduced form approach (GKX); **(3)** routines that implement stock-level analysis (GKX, CPZ, IPCA, and CA) along with KNS that focuses on equity portfolios, and **(4)** models that account for expected return variations with both firm characteristics and macro conditions (GKX and CPZ) versus firm characteristics only (IPCA, CA, and KNS).

In what follows, we describe the sample employed to implement the GKX, IPCA, and CA methods. The investment universe consists of all NYSE/AMEX/Nasdaq stocks, with daily and monthly stock data obtained from the Center for Research in Security Prices (CRSP). Quarterly and annual financial statement data come from the COMPUSTAT database. We construct 94 firm characteristics that have

[8] Kelly, Pruitt, and Su (2019) show that IPCA models with five or six latent factors fail to reject the null hypothesis that characteristics explain expected returns only because they proxy for systematic risk exposures. In addition, the total R^2 and predictive R^2 tend to converge in the restricted and unrestricted IPCA models, and the incremental explanatory power from observable factors is negligible if we include five or six latent factors. We consider six latent factors to be in line with the most recent Fama-French 6-factor model (Fama and French 2018).

[9] We thank Shihao Gu, Bryan Kelly, and Dacheng Xiu for generously sharing the data on stock-level predicted returns from 1987 to 2016.

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been documented as significant cross-sectional return predictors, including annually updated predictors such as absolute accruals and asset growth, quarterly updated predictors such as cash holdings and corporate investment, and monthly updated predictors such as 12-month momentum and idiosyncratic volatility.[10] To avoid forward-looking biases, monthly characteristics are delayed by at most one month, and quarterly and annual characteristics are delayed by at least four months and six months, respectively. We also account for 74 industry dummies based on the first two digits of Standard Industrial Classification (SIC) codes and eight monthly macroeconomic predictors, as in Welch and Goyal (2008), including the dividend price ratio, earnings price ratio, stock variance, book-to-market ratio, net equity expansion, T-bill rate, term spread, and default yield spread.[11] We consider not only stock-level and industry-level predictors but also interactions between stock characteristics and macroeconomic state variables (94 × 8), resulting in 920 predictors in total.

The full sample period ranges from 1957 to 2017. The full sample for GKX and CA is then divided into three subperiods: 18 years for the training sample (1957 to 1974), 12 years for the validation sample (1975 to 1986), and the remaining 31 years (1987 to 2017) for out-of-sample testing. We train the model every year so that every year, the training sample expands. The size of the validation sample remains fixed while we roll it forward by one year. For the out-of-sample estimation, we average across an ensemble of nine models with the same neural network architecture but distinct initial values. The final sample for out-of-sample tests consists of 21,882 stocks, with the number of stocks per month ranging between 5,117 and 7,877. In addition to the expanding window scheme for the training sample, we also conduct an experiment using a rolling window for the training sample (18 years) as well as for the validation sample (12 years). The results of using either a rolling or an expanding window for the training sample are qualitatively identical. Hence, there is no particular effect of increased learning in the training sample. IPCA skips the validation procedure and requires at least 120 months for the in-sample estimation, and forward rolling is performed on a monthly basis.

[10] Details on each of the 94 firm characteristics can be found in the Appendix in Green, Hand, and Zhang (2017) and Internet Appendix Table A.6 in Gu, Kelly, and Xiu (2020). We thank Jeremiah Green for making the SAS code used by Green, Hand, and Zhang (2017) available via his website, https://sites.google.com/site/jeremiahrgreenacctg/home.
[11] We thank Amit Goyal for making the data available via his website, http://www.hec.unil.ch/agoyal/.

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The CPZ sample consists of all U.S. stocks from CRSP with available data on 46 firm characteristics related to past returns, investment, profitability, intangibles, value, and trading frictions.[12] CPZ further include 178 macroeconomic predictors, as well as nonlinear interactions among firm characteristics and between firm characteristics and macroeconomic states. The full sample period ranges from 1967 to 2016, and it is divided into 20 years for the training sample (1967 to 1986), 5 years for the validation sample (1987 to 1991), and the remaining 25 years (1992 to 2016) for out-of-sample tests. The final sample includes 7,904 stocks, with the number of stocks per month ranging between 1,933 and 2,755. The CPZ sample is populated with fewer stocks, particularly due to the requirement to have full data records for all firm characteristics.[13] For comparison, GKX set missing characteristics to be equal to their corresponding median values across all stocks.

While the four machine learning methods we analyze could differ in their sample coverage, input variables, and optimization objectives, we do not take a stand on which is the most appropriate objective, and we do not aim to identify the best-performing method. Despite the distinct objectives of machine learning methods, such as estimating the expected returns, variances, and covariances or maximizing the ex ante Sharpe ratio, they all provide predictions for future stock returns that can be used to form trading strategies. If machine learning methods competently predict stock returns or estimate the pricing kernel, the predicted returns or loadings on the SDF should be sufficient to rank stocks and thus can be used to form outperforming investment strategies. The choice of objective is essentially a parameterization and implementation choice and relies on specific assumptions on the economic mechanism underlying the data generating process. The model performance depends on which assumption is closer to the true, unknown data generating process and remains an empirical question. Rather than comparing the models or assessing the corresponding objectives, we aim to comprehensively examine whether machine learning signals could provide value that extends to economic significance.

B. Subsamples with Economic Restrictions

[12] We thank Markus Pelger for generously sharing the data on stochastic discount factor and stock-level factor loadings from 1967 to 2016. Details on each of the 46 firm characteristics can be found in the Appendix C in Chen, Pelger, and Zhu (2020).
[13] Unreported results confirm our main findings in an overlapping sample for all machine learning methods.

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To set the stage, we replicate all machine learning methods using the full sample, as in the original studies. We then move on to restrict the sample to the universe of relatively cheap-to-trade stocks. In particular, the extant literature highlights that anomalous patterns in the cross-section of asset returns are concentrated in microcap stocks.[14] For instance, Novy-Marx and Velikov (2016) show that microcaps display high gross Sharpe ratios in most anomalies relative to other size groups, but the difference is much smaller in net Sharpe ratios after accounting for transaction costs. In the same vein, Hou, Xue, and Zhang (2020) document that 65% (82%) of anomalies are statistically insignificant after excluding microcaps based on NYSE breakpoints as well as when employing value-weighted returns and the cutoff *t*-statistic of 1.96 (2.78), suggesting that capital markets might be more efficient than previously thought.

Thus, on the one hand, the collective evidence indicates that microcaps are costly to trade to the extent that anomalies in those firms could easily become unprofitable for marginal investors. On the other hand, machine learning techniques are particularly useful for uncovering complex patterns and hidden relationships and for combining multiple weak sources of information into a composite signal, and they are often more effective than linear regressions in handling multicollinearity (e.g., Rasekhschaffe and Jones 2019; Gu, Kelly, and Xiu 2020; Karolyi and Van Nieuwerburgh 2020). Altogether, a natural question to explore is whether machine learning techniques are able to predict cross-sectional stock returns for stocks other than microcaps. Thus, our first subsample excludes microcaps.

In the same vein, our second subsample includes only rated firms, i.e., firms with data on Standard & Poor's (S&P) long-term issuer credit ratings.[15] In a given month, approximately 90% of the rated firms are above the 20th NYSE size percentile. Unreported results also show that rated firms tend to be large, value firms with considerably higher past returns and liquidity, lower idiosyncratic volatility, and

[14] Fama and French (2008) recognize microcaps as stocks with a market capitalization smaller than the 20th NYSE size percentile.

[15] We obtain the monthly S&P long-term issuer credit ratings from the COMPUSTAT database. We follow Avramov, Chordia, Jostova, and Philipov (2009) in creating the numeric rating score, which transforms the S&P ratings into ascending numbers as follows: AAA = 1, AA+ = 2, AA = 3, AA– = 4, A+ = 5, A = 6, A– = 7, BBB+ = 8, BBB = 9, BBB– = 10, BB+ = 11, BB = 12, BB– = 13, B+ = 14, B = 15, B– = 16, CCC+ = 17, CCC = 18, CCC– = 19, CC = 20, C = 21, and D = 22. A higher credit rating score implies higher credit risk.

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more analyst coverage than nonrated firms. Thus, we focus on relatively cheap-to-trade stocks by excluding nonrated firms.

The third subsample imposes an additional filter on the universe of rated firms. In particular, Avramov, Chordia, Jostova, and Philipov (2013, 2018) show that market anomalies among stocks and corporate bonds tend to concentrate in financially distressed firms and particularly around credit rating downgrades. Their suggested mechanism is straightforward. Distressed firms display extreme values of predictive characteristics such as more negative past returns, high idiosyncratic volatility, a high fraction of negative earnings surprises, and high analyst dispersion; thus, they are sorted into the short leg of anomaly portfolios. The sluggish response to financial distress by retail and institutional investors leads to a wide range of anomalous patterns in the cross-section of stock and bond returns. Collectively, on the one hand, investors tend to overprice financially distressed stocks. On the other hand, credit rating downgrades are associated with substantially elevated trading frictions, and therefore, overpricing cannot be easily arbitraged away. Thus, the third subsample excludes distressed firms around credit rating downgrades. Specifically, among rated firms, we further exclude stock-month observations from 12 months before to 12 months after an issuer credit rating downgrade.

We report the number of stocks in each year for the full sample and three subsamples in Internet Appendix Table IA1. Notably, applying sensible restrictions significantly reduces the number of stocks. The average monthly GKX (CPZ) sample is reduced by 49% (43%) after excluding microcaps, 78% (71%) after excluding nonrated firms, and 83% (78%) after excluding financially distressed firms around credit rating downgrades. Consequently, the existing evidence on machine learning-based investments could be dominated by stocks that are plentiful albeit low in aggregate market value and difficult to value and arbitrage.[16] This preliminary finding further motivates us to explore whether machine learning methods can clear common economic restrictions in empirical finance.

[16] While some prior work account for size-based adjustments, the market cap considerations in existing studies are not unified. For instance, GKX report equal-weighted performance when microcaps are excluded (Table A.10) and value-weighted performance without market cap restrictions (Table 7); CPZ consider the market cap subsample threshold as a parameter; KPS partition the sample into large and small stocks, where the former is the top 1,000 stocks in terms of market cap and the latter consists of the rest; and KNS restrict the investment universe to stocks with market caps greater than 0.01% of the aggregate stock market capitalization. Our approach is unified across all methods, i.e., excluding microcaps and value weighting, as proposed by Hou, Xue, and Zhang (2020) in the context of individual anomalies. We also consider restrictions that go beyond the market cap in both the cross-section and time series. Resorting to such a unified approach delivers new and previously undocumented evidence on economic significance.

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III. Return Predictability in the Presence of Economic Restrictions

We assess return predictability using conventional portfolio sorts. In particular, at the end of each month t, we construct portfolios using the four proposed machine learning signals: **(1)** the one-month-ahead out-of-sample stock return prediction using the NN3 model (GKX); **(2)** the risk loadings on the SDF estimated from a combination of deep neural networks, as noted earlier (CPZ); **(3)** the one-month-ahead out-of-sample stock return prediction using the IPCA model (KPS), and **(4)** the one-month-ahead out-of-sample stock return prediction using the CA model (Gu, Kelly, and Xiu 2021). A higher value of predicted returns and risk loadings indicates higher expected returns in the holding period. We sort stocks into decile portfolios based on predicted returns or risk loadings and evaluate portfolio returns in month $t + 1$. The bottom (top) decile consists of stocks with the lowest (highest) expected returns in the next month. We compute equal-weighted and value-weighted holding period returns for each decile portfolio. We also implement the zero-cost trading strategy by taking long positions in the top decile of stocks (highest expected returns) and selling short stocks in the bottom decile (lowest expected returns). The payoff of the long-short investment strategy is computed as the high (top decile) minus low (bottom decile) portfolio return ("HML").[17]

In addition to raw portfolio returns, we report risk-adjusted returns from **(1)** the CAPM, i.e., only adjusting for the market factor (MKT, defined as the excess return on the value-weighted CRSP market index over the 1-month T-bill rate); **(2)** the Fama-French-Carhart 4-factor plus liquidity factor model (FFC+PS) consisting of the market factor (MKT), the size factor (SMB, defined as small minus big firm return premium), the book-to-market factor (HML, defined as high book-to-market minus low book-to-market return premium) (Fama and French 1993), Carhart (1997) momentum factor (MOM, defined as winner minus loser return premium), and Pástor-Stambaugh (2003) liquidity factor; **(3)** the Fama-French 5-factor model (FF5) consisting of the market factor (MKT), the size factor (SMB), the book-to-market factor (HML), the profitability factor (RMW, defined as robust minus weak return

[17] We follow the existing literature and report performance based on decile portfolios, e.g., Table 7 in GKX, Figure 7 and Table II in CPZ, Table 3 in Gu, Kelly, and Xiu (2021) for both IPCA and CA methods. In addition, we report out-of-sample Sharpe ratios in Tables 5 and 6 for all machine learning signals in both the full sample and three subsamples. As a robustness check, we also report investment payoff based on *all* stocks (instead of the extreme deciles) in Table 10 and Internet Appendix Table IA9.

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premium), and the investment factor (CMA, defined as conservative minus aggressive return premium) (Fama and French 2015); **(4)** the Fama-French 6-factor model (FF6) that adds the momentum factor (MOM) to FF5 (Fama and French 2018), and **(5)** the Stambaugh-Yuan 4-factor model (SY) consisting of the market factor (MKT), the size factor (SMB), and two mispricing factors arising from the cluster of anomalies related to firm management (MGMT) and performance (PERF) (Stambaugh and Yuan 2017).[18] We rely on a single time series regression to estimate the out-of-sample alpha, while our main findings remain unchanged if we estimate time-varying beta for the long-short portfolio using a five-year rolling window. The standard errors in all estimations are corrected for autocorrelation with four lags using the Newey and West (1987) method.[19]

A. Evidence from NN3-Predicted Returns

We start by assessing the out-of-sample return predictability of the GKX method. Table 1 reports the value-weighted results. In the interest of brevity, we tabulate the equal-weighted results in Internet Appendix Table IA2 and only discuss the main findings in this subsection. As shown in Panel A of Table 1, using all stocks in our sample, the value-weighted long-short portfolio return is 1.56% per month (*t*-statistic = 4.53) over the 1987–2017 sample period, and the risk-adjusted return ranges between 0.77% and 1.89% with a *t*-statistic above 3.03 across all factor models. In contrast to the equal-weighted results, the profits weaken considerably in value-weighted portfolios, and the average decline in economic magnitude is 48% across all performance measures. In addition, the long position on stocks with the highest expected returns generates a significant and economically larger payoff than the short position. For instance, the long leg yields a significant FF6-adjusted return of 0.77% per month, while the corresponding payoff on the short leg falls to a statistically insignificant -0.15%.

The evidence on the relative strength of the long versus short leg appears to be at odds with the literature documenting that the profitability of anomaly-based trading strategies is attributable primarily to the short leg of the trade (e.g., Hong, Lim, and Stein 2000; Stambaugh, Yu, and Yuan 2012; Avramov,

[18] We thank Kenneth French and Robert Stambaugh for making the common factor returns available via their websites: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html; https://fnce.wharton.upenn.edu/profile/stambaug/. The results using Stambaugh and Yuan (2017) factors end in 2016 due to data availability.

[19] Newey and West (1987) suggest using lags corresponding to $N^{1/4}$, where *N* refers to the number of observations. Our sample includes 372 monthly observations; thus, we use four lags.

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Chordia, Jostova, and Philipov 2013, 2018). The evidence, however, supports the concept that machine learning routines possess superior ability to detect complex features in the data that otherwise remain unnoticed. The empirical success of long positions could be particularly valuable for institutions, such as mutual funds and pension funds, that focus primarily on long positions.

We next exclude microcaps, i.e., stocks with a market capitalization smaller than the 20th NYSE size percentile at the end of the portfolio formation month t. As shown in Panel A of Table 1, the value-weighted long-short portfolio return is significant at 1.05% per month. The performance further deteriorates after adjusting for risk exposures using the FF6 model, i.e., a statistically insignificant 0.31% per month, in contrast to 0.92% in the full sample. In addition, the predictive power of the GKX method in cross-sectional stock returns is adequately captured by the Stambaugh and Yuan (2017) four factors, resulting in a statistically insignificant SY-adjusted return for both equal-weighted and value-weighted long-short portfolios. Relative to the full sample, the equal-weighted (value-weighted) payoff is 65% (49%) lower after excluding microcaps across all performance measures.

The second subsample considers only rated firms, i.e., firms with data on S&P long-term issuer credit ratings. We sort all rated firms into decile portfolios based on NN3-predicted returns and calculate holding period returns. We tabulate the results in Panel B of Table 1. The value-weighted long-short portfolio yields a significant return of 1.02% per month and an FF6-adjusted return of 0.43% per month. Excluding nonrated firms creates a 57% (46%) decline in economic magnitude across all equal-weighted (value-weighted) performance measures relative to the full sample.

The third subsample excludes observations on distressed firms around credit rating downgrades. In particular, among rated firms, we further exclude stock-month observations from 12 months before to 12 months after an issuer credit rating downgrade. This is not a real-time trading strategy, as we look ahead when discarding the 12-month period prior to the downgrade. However, we aim to investigate whether the trading profits generated from machine learning algorithms go beyond a small subset of firm-months (i.e., distressed firms around credit rating downgrades).[20] As shown in Panel B of Table 1, during periods of improving or stable credit conditions, the value-weighted long-short portfolio yields

[20] The stock-month observations from 12 months before to 12 months after an issuer credit rating downgrade account for 23% of the credit rating sample.

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a significant return of 0.72% per month, while the FF6-adjusted return is no longer statistically significant. Moreover, only one (two) of five risk-adjusted returns remains significant at the 5% threshold for the equal-weighted (value-weighted) portfolio. Excluding distressed firms creates an 86% (70%) decline in economic magnitude across all equal-weighted (value-weighted) performance measures relative to the full sample.

For perspective, we consider two traditional methods as benchmarks. First, we apply the standard OLS regression methodology with all 920 predictors used in the GKX estimation to predict the one-month-ahead out-of-sample stock returns. For consistency, the sample starts from 1957, and the out-of-sample test ranges from 1987 to 2017. We estimate the OLS regression every month, and the sample expands over time. Next, we sort all firms into decile portfolios based on OLS-predicted returns and calculate holding period returns. Given the multidimensional challenge in the OLS regression, our second benchmark relies on portfolio sorts while combining the payoff on individual anomalies. Specifically, we sort stocks into deciles according to each of the 94 firm characteristics. We adjust the sign of those characteristics so that a higher value predicts higher future performance, in accordance with the literature. We compute the equal- and value-weighted holding period returns for each characteristic and then average the portfolio returns across 94 characteristics.

We tabulate the results in Internet Appendix Table IA3, where Panel A shows the results for portfolios sorted by OLS-predicted returns and Panel B shows those for the combination of individual anomalies. We report only the performance of long-short portfolios for brevity. As expected, the GKX method substantially improves the investment payoff relative to the traditional methods. It outperforms the standard OLS by 82% (99%) across all equal-weighted performance measures for the full sample (after excluding microcaps), and the OLS method fails to deliver significant value-weighted payoffs for the full sample as well as three subsamples. Relative to the full sample, the equal-weighted payoff based on OLS-predicted returns is 66% (64%, 64%) lower after excluding microcaps (nonrated firms, distressed firms) across all performance measures, while the corresponding value-weighted payoff is 40% (43%, 76%) lower.

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Moving to the combination of individual anomalies as an alternative benchmark, the GKX method generates a payoff that is approximately seven (nine) times higher across all equal-weighted (value-weighted) performance measures for the full sample and continues to outperform in all three subsamples. Relative to the full sample, the equal-weighted payoff based on individual anomalies is 44% (25%, 77%) lower after excluding microcaps (nonrated firms, distressed firms) across all performance measures, while the corresponding value-weighted payoff is 38% (55%, 106%) lower.

In summary, when we apply standard economic restrictions and focus exclusively on the subset of relatively cheap-to-trade stocks, the long-short portfolio performance weakens considerably in terms of both statistical significance and economic magnitude. On the one hand, machine learning substantially improves the investment payoff compared to traditional methods such as OLS regression and portfolio sorts based on individual anomalies. On the other hand, both machine learning and traditional methods deliver lower payoffs in the presence of economic restrictions, and their performance deteriorates by a similar proportional magnitude.

One may argue that since we train the NN3 model using the entire universe of stocks, this training scheme could be biased in favor of detecting predictive patterns, especially for microcaps, nonrated firms and distressed firms. However, as long as there are shared attributes between the subsamples, training based on the comprehensive universe could benefit from the rich return structures in big data and capture subsample patterns better. Nevertheless, as a robustness check, we retrain the NN3 model in each subsample separately and assess the out-of-sample return predictability.

The results are tabulated in Table 2. We report only the value-weighted performance of long-short portfolios for brevity and tabulate the equal-weighted results in Internet Appendix Table IA4. First, we exclude microcaps in both the NN3 estimations and the portfolio sorts. Relative to the full sample, the equal-weighted (value-weighted) payoff is 54% (37%) lower after excluding microcaps across all performance measures. Moreover, there is a modest improvement in trading profits when we exclude microcaps in the training and validation samples. The value-weighted return (FF6-adjusted return) is 1.19% (0.49%) per month, in contrast to 1.05% (statistically insignificant at 0.31%) when the machine

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learning algorithm is trained with the full sample, while the SY-adjusted return remains statistically insignificant.

Next, we exclude nonrated firms from both the NN3 estimations and the portfolio sorts. Since the data on S&P long-term issuer credit ratings are sparse before December 1985, the out-of-sample test starts in 1999. The training and validation samples are defined as in GKX. To put our findings in perspective, we repeat the analysis from Table 1 during the post-1999 period and show that excluding nonrated firms creates a 50% (52%) decline in economic magnitude across all equal-weighted (value-weighted) performance measures relative to the full sample. In addition, for investors interested in trading rated firms, training the machine learning algorithm with the full sample could increase the trading profits. When we train the NN3 model using the entire universe of stocks and exclude nonrated firms in portfolio construction, the value-weighted long-short portfolio yields a significant FF6-adjusted return of 0.72% per month. In contrast, training with the subset of rated firms generates a statistically insignificant monthly FF6-adjusted return of 0.41%.

We further exclude distressed firms around credit rating downgrades from the subsample of rated firms in both the NN3 estimations and the portfolio sorts. Excluding distressed firms creates an 82% (84%) decline in economic magnitude across all equal-weighted (value-weighted) performance measures relative to the full sample, and the FF6-adjusted return is statistically insignificant for both equal-weighted and value-weighted portfolios. Furthermore, training the machine learning algorithm with the full sample also generates higher value-weighted trading profits for nondistressed firms.

Overall, training the machine learning algorithm in subsamples with economic restrictions does not alter our main findings, i.e., investment payoffs on machine learning-based trading strategies deteriorate in the presence of sensible economic restrictions. In addition, training the machine learning algorithm using a selective set of stocks of interest could adversely affect the out-of-sample return predictability potentially as a result of data scarcity.

Since we focus on value-weighted portfolio returns on a risk-adjusted basis in the out-of-sample test, we also employ an alternative objective function to train the NN3 model. Instead of minimizing the equal-weighted forecast errors in predicting the raw returns, we train the model to minimize the value-weighted forecast errors in predicting the FF6-adjusted returns. This allows us to tilt the estimates

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towards large stocks and focus on predicting risk-adjusted returns in training and validating the machine learning algorithm.

The results are tabulated in Internet Appendix Table IA5, where Panel A shows the results for portfolios sorted by NN3-predicted alphas and Panel B shows the results for portfolios sorted by NN3-predicted returns (as in GKX) using the same universe of stocks.[21] In Panel A, the monthly value-weighted FF6-adjusted return is 0.61% for the full sample, 0.38% after excluding microcaps, and becomes statistically insignificant after excluding nonrated firms or credit rating downgrades. In addition, the trading profits based on NN3-predicted alphas using the value-weighted loss function do not outperform the original GKX method (i.e., NN3-predicted returns using the equal-weighted loss function) for the full sample or for most subsamples with economic restrictions, implying that the seemingly more aligned objective function does not necessarily improve the predictive performance. Collectively, our main findings are robust to the alternative objective function that tilts towards predicting risk-adjusted returns for large stocks.

In short, our findings are robust to alternative ways to implement the neural network model, such as imposing economic restrictions on the training and validation samples (rather than using the entire universe of stocks), as well as considering an alternative loss function that value weights (rather than equal weights) forecast errors of risk-adjusted returns (rather than raw returns). As Karolyi and Van Nieuwerburgh (2020) point out, our experiments also call for more research on the best practices to make implementation choices when applying machine learning models to financial data, such as how to balance the trade-off between completeness and relevance in sample selection and how to set an appropriate objective function.

B. Evidence from the Adversarial Approach

We implement the analysis from Table 1 but employ the deep learning signal based on the adversarial approach of CPZ. Table 3 has the same layout as Table 1, where Panel A shows the results for the full sample and the subsample excluding microcaps, and Panel B shows those for the subsample

[21] Given that we require at least 24 observations for alpha estimation, the sample size is slightly smaller than the main results. However, as shown in Internet Appendix Table IA5, Panel B, the findings are qualitatively and quantitatively comparable to those in Table 1.

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of rated firms and the subsample excluding credit rating downgrades. For brevity, we tabulate the equal-weighted results in Internet Appendix Table IA6.

As shown in Panel A, over the 1987–2016 sample period, the long-short portfolio return (FF6-adjusted return) across all stocks is highly significant and economically large, i.e., 2.18% (1.87%) per month in the value-weighted portfolio. Similar to our prior findings, the economic magnitude is considerably attenuated after excluding microcaps, i.e., the long-short portfolio return (FF6-adjusted return) declines to 1.08% (0.55%). Across all performance measures, the equal-weighted (value-weighted) trading profit is 60% (61%) lower for the subsample excluding microcaps relative to the full sample.

As shown in Panel B, the value-weighted long-short strategy across rated firms yields a monthly return of 0.81%, and the FF6-adjusted return is statistically insignificant. If we further exclude credit rating downgrades, the long-short portfolio return (FF6-adjusted return) is 0.92% (0.57% with *t*-statistic = 1.82). Across all performance measures, the equal-weighted (value-weighted) trading profit is 60% (72%) lower for the subsample of rated firms relative to the full sample and 65% (64%) lower when we further exclude firms with deteriorating credit conditions. Thus, investment payoffs based on the CPZ signal also deteriorate in the presence of sensible economic restrictions.

C. Evidence from IPCA and the Conditional Autoencoder

In this subsection, we employ alternative machine learning signals based on the instrumented principal component analysis (IPCA) approach of KPS and the conditional autoencoder (CA) model of Gu, Kelly, and Xiu (2021). Both IPCA and CA extract latent factors consistent with the notion of principal component analysis (PCA). However, unlike the standard PCA, both IPCA and CA allow factor loadings to vary with predictive characteristics either linearly (IPCA) or nonlinearly through neural networks (CA).[22]

We implement similar portfolio analysis as before. Table 4 reports the results, where Panel A shows the results for portfolios sorted by IPCA-predicted returns and Panel B shows those for portfolios sorted

[22] Gomes, Kogan, and Zhang (2003) formulate an equilibrium model in which beta could vary with firm-level predictors, such as size and book-to-market, while Avramov and Chordia (2006) show empirically that such conditional formulations improve the pricing ability of the model.

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by CA2-predicted returns. For brevity, we present only raw returns and FF6-adjusted returns for the full sample as well as for subsamples that exclude microcaps, nonrated firms, and credit rating downgrades.[23]

As shown in Panel A, the value-weighted long-short portfolio return (FF6-adjusted return) across all stocks is 0.95% (0.62%) per month based on the IPCA signal, which is 39% (32%) lower than that based on the GKX signal, as shown in Table 1, and 57% (67%) lower than that based on the CPZ signal, as shown in Table 3. More importantly, there is no material deterioration of performance in subsamples with economic restrictions. The long-short strategy based on the IPCA signal generates significant FF6-adjusted returns for all subsamples, ranging between 0.43% and 0.61% per month. Although the IPCA signal underperforms GKX and CPZ for the full sample, it outperforms the GKX signal by 82% and the CPZ signal by 10% based on FF6-adjusted returns across the three subsamples with economic restrictions. While IPCA draws on linear dependence between average returns and firm characteristics, deep learning models already facilitate nonlinearities. The evidence is thus consistent with the concept that accounting for nonlinearities is especially useful for microcaps and difficult-to-arbitrage stocks.[24]

In Panel B of Table 6, the value-weighted long-short portfolio return (FF6-adjusted return) across all stocks is 1.16% (0.75%) per month based on the CA signal, and the performance weakens considerably in subsamples with economic restrictions. The monthly FF6-adjusted return for the long-short portfolio declines to 0.39% after we exclude microcaps and becomes statistically insignificant after we exclude nonrated firms or distressed firms.

Finally, we conduct two sets of robustness checks. First, to address the potential inconsistency between the in-sample estimations (based on the full sample) and the portfolio sorts (based on subsamples with economic restrictions), we re-estimate the IPCA for each subsample separately. When credit rating data are required for the in-sample estimation, the out-of-sample test begins in 1996. The results are reported in Internet Appendix Table IA7, Panel A. Excluding microcaps from the IPCA

[23] Stocks that are plentiful but low in aggregate market value may dominate equal-weighted portfolio returns, and value-weighting accurately captures the total wealth effect experienced by investors (Fama 1998); therefore, we focus on value-weighted results in all subsequent analyses. Unreported results indicate similar findings for equal-weighted portfolios.
[24] Without an exhaustive analysis of candidate machine learning models, it is premature to identify the best-performing method in the presence of economic restrictions. We leave this issue for future research.

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estimation yields a value-weighted FF6-adjusted return of 0.76% per month, in contrast to 0.61% when IPCA is estimated for the full sample. If we exclude microcaps when training the NN3 model and IPCA estimation, the IPCA signal still outperforms the GKX signal (the corresponding FF6-adjusted return is 0.49%, as shown in Table 2). Moreover, estimating IPCA for the subsample of rated or nondistressed firms considerably reduces the model performance, and all long-short portfolio payoffs are no longer significant at the 5% threshold. Consistent with our early experiment on NN3 estimation (Table 2), restricting the in-sample estimation to a selective set of stocks of interest could adversely affect the out-of-sample return predictability.

Another robustness check is to estimate the unrestricted version of IPCA that allows for intercepts as functions of the instruments. As shown in Internet Appendix Table IA7, Panel B, we do not find a material deterioration of performance for subsamples excluding microcaps and nonrated firms, while the FF6-adjusted return is no longer statistically significant once we exclude distressed firms.

Collectively, the innovative machine learning (especially deep learning) techniques that we analyze face the usual challenge of cross-sectional return predictability, and anomalous return patterns are concentrated in stocks that are relatively difficult to value and difficult to arbitrage. The trading profits based on deep learning signals in GKX, CPZ, and CA often disappear on a risk-adjusted basis after we impose common economic restrictions in empirical finance, such as excluding microcaps, nonrated firms or distressed firms. Although IPCA underperforms nonlinear deep learning models for the full sample, it delivers consistent risk-adjusted performance even in the presence of economic restrictions. Despite the challenge of generating sizable trading profits using the existing machine learning tools, all machine learning methods substantially improve the investment payoff compared to the traditional methods with and without economic restrictions.

D. Nonnormality, Turnover, and Trading Costs of Machine Learning Portfolios

Beyond out-of-sample return predictability, investors should be concerned with other potential risks and costs in applying investment strategies. Focusing on individual anomalies, past work shows that anomaly payoffs are prone to large drawdowns. Daniel and Moskowitz (2016) document that momentum strategies are characterized by occasional large crashes. A recent work by Arnott, Harvey,

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Kalesnik, and Linnainmaa (2019) shows that nine out of 14 popular factors are fat-tailed and asymmetric on the downside. In addition, transaction costs could significantly reduce the anomaly payoff. Novy-Marx and Velikov (2016) find that most low-turnover and mid-turnover strategies remain profitable after accounting for transaction costs, while no high-turnover strategy achieves significantly positive net excess return.

In this subsection, we explore the downside risk and turnover associated with machine learning portfolios. As described before, we sort stocks into decile portfolios according to four machine learning signals. We compute the value-weighted holding period return for each decile portfolio and implement a zero-cost trading strategy by taking long positions in the top decile of stocks and selling short stocks in the bottom decile. We also include the market portfolio as a benchmark, and market excess return is defined as value-weighted CRSP market return in excess of the 1-month T-bill rate. We report the annualized Sharpe ratio, the skewness and excess kurtosis of the monthly returns,[25] the maximum drawdown, the average monthly return during the crisis period, and the monthly turnover for the long-short machine learning portfolios and market portfolio.

In particular, we follow GKX to define the maximum drawdown of a strategy as $MDD = \max_{0 \le t_1 \le t_2 \le T} (Y_{t_1} - Y_{t_2})$, where Y_{t_1} and Y_{t_2} refer to the cumulative log return from month 0 (i.e., January 1987) to t_1 and t_2, respectively. In addition, the crisis period includes the market crash in October 1987, the Russian default in August 1998, the bursting of the tech bubble in April 2000, the Quant crisis in August 2007, and the Bear Stearns bailout and Lehman bankruptcy during the recent financial crisis, i.e., March, September, and October 2008 (Griffin, Harris, Shu, and Topaloglu 2011; Cella, Ellul, and Giannetti 2013).

We also follow GKX to define the turnover in month t as $TO_t = \frac{1}{2}\sum_{i \in L} \left| w_{i,t} - \frac{w_{i,t-1}(1+r_{i,t})}{\sum_i w_{i,t-1}(1+r_{i,t})} \right| + \frac{1}{2}\sum_{j \in S} \left| w_{j,t} - \frac{w_{j,t-1}(1+r_{j,t})}{\sum_j w_{j,t-1}(1+r_{j,t})} \right|$, where $i \in L$ ($j \in S$) indicates that stock i (j) belongs to the entire universe of long positions (short positions) in months t and $t-1$, $w_{i,t}$ ($w_{j,t}$) refers to the weight of stock i (j) in the portfolio in month t, and $r_{i,t}$ ($r_{j,t}$) refers to the return of stock i (j) in month t. By

[25] For perspective, the skewness and excess kurtosis are equal to zero under a normal distribution.

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construction, the one-side turnover for long positions or short positions ranges between 0 and 1, and the turnover in the long-short portfolio, i.e., TO_t, ranges between 0 and 2.

We tabulate the results in Table 5, where Panel A shows the results for portfolios sorted by the NN3-predicted returns (GKX), Panel B shows those for portfolios sorted by the risk loadings on the SDF (CPZ), Panel C shows those for portfolios sorted by the IPCA-predicted returns (KPS), Panel D shows those for portfolios sorted by the CA2-predicted returns (Gu, Kelly, and Xiu 2021), and Panel E shows those for the market portfolio. We report results for the full sample as well as subsamples that exclude microcaps, nonrated firms, and credit rating downgrades. Following Arnott, Harvey, Kalesnik, and Linnainmaa (2019), all returns are scaled to 10% volatility per year to facilitate comparisons across various samples and methods.

First, machine learning portfolios earn an annualized Sharpe ratio ranging between 0.78 and 1.23 for the full sample, compared to 0.53 for the market portfolio. Imposing economic restrictions significantly reduces the Sharpe ratio, although machine learning methods still outperform the market in most subsamples except for the one that excludes credit rating downgrades.

Second, both the GKX and CPZ methods display positive skewness and excess kurtosis for the full sample and all three subsamples, while the market portfolio is negatively skewed. Additionally, both IPCA and CA methods exhibit negative skewness for the full sample and most subsamples, although they are less negatively skewed than the market portfolio.

Third, machine learning methods are able to mitigate downside risk and protect investors from extreme crashes. The maximum drawdown for machine learning portfolios ranges between 20% and 35% for the full sample, while the market portfolio experiences a larger drawdown at 49%. Among most subsamples, machine learning methods also experience comparatively smaller drawdowns than the market portfolio. In addition, the average monthly returns on machine learning-based trading strategies are mostly positive during the crisis period, i.e., 2.93% to 4.10% for the GKX method, -0.02% to 0.90% for the CPZ method, -0.64% to 1.49% for the IPCA method, and -1.80% to -0.05% for the

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CA method. All machine learning methods exhibit a significant improvement from the average market return of nearly -7% contemporaneously.[26]

Finally, all machine learning methods require high turnover in portfolio rebalancing. The monthly turnover in the long-short portfolio ranges between 87% and 98% for the GKX method, between 163% and 168% for the CPZ method, between 113% and 119% for the IPCA method, and between 148% and 157% for the CA method.[27] This creates a one-side turnover (average turnover on the long and short sides) of at least 43% for the GKX method, 81% for the CPZ method, 56% for the IPCA method, and 74% for the CA method. To put this in perspective, low-turnover strategies such as size and value typically display monthly one-side turnover of below 10%; the corresponding number is between 14% and 35% for mid-turnover strategies such as failure probability and idiosyncratic volatility and above 90% for high-turnover strategies such as short-run reversals and seasonality (Novy-Marx and Velikov 2016). Recall that the GKX (CPZ, IPCA, CA) method generates a value-weighted monthly FF6-adjusted return of 0.92% (1.87%, 0.62%, 0.75%) for the full sample and 0.31% (0.55%, 0.61%, 0.39%) after excluding microcaps; we can therefore infer a break-even transaction cost of 0.94% (1.12%, 0.53%, 0.48%) for the long-short portfolio in the full sample and 0.36% (0.34%, 0.54%, 0.26%) for those in the subsample excluding microcaps to completely absorb the FF6-adjusted return in the GKX (CPZ, IPCA, CA) method.[28]

Such break-even transaction cost estimates seem moderate relative to those in the existing literature. Novy-Marx and Velikov (2016) focus on value-weighted portfolios sorted by the signals using NYSE breakpoints between 1963 and 2013 and suggest that transaction costs account for more than 1% of the

[26] For perspective, we assess the performance of other long-short portfolios using the FF6 model for the crisis period. Both the size factor (SMB) and the momentum factor (MOM) yield negative returns, i.e., -3.16% for SMB and -0.20% for MOM, while some other factors provide a good hedge during the crisis, i.e., 2.44% for the book-to-market factor (HML), 3.21% for the profitability factor (RMW), and 3.46% for the investment factor (CMA). All returns are scaled to 10% annual volatility.

[27] Unreported results show that the monthly turnover in the equal-weighted long-short portfolio ranges between 79% and 82% for the GKX method, between 141% and 147% for the CPZ method, between 99% and 103% for the IPCA method, and between 130% and 136% for the CA method.

[28] We compute the break-even transaction cost as the payoff to the long-short investment strategy divided by its turnover. For instance, the break-even transaction cost for GKX method for the full sample equals 0.916%/0.976 = 0.94%, where 0.916% is the value-weighted FF6-adjusted return (Table 1, Panel A), and 0.976 is the corresponding turnover (Table 5, Panel A). Unreported results show that if we exclude microcaps from the training sample (as shown in Table 2), the monthly turnover in the value-weighted long-short portfolio remains at 99% for the GKX method, which further creates a break-even transaction cost of 0.49%. In addition, when we employ NN3 with the value-weighted loss function to predict alpha (as shown in Internet Appendix Table IA5, Panel A), the monthly turnover in the value-weighted long-short portfolio in the full sample remains at 106%, which further creates a break-even transaction cost of 0.57%.

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monthly one-sided turnover (equivalent to 0.5% of the long-short portfolio turnover as in our estimates) and that the statistical significance of the return spread decreases proportionately. Since Novy-Marx and Velikov (2016) sort portfolios based on NYSE breakpoints, the long-short portfolios are less likely to be dominated by small firms, and their estimates could be more relevant for our subsample excluding microcaps.[29]

Alternatively, several papers argue that investors face proportional transaction costs that decrease with firm size and over time (e.g., Brandt, Santa-Clara, and Valkanov 2009; Hand and Green 2011). We follow Brandt, Santa-Clara, and Valkanov (2009) and define the one-way transaction cost of stock i in month t as $c_{i,t} = z_{i,t} \times T_t$, where $z_{i,t} = 0.006 - 0.0025 \times ME_{i,t}$, $ME_{i,t}$ refers to the market capitalization of stock i in month t and is normalized to be between 0 and 1. T_t decreases linearly from 2.6 in 1987 to 1 in 2002 and remains at 1 thereafter.[30] As a result, the transaction cost is approximately 1.56% for the smallest firms and 0.91% for the largest firms in 1987 and approximately 0.6% for the smallest firms and 0.35% for the largest firms after 2002. This indicates an average transaction cost of 0.67% for the full sample and 0.64% for the subsample excluding microcaps over the entire sample period from 1987 to 2017.

For all machine learning methods, our break-even transaction costs, which range between 0.48% and 1.12% for the full sample and between 0.26% and 0.54% for the subsample excluding microcaps, appear to be mostly below or approximately the same as the estimated costs, i.e., 0.5% to 0.67%. Therefore, accounting for reasonable transaction costs would make it difficult for most machine learning signals to leave alpha on the table. However, our findings do not imply that machine learning-based trading strategies are unprofitable for *all* traders after accounting for transaction costs. Instead, we show that it is challenging for an average investor to achieve statistically and economically meaningful risk-adjusted performance in the presence of reasonable transaction costs. Investors may thus need to adjust their expectations of the actual investment returns they can receive. While we

[29] One caveat of applying the transaction costs estimates from Novy-Marx and Velikov (2016) is that the transaction costs are based on effective bid-ask spread and discard the price impact, and therefore underestimate the true costs faced by large traders in the market. On the other hand, the transaction costs are computed from aggregate data including multiple trade types and traders; hence, they overestimate the actual trading costs of a sophisticated institutional trader (Frazzini, Israel, and Moskowitz 2018).

[30] Brandt, Santa-Clara, and Valkanov (2009) define T_t so that transaction costs in 1974 are four times larger than those in 2002. We follow their approach to compute a starting value of 2.6 in 1987.

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explicitly focus on the universe of cheap-to-trade stocks to assess economic significance, from the modeling perspective, investors who are sensitive to transaction costs could modify the existing off-the-shelf machine learning models and construct portfolios under constraints, hence optimizing after-trading-cost performance. Potentially promising examples are presented in Bryzgalova, Pelger, and Zhu (2020), Chen, Pelger, and Zhu (2020), Allena (2021), and Cong, Tang, Wang, and Zhang (2021). Their findings reinforce the notion that machine learning signals based on the entire sample do not sufficiently characterize the investment universe beyond difficult-to-arbitrage stocks; thus, accounting for trading costs in the optimization improves performance.

E. Performance and Weights of the SDF-Implied Tangency Portfolio

In this subsection, we consider the method proposed by KNS. Similar to CPZ, the KNS approach incorporates a no-arbitrage condition to estimate the SDF. While CPZ estimate the SDF using individual stocks, KNS focus on equity portfolios that represent characteristics-based trading strategies. We first form portfolios based on the 94 predictive characteristics employed by GKX.[31] We follow KNS and perform a rank transformation for each characteristic and normalize each rank-transformed characteristic. We then construct long-short portfolios and compute characteristics-weighted portfolio returns. We rely on daily returns and split the sample into two subperiods. The first period, from September 1964 to December 2004, is used for in-sample estimation.[32] The second period, from January 2005 to December 2017, establishes the out-of-sample testing period.

Focusing on the first period, we estimate market loadings and then orthogonalize portfolio returns with respect to the market. We next estimate SDF slope coefficients for each characteristic using a ridge regression. That is, we minimize the Hansen-Jagannathan distance (Hansen and Jagannathan 1991) measure subject to an L^2-penalty, while the penalty parameter is chosen by a three-fold cross-validation

[31] KNS rely on two sets of characteristics. One includes 50 firm characteristics underlying common anomalies, and the other is based on 80 predictive characteristics consisting of 68 financial ratios from WRDS and 12 variables based on past monthly returns. They supplement the two sets of raw characteristics with characteristics based on second and third powers and linear first-order interactions of characteristics. We adopt the same set of 94 firm characteristics as in GKX to be consistent with other results reported in this paper.

[32] The in-sample estimation ranges from September 1964 to December 2004 for the full sample and for the subsample excluding microcaps and ranges from January 1986 to December 2004 for the subsamples excluding nonrated firms and distressed firms due to data availability.

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method. To incorporate our proposed economic restrictions, we separately estimate the SDF for the full sample as well as three subsamples that exclude microcaps, nonrated firms, and financially distressed firms.[33] Since SDF slope coefficients indicate weights of the MVE portfolio, we use SDF coefficients estimated from the pre-2005 sample to compute the implied out-of-sample MVE portfolio return (orthogonalized with respect to the market portfolio). Portfolio returns are *rescaled* to have standard deviations equal to the in-sample standard deviation of the excess return on the aggregate market index, allowing us to quantify the investment weights in the tangency portfolio conditional on a predetermined level of volatility.[34]

The SDF-implied MVE portfolio return is already CAPM-adjusted because the portfolio is orthogonal to the market. In addition, the FF6-adjusted return is estimated by regressing SDF-implied MVE portfolio returns on benchmark portfolio returns, where the benchmark portfolio return is estimated from unregularized MVE portfolio weights based on five nonmarket factors in the pre-2005 period. We also report the annualized Sharpe ratio and the quantile distribution of MVE portfolio weights (i.e., SDF slope coefficients) across the 94 characteristics.

The results are tabulated in Table 6. First and foremost, the evidence in KNS clearly applies to our universe of 94 test assets. The performance of the SDF-implied MVE portfolio is notable, with an annual Sharpe ratio of 2.32 and both CAPM-adjusted and FF6-adjusted returns exceeding 3% monthly for the full sample. Next, we confirm our finding that imposing economic restrictions reduces the performance of machine learning methods. For instance, the SDF-implied MVE portfolio yields an FF6-adjusted return of 0.90% per month after excluding microcaps, and the FF6-adjusted return is no longer significant at the 5% level after excluding nonrated firms or distressed firms. The annualized Sharpe ratio also declines to 0.98 (0.90, 0.83) after we exclude microcaps (nonrated firms, distressed firms).

[33] KNS exclude stocks with market caps below 0.01% of aggregate stock market capitalization at each point in time. For perspective, this sample selection criterion excludes all microcaps and approximately 60% of the nonmicrocaps in our sample. We nevertheless implement their analysis for the full sample as well as three subsamples to demonstrate the role of economic restrictions. In addition, we also examine the important role of financial distress at the firm level and business conditions in the aggregate.

[34] This rescaling exercise addresses the concern of SDF scalability. Otherwise, portfolio weights are not uniquely identified because they are *proportional* to the inverse of the covariance matrix times the mean vector.

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Notably, portfolio weights (or pricing kernel slope coefficients) display high dispersion across the 94 predictors and often exhibit extreme, possibly infeasible, values. To achieve the same volatility as the market, the SDF-implied MVE portfolio requires taking a -199% (-91%) short position for an individual predictor at the 10th (25th) percentile and a 169% (96%) long position at the 90th (75th) percentile. Portfolio positions are obviously more extreme at the 5th and 95th percentiles, rising to -234% and 190%, respectively.[35] Thus, the pricing kernel-based optimal portfolio may be an inadmissible investment from a practical perspective. However, imposing economic restrictions significantly lowers the odds of extreme positions. For instance, the SDF-implied MVE portfolio requires taking a -24% (-14%, -22%) short position for an individual predictor at the 25th percentile and a 41% (19%, 14%) long position at the 75th percentile after microcaps (nonrated firms, distressed firms) are excluded.

The collective evidence suggests that the machine learning (especially deep learning) techniques we examine, face the usual challenge of cross-sectional return predictability. We analyze the out-of-sample trading profits using predictive signals generated from five machine learning methods advocated by the recent literature, including three deep learning signals (i.e., GKX, CPZ, CA) and the IPCA and KNS methodologies. We consider both the SDF methodology adopted by CPZ (stock-level) and KNS (portfolio-level) as well as beta pricing formulations per IPCA and CA. Once we apply NYSE breakpoints to exclude microcaps, the value-weighted FF6-adjusted return is 66% (71%, 48%) lower than that for the full sample based on the GKX (CPZ, CA) signal. Similarly, excluding distressed firms, the value-weighted FF6-adjusted return is 78% (69%, 94%) lower than that for the full sample based on the GKX (CPZ, CA) signal. In addition, the value-weighted FF6-adjusted return is significant in only one of three subsamples (i.e., excluding microcaps, nonrated firms, and distressed firms) at the 5% threshold for all three deep learning signals. Moreover, IPCA underperforms deep learning models for the full sample but does not display a material deterioration of performance in subsamples with economic restrictions. As machine learning-based trading strategies require relatively high portfolio

[35] A plausible way to mitigate extreme stock positions would be to add an L^1-penalty (as in the least absolute shrinkage and selection operator, namely, LASSO) in addition to L^2. While the tuning parameter in LASSO can be found through a model selection criterion or cross-validation, its optimal value can still indicate extreme positions. To mitigate that concern, the tuning parameter value must be bounded. The addition of an L^1-penalty could challenge the nonsparse representation advocated by KNS.

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turnover and take extreme long-short positions in the SDF-implied tangency portfolio, investors may need to further lower their expectations of achievable performance.

Our findings suggest that economic restrictions play an important role in assessing whether newly proposed machine learning methods are effective and exploitable in real time, especially for investors who are sensitive to transaction costs and prefer to avoid microcaps and distressed firms in portfolio management. In line with Arnott, Harvey, and Markowitz (2018) and Karolyi and Van Nieuwerburgh (2020), our findings highlight the importance of adopting back-testing protocols (such as economic restrictions, in our context) in evaluating machine learning methods and verifying their external validity when applying to other settings, e.g., different universes of stocks, markets, asset classes, and sample periods.

IV. Time-Varying Return Predictability

The above results demonstrate that the cross-sectional return predictability of machine learning signals deteriorates among relatively cheap-to-trade stocks. We investigate whether return predictability through machine learning methods varies over time. Specifically, we first link trading profits to variations in market conditions and then examine predictable patterns over the most recent years.

A. Analysis Based on Portfolio Returns

Economic theory implies that less trading friction and more arbitrage activity should improve price efficiency. Consequently, anomaly-based trading strategies should be more profitable in the presence of binding limits to arbitrage. The existing evidence concerning many anomalies has typically supported this expectation. First, Stambaugh, Yu, and Yuan (2012) find stronger market anomalies following high-sentiment periods. They attribute the sentiment effect to binding short-sale constraints, which are particularly prominent during episodes of high investor sentiment, because stock prices reflect the views of more optimistic investors in the presence of heterogeneous beliefs about fundamental values (Miller 1977). Avramov, Chordia, Jostova, and Philipov (2018) further show that market-wide sentiment and firm-level financial distress jointly drive overpricing among stocks and corporate bonds. Second, a great deal of theoretical work predicts that higher volatility reduces the liquidity-provision capacity of market

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makers due to tightened funding constraints and reduced risk appetite (Gromb and Vayanos 2002; Brunnermeier and Pedersen 2009; Adrian and Shin 2010). Thus, anomaly payoffs (especially those related to liquidity provision) could increase due to liquidity dry-up during times of financial market turmoil. Finally, Chordia, Subrahmanyam, and Tong (2014) find that the recent regime of increased stock market liquidity is associated with the attenuation of equity return anomalies due to increased arbitrage activity.

In our empirical experiments, we examine the payoff of machine learning portfolios in subperiods depending on the state of investor sentiment, market volatility, and market liquidity. We consider the following market state variables: (1) investor sentiment, defined as the monthly investor sentiment from Baker and Wurgler (2007) (*SENT*), or the aligned investor sentiment using the partial least squares approach from Huang, Jiang, Tu, and Zhou (2015) (*PLS SENT*);[36] (2) realized market volatility (*MKTVOL*), defined as the standard deviation of daily CRSP value-weighted index returns in a month; (3) implied market volatility (*VIX*), defined as the monthly VIX index of implied volatilities of S&P 500 index options,[37] and (4) market illiquidity (*MKTILLIQ*), defined as the value-weighted average of stock-level Amihud (2002) illiquidity for all NYSE/AMEX stocks in a month (Avramov, Cheng, and Hameed 2016). We divide the full sample into two subperiods, i.e., high versus low investor sentiment (aligned investor sentiment, realized market volatility, implied market volatility), according to the median breakpoint of *SENT* (*PLS SENT*, *MKTVOL*, *VIX*) over the entire sample period. Unlike other market state variables, we obtain the median breakpoint for market illiquidity in the pre- and post-2001 periods separately, as the decimalization in January 2001 considerably reduced trading costs.

We repeat the portfolio analysis described before and sort stocks into decile portfolios according to machine learning signals. We compute the value-weighted holding period return for each decile portfolio and implement a zero-cost trading strategy by taking long positions in the top decile of stocks and shorting stocks in the bottom decile. We report the results for the full sample as well as subsamples that exclude microcaps, nonrated firms, and credit rating downgrades. For brevity, we present only FF6-

[36] We thank Jeffrey Wurgler and Guofu Zhou for making the index of investor sentiment available via their websites, http://people.stern.nyu.edu/jwurgler/; http://apps.olin.wustl.edu/faculty/zhou/zpublications.html.
[37] We obtain the monthly VIX index from the CBOE website, http://www.cboe.com/products/vix-index-volatility/vix-options-and-futures/vix-index/vix-historical-data.

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adjusted returns in the long-short trading strategy, while our findings are robust to alternative performance measures, such as raw returns and various risk-adjusted returns, which are documented previously.

We tabulate the results in Table 7, where Panel A shows the results for portfolios sorted by the NN3-predicted returns (GKX), Panel B shows those for portfolios sorted by the risk loadings on the SDF (CPZ), Panel C shows those for portfolios sorted by the IPCA-predicted returns (KPS), and Panel D shows those for portfolios sorted by the CA2-predicted returns (Gu, Kelly, and Xiu 2021). Starting with Panel A, several findings are worth noting. First, the long-short trading profit across all stocks is significant at the 5% level in all subperiods except for the low-*VIX* period. The investment strategy is also more profitable during periods of high investor sentiment, high market volatility, and low market liquidity. Second, among all market state variables, both realized and implied market volatility play an important role in explaining time-varying return predictability. The value-weighted FF6-adjusted return is 0.64% (statistically insignificant at 0.22%) per month at times of low *MKTVOL* (*VIX*) and dramatically increases to 1.28% (1.66%) at times of high *MKTVOL* (*VIX*), while the full sample average is 0.92% from Table 1, Panel A. Third, if we focus on the subsamples excluding microcaps or nonrated firms, the investment strategy remains more profitable during periods of high market volatility (in terms of both realized and implied volatility). Finally, considering the subsample excluding credit rating downgrades, none of the subperiods displays significant long-short trading profit at the 5% level. This last finding reinforces the concept that after excluding distressed firms around credit rating downgrades, the GKX signal does not deliver meaningful FF6-adjusted returns, i.e., the full sample average is statistically insignificant at 0.20% per month (Table 1, Panel B).

In Panel B, decile portfolios are sorted by risk loadings on the SDF (CPZ). For the full sample, we observe significant long-short trading profit at the 5% level in all subperiods, and the investment strategy outperforms during periods of high investor sentiment, high market volatility (in terms of both realized and implied volatility), and low market liquidity. In line with the full sample results shown in Table 3, return predictability deteriorates upon imposing economic restrictions, and only two subperiods display significant long-short trading profit at the 5% level.

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In Panel C, the trading strategy based on the IPCA signal remains profitable at the 5% level for the full sample as well as for the subsample excluding microcaps in all subperiods except for the low-*PLS SENT* period. More importantly, the trading profit based on IPCA does not change materially during high limits-to-arbitrage market states. For instance, the monthly value-weighted FF6-adjusted return is 0.51% (0.70%) for the full sample and 0.56% (0.62%) for the subsample excluding microcaps at times of low (high) limits to arbitrage (proxied by investor sentiment, market volatility, and market liquidity). Shifting the focus to subsamples excluding nonrated firms and credit rating downgrades, the IPCA signal outperforms in market states with low limits to arbitrage. The monthly value-weighted FF6-adjusted return is 0.59% (0.59%) when we exclude nonrated firms and 0.47% (statistically insignificant at 0.37%) when we exclude distressed firms around credit rating downgrades at times of low (high) limits to arbitrage.

In Panel D, unlike the GKX and CPZ methods, the CA signal shows mixed evidence across market states for the full sample. Considering the subsample excluding microcaps, the CA signal generates higher profit during periods of low limits to arbitrage. Unreported results further suggest that machine learning portfolios display similar turnover across various market states.

Overall, we find that GKX and CPZ signals predict cross-sectional returns across all stocks, especially during periods of high investor sentiment, high market volatility, and low market liquidity—consistent with the economic concept of limits to arbitrage. In contrast, the machine learning signals generated from conditional beta pricing models, i.e., IPCA and CA models, display low time series variation in trading profits and could even outperform in low limits-to-arbitrage market states. However, restricting the investment universe to cheap-to-trade stocks attenuates return predictability across all market states, and the payoff of deep learning signals (GKX, CPZ, and CA) often becomes statistically insignificant on a risk-adjusted basis over the entire sample period as well as in various market states.

B. Time Series Regressions

We next perform regression analysis to jointly consider all market state variables. We also explicitly control for other proxies for market states and macroeconomic conditions. Since the return predictability

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of deep learning signals weakens considerably in subsamples with economic restrictions, we focus on the full sample including all stocks to conduct time series analysis.

First, we examine whether the payoff of machine learning portfolios varies by market state. Specifically, we estimate the following monthly time series regression:

$$\begin{aligned} HML_t = \alpha_0 &+ \beta_1 High\ SENT_{t-1} + \beta_2 High\ MKTVOL_{t-1} + \beta_3 High\ MKTILLIQ_{t-1} \\ &+ \beta_4 M_{t-1} + c_1' F_t + c_2' F_t \times High\ SENT_{t-1} + c_3' F_t \times High\ MKTVOL_{t-1} \\ &+ c_4' F_t \times High\ MKTILLIQ_{t-1} + e_t \end{aligned} \quad (1)$$

where HML_t refers to the value-weighted return on the high minus low decile in machine learning portfolios across all stocks in month t. $High\ SENT_{t-1}$ refers to a dummy variable that takes a value of 1 if the Baker and Wurgler (2007) investor sentiment (*SENT*) is above the median over the entire sample period and 0 otherwise; $High\ SENT_{t-1}$ is further replaced with $High\ PLS\ SENT_{t-1}$, defined as a dummy variable that takes a value of 1 if the aligned investor sentiment from Huang, Jiang, Tu, and Zhou (2015) is above the median over the entire sample period and 0 otherwise; $High\ MKTVOL_{t-1}$ refers to a dummy variable that takes a value of 1 if the market volatility (*MKTVOL*) is above the median over the entire sample period and 0 otherwise; $High\ MKTVOL_{t-1}$ is further replaced with $High\ VIX_{t-1}$, defined as a dummy variable that takes a value of 1 if the implied market volatility (*VIX*) is above the median over the entire sample period and 0 otherwise; and $High\ MKTILLIQ_{t-1}$ refers to a dummy variable that takes a value of 1 if market illiquidity (*MKTILLIQ*) is above the median and 0 otherwise. We obtain the median breakpoint for market illiquidity in the pre- and post-2001 periods separately. All market state variables are defined as in Table 7. M_{t-1} refers to a set of other proxies for market conditions: down market state (*DOWN*), defined as a dummy variable that takes the value of 1 if the CRSP value-weighted index return is negative and 0 otherwise; term spread (*TERM*), defined as the difference between the average yield of 10-year Treasury bonds and three-month T-bills; and default spread (*DEF*), defined as the difference between the average yield of bonds rated BAA and AAA by Moody's. The vector *F* stacks the six common risk factors identified in Fama and French (2018), including the market factor (MKT), the size factor (SMB), the book-to-market factor (HML), the profitability factor (RMW), the investment factor (CMA), and the momentum factor (MOM). We also report Newey and West (1987) adjusted *t*-statistics with four lags.

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The results are presented in Internet Appendix Table IA8. In Panels A and B, decile portfolios are sorted by the NN3-predicted returns (GKX) and the risk loadings on the SDF (CPZ), respectively. We find that the trading profit based on the GKX signal is higher during periods of high market volatility in terms of both realized and implied volatility after controlling for investor sentiment, market liquidity, macroeconomic variables, and risk factors (models 1 to 4). Models 5 to 8 further consider the time-varying factor risk exposures by interacting the six common risk factors with market state variables. We continue to find that the trading profit based on the GKX signal is higher during high-*VIX* periods. Similarly, the long-short portfolio payoff based on the CPZ signal is higher during periods of high investor sentiment, high implied market volatility, and low market liquidity (models 1 to 4). In addition, the implied market volatility appears to be the most informative market state variable and is highly significant in all specifications after controlling for macroeconomic variables and time-varying risk exposures. Overall, market-wide predictive variables capture a large variation in investment payoffs based on GKX and CPZ signals.

In Panels C and D, decile portfolios are sorted by the IPCA-predicted returns (KPS) and the CA2-predicted returns (Gu, Kelly, and Xiu 2021), respectively. The findings are in line with the portfolio analysis shown in Table 7, i.e., the trading profits generated from conditional beta pricing models do not exhibit significant time series variation across market states after controlling for macroeconomic variables and risk factors as well as time-varying risk exposures.

Overall, this joint predictive regression specification further reinforces our finding that machine learning payoffs based on GKX and CPZ signals display substantial time variation and, more importantly, that superior performance characterizes market states that are associated with high trading frictions.

C. Return Predictability in Recent Years

The U.S. equity market has undergone substantial structural changes since the 2000s, such as the introduction of decimalization and improved market liquidity, the greater participation of institutional investors, better access to a broader range of data, developments in financial technology, and the adoption of advanced quantitative analysis. Chordia, Subrahmanyam, and Tong (2014) show that the

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majority of anomalies were attenuated after decimalization in January 2001, with the average return and Sharpe ratio from a trading strategy consisting of 12 anomalies more than halved.

In the presence of the growing popularity of exploiting big data and quantitative models in asset management, we examine whether machine learning techniques have remained meaningful in recent years. We repeat the portfolio analysis described before and sort stocks into decile portfolios according to machine learning signals. We implement a zero-investment trading strategy by taking long positions in the top decile of stocks and shorting stocks in the bottom decile and compute the holding period return in the post-2001 period. We report only the value-weighted performance of long-short portfolios in Table 8 for brevity, where Panel A shows the results for portfolios sorted by the NN3-predicted returns (GKX), Panel B shows those for portfolios sorted by the risk loadings on the SDF (CPZ), Panel C shows those for portfolios sorted by the IPCA-predicted returns (KPS), and Panel D shows those for portfolios sorted by the CA2-predicted returns (Gu, Kelly, and Xiu 2021).

Our main findings from 1987 to 2017 remain intact in the post-2001 period for the GKX, IPCA, and CA signals. The value-weighted long-short portfolio return based on the GKX (IPCA, CA) signal across all stocks is significant at 1.57% (0.88%, 1.65%) per month and 1.18% (0.69%, 1.12%) after adjusting for the FF6 model. The economic magnitude is also comparable with that for the entire sample period, i.e., 1.56% (0.95%, 1.16%) for raw returns and 0.92% (0.62%, 0.75%) for the FF6-adjusted returns in Tables 1 and 4. Moreover, the machine learning signal continues to predict the cross-section of stock returns among nonmicrocaps and rated firms in recent years. Shifting the focus to the most restrictive subsample that excludes credit rating downgrades, the FF6-adjusted return is significant only at the 5% threshold for the IPCA method.[38]

As shown in Panel B of Table 8, the value-weighted long-short portfolio return based on the CPZ signal across all stocks is significant at 1.86% per month and 1.02% after adjusting for the FF6 model. The economic magnitude is slightly lower than that for the 1987 to 2016 period, i.e., 2.18% for raw

[38] Unreported results show that the machine learning-based investments remain profitable in the post-2013 period, i.e., the last five years in our sample, and the monthly value-weighted long-short portfolio return is 0.72% (1.30%, 0.98%) across all stocks and 0.46% (1.31%, 1.26%) after excluding microcaps for the GKX (IPCA, CA) signal. In addition, unrestricted IPCA also predicts the cross-section of stock returns for the full sample as well as for subsamples with economic restrictions in the post-2001 period (Internet Appendix Table IA7, Panel C).

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returns and 1.87% for FF6-adjusted returns (Table 3, Panel A), but the recent performance is comparable with other deep learning signals. Imposing economic restrictions further weakens out-of-sample return predictability in recent years, and we do not detect significant FF6-adjusted returns in any of the three subsamples.[39]

The overall evidence suggests that machine learning signals continue to predict cross-sectional stock returns in recent years for the full sample. Unlike for individual anomalies, there is no vast drop in trading profits of machine learning signals. This supports the concept that machine learning methods are able to combine multiple, presumably weak, signals into a meaningful set of information. On the other hand, anomalous return patterns are still confined within difficult-to-arbitrage stocks, and thus, practitioners should remain cautious in utilizing machine learning algorithms for real-time trading.

V. Economic Foundations of Machine Learning

Cochrane (2011) points out that traditional regression analysis and portfolio sorts could be insufficient for handling a large number of predictive variables. In response, machine learning offers a natural way to accommodate a high-dimensional predictor set and flexible functional forms and employs "regularization" methods to select models and mitigate overfitting biases. However, deep learning models are opaque in nature and are often referred to as "black boxes." As emphasized by Karolyi and Van Nieuwerburgh (2020), understanding the relevant economic mechanisms is essential for machine learning tools, especially if the goal is robust and credible out-of-sample predictability.

In this section, we focus on the full sample and provide evidence on the economic driving forces of return predictability in machine learning methods. Specifically, we examine whether stocks with similar machine learning signals also share other characteristics that predict future returns. Examining the common features of stocks selected by machine learning methods allows us to determine whether the investment decision is economically interpretable. We further control for industry benchmarks and investigate the source of return predictability.

[39] In unreported tests, we run Fama and MacBeth (1973) regressions of realized excess returns on predictive signals derived from all four machine learning methods. Evidence indicates considerably lower slope coefficients in the post-2001 period. Thus, considering all stocks (beyond the extreme long and short portfolios), machine learning-based predictability seems to decrease in recent years.

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A. Stock Characteristics of Machine Learning Portfolios

At the end of each month t, stocks are sorted into deciles per machine learning signal. We then compute the equal-weighted average of a comprehensive set of stock characteristics at the end of month t for each portfolio. Most stock characteristics come from the firm-level predictors used in GKX, including *Absolute Accruals*, *Log(Age)*, *Assets Growth*, *Beta*, *Book-to-Market*, *ΔShares Outstanding*, *Corporate Investment*, *Dividend-to-Price*, *Gross Profitability*, idiosyncratic return volatility (*IdioVol*), *Log(Illiquidity)*, *Leverage*, *12M Momentum*, return on assets (*ROA*), and return on equity (*ROE*). We also consider other firm characteristics, such as *Log(Price)*, *Log(Size)*, *1M Return*, the percentage of rated firms (*%Rated*), *Credit Rating*, *Analyst Coverage*, *Analyst Dispersion*, and standardized unexpected earnings (*SUE*). We obtain analyst forecast data from the Institutional Brokers' Estimate System (I/B/E/S). Internet Appendix A provides detailed definitions of each variable.

While most stock characteristics examined in this subsection are also used as input variables for the machine learning models, whether the same features are preserved in machine learning portfolios is unclear. For instance, if machine learning portfolios capture mainly the nonlinearities and interaction effects, the stocks identified by machine learning signals may not align with those in standard anomaly-based trading strategies relying on a single characteristic.

The results are presented in Table 9, where Panel A shows the results for portfolios sorted by the NN3-predicted returns (GKX) and the risk loadings on the SDF (CPZ) and Panel B shows the results for portfolios sorted by the IPCA-predicted returns (KPS) and the CA2-predicted returns (Gu, Kelly, and Xiu 2021). For brevity, we tabulate only stock characteristics for the bottom and top decile portfolios, as well as the difference in values between high- and low-decile portfolios ("HML"). We report Newey and West (1987) adjusted *t*-statistics with four lags. Several findings are worth noting. First, all machine learning methods identify stocks in line with most anomaly-based trading strategies. For instance, stocks in the long positions of a machine learning-based trading strategy are typically small, value, illiquid, and old stocks with low price, low beta, high 11-month return (medium-term winners), low asset growth, low equity issuance, low credit rating coverage, and low analyst coverage.

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Therefore, despite their opaque nature, machine learning techniques successfully identify mispriced stocks with solid economic foundations.[40]

Second, there are two notable incidences in which machine learning signals trade in the opposite direction of individual anomaly characteristics. First, all machine learning methods take long positions in stocks with high corporate investment, which, on an individual basis, predicts lower future returns on average (Titman, Wei, and Xie 2004). Second, all machine learning methods except for CPZ take long positions in stocks with high idiosyncratic volatility. Future returns may not be linear in either corporate investment or idiosyncratic volatility and return predictability could be affected by other related firm characteristics or macro conditions. As shown by Titman, Wei, and Xie (2004), the negative investment-return relation is more prominent among firms with higher cash flows and lower debt ratios. Stambaugh, Yu, and Yuan (2015) also document that the idiosyncratic volatility-return relation is negative for overpriced stocks but turns positive for underpriced stocks. Such complex and often ambiguous patterns in the cross-section highlight the merits of employing machine learning techniques because they can distill information from a large set of correlated characteristics.

An advantage of machine learning methods is that they accommodate high dimensionality and complex patterns in the data without preselection of truly useful characteristics and models, hence avoiding the data snooping problem that challenges the credibility of the anomaly literature (e.g., Harvey, Liu, and Zhu 2016; McLean and Pontiff 2016; Harvey 2017; Hou, Xue, and Zhang 2020). Our findings support this concept by showing that machine learning signals indeed identify mispriced stocks that are in line with well-established empirical facts without requiring any prior knowledge. Despite their opaque nature, machine learning models generate economically interpretable trading strategies, which is essential for robust and credible out-of-sample predictability (Karolyi and Van Nieuwerburgh 2020).

Notably, several other papers also investigate the economic interpretability of machine learning models. For instance, Kelly, Pruitt, and Su (2019) and Gu, Kelly, and Xiu (2020) identify important characteristics according to their contribution to overall model fit. Cong, Tang, Wang, and Zhang (2021)

[40] Unreported results confirm that our main findings are robust in the post-2001 period and various market states such as investor sentiment, market volatility, and market liquidity.

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propose an "economic distillation" procedure that projects complex AI models onto linear modeling or natural language spaces to identify the dominant characteristics (including their higher-order terms and interaction terms). Similar to our approach, Sak, Huang, and Chng (2021) employ a characteristic sort to identify dominant features. These studies aim to identify the influential covariates for the cross-sectional return predictability and rank all predictors by their importance. In contrast, we do not implement a horse race across all predictors in the machine learning model. Instead, we focus on the ex post interpretability of machine learning signals and associate them with a list of widely adopted and economically motivated trading signals. While the important characteristics identified in prior work are model-specific, we complement their results by providing broader implications for the economic interpretability of machine learning methods.

B. Intra-Industry vs. Inter-Industry Return Predictability

Returns on firms within the same industry are highly correlated, as they could be affected by common technological shocks, changes in operational and regulatory environments, and industry-specific demand and supply for certain products and services. Our prior findings suggest that deep learning signals mostly predict future returns in difficult-to-arbitrage stocks. If such trading signals capture temporary mispricing and subsequent correction due to market frictions, matching similar firms within the same industry provides a natural framework to control for firm fundamentals and understand the source of return predictability. Controlling for industry benchmarks can also clarify whether machine learning methods specialize in stock picking or industry rotation, i.e., whether they cluster similar firms and trade the industry portfolio.

To pursue the analysis, we first implement an unconditional trading strategy based on the NN3-predicted return of stock i in month t (GKX), denoted by $\hat{R}_{i,t}$. We take a \$1 long position on stocks that are expected to outperform the market (market winners), i.e., $\hat{R}_{i,t} - \hat{R}_{m,t} > 0$ and \$1 short position on stocks that are expected to underperform the market (market losers), i.e., $\hat{R}_{i,t} - \hat{R}_{m,t} < 0$, where $\hat{R}_{m,t}$ refers to the equal-weighted average of $\hat{R}_{i,t}$ across all stocks in the market. That is, $\hat{R}_{m,t} = \frac{1}{N_t}\sum_{i=1}^{N_t} \hat{R}_{i,t}$, where N_t refers to the number of stocks in the market. We hold the portfolio over the next month. The

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winner minus loser profit from the unconditional strategy at month $t+1$, denoted as WML_{t+1}, is given by

$$WML_{t+1} = \frac{1}{H_t}\sum_{i=1}^{N_t}(\hat{R}_{i,t} - \hat{R}_{m,t})R_{i,t+1} \tag{2}$$

$$H_t = \frac{1}{2}\sum_{i=1}^{N_t}|\hat{R}_{i,t} - \hat{R}_{m,t}| \tag{3}$$

where $R_{i,t+1}$ refers to the return of stock i in month $t+1$, and all other variables are defined as above. The portfolio weighting scheme is similar to that in Nagel (2012), and the long-short portfolio includes all stocks in the investment universe. The weight of each stock is proportional to the stock's NN3-predicted return on a market-adjusted basis, with higher weights for better performers in the long leg and more negative weights for worse performers in the short leg. The investment in each security is scaled by the inverse of the sum of absolute deviations of stock returns from the market average so that the strategy is \$1 long in market winner stocks and \$1 short in market loser stocks. This unconditional strategy also provides a robustness check of our main results, where we focus only on extreme decile portfolios to construct a long-short strategy rather than on all investable stocks.

Next, as in Hameed and Mian (2015), we decompose the unconditional strategy into two components. In particular, Equation (2) can be rewritten as follows:

$$\begin{aligned} WML_{t+1} &= \frac{1}{H_t}\sum_{i=1}^{N_t}(\hat{R}_{i,t} - \hat{R}_{j,t} + \hat{R}_{j,t} - \hat{R}_{m,t})R_{i,t+1} \\ &= \frac{1}{H_t}\sum_{i=1}^{N_t}(\hat{R}_{i,t} - \hat{R}_{j,t})R_{i,t+1} + \frac{1}{H_t}\sum_{i=1}^{N_t}(\hat{R}_{j,t} - \hat{R}_{m,t})R_{i,t+1} \\ &= \frac{1}{H_t}\sum_{i=1}^{N_t}(\hat{R}_{i,t} - \hat{R}_{j,t})R_{i,t+1} + \frac{1}{H_t}\sum_{j=1}^{L_t}(\hat{R}_{j,t} - \hat{R}_{m,t})N_{j,t}R_{j,t+1} \end{aligned} \tag{4}$$

where $\hat{R}_{j,t}$ refers to the equal-weighted average of $\hat{R}_{i,t}$ across all stocks in industry j. That is, $\hat{R}_{j,t} = \frac{1}{N_{j,t}}\sum_{i=1}^{N_{j,t}}\hat{R}_{i,t}$, where $N_{j,t}$ refers to the number of stocks in industry j. L_t refers to the number of

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industries, and $R_{j,t+1}$ refers to the equal-weighted average of stock returns across all stocks in industry j in month $t+1$. That is, $R_{j,t+1} = \frac{1}{N_{j,t}} \sum_{i=1}^{N_{j,t}} R_{i,t+1}$. All other variables are defined as in Equation (2).[41]

The first term in Equation (4) represents returns to an intra-industry strategy that buys stocks that are expected to outperform the industry portfolio (industry winners), i.e., $\hat{R}_{i,t} - \hat{R}_{j,t} > 0$, and sells stocks that are expected to underperform the industry portfolio (industry losers), i.e., $\hat{R}_{i,t} - \hat{R}_{j,t} < 0$. The second term represents returns to an inter-industry strategy that buys the industry portfolio if the industry is expected to outperform the overall market (winner industries), i.e., $\hat{R}_{j,t} - \hat{R}_{m,t} > 0$, and sells the industry portfolio if the industry is expected to underperform the overall market (loser industries), i.e., $\hat{R}_{j,t} - \hat{R}_{m,t} < 0$. To scale the investment in each component to \$1 long and \$1 short, we multiply the profits by the factor of proportionality as follows:

$$
\begin{aligned}
WML_{t+1} &= \frac{H_t^{INTRA}}{H_t} \frac{1}{H_t^{INTRA}} \sum_{i=1}^{N_t} \left(\hat{R}_{i,t} - \hat{R}_{j,t}\right) R_{i,t+1} \\
&+ \frac{H_t^{INTER}}{H_t} \frac{1}{H_t^{INTER}} \sum_{j=1}^{L_t} \left(\hat{R}_{j,t} - \hat{R}_{m,t}\right) N_{j,t} R_{j,t+1} \\
&= \frac{H_t^{INTRA}}{H_t} \times WML_{t+1}^{INTRA} + \frac{H_t^{INTER}}{H_t} \times WML_{t+1}^{INTER}
\end{aligned} \tag{5}
$$

$$
H_t^{INTRA} = \frac{1}{2} \sum_{i=1}^{N_t} |\hat{R}_{i,t} - \hat{R}_{j,t}| \tag{6}
$$

$$
H_t^{INTER} = \frac{1}{2} \sum_{j=1}^{L_t} |\hat{R}_{j,t} - \hat{R}_{m,t}| N_{j,t} \tag{7}
$$

where all variables are defined as in equations (2) and (4). In particular, the winner minus loser profit from the intra-industry strategy at month $t+1$, denoted as WML_{t+1}^{INTRA}, is given by

$$
WML_{t+1}^{INTRA} = \frac{1}{H_t^{INTRA}} \sum_{i=1}^{N_t} \left(\hat{R}_{i,t} - \hat{R}_{j,t}\right) R_{i,t+1} \tag{8}
$$

Similarly, the winner minus loser profit from the inter-industry strategy at month $t+1$, denoted as WML_{t+1}^{INTER}, is given by

[41] Given the nature of this decomposition, we analyze machine learning signals based only on predicted returns in this subsection, i.e., GKX, IPCA, and CA.

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$$WML_{t+1}^{INTER} = \frac{1}{H_t^{INTER}} \sum_{j=1}^{L_t} (\hat{R}_{j,t} - \hat{R}_{m,t}) N_{j,t} R_{j,t+1} \quad (9)$$

As shown in Equation (5), the unconditional reversal profit (WML_{t+1}) is a weighted average of WML_{t+1}^{INTRA} and WML_{t+1}^{INTER}, and the weights depend on the scaling factors for the different strategies, i.e., $\frac{H_t^{INTRA}}{H_t}$ and $\frac{H_t^{INTER}}{H_t}$.

The results are tabulated in Table 10, where Panel A shows the results for the full sample and subsample excluding microcaps and Panel B shows those for the subsamples excluding nonrated firms or credit rating downgrades. In Panels A1 and B1, we report the returns for winner and loser portfolios obtained using the unconditional strategy, where winners (losers) consist of stocks that are expected to outperform (underperform) the market average according to the NN3-predicted returns in GKX. We also implement the trading strategy by taking long positions in winner stocks and shorting loser stocks. The zero-investment trading profit is computed as the winner minus loser portfolio returns ("WML"). Over the 1987–2017 sample period, the average long-short portfolio return is a significant 1.81% per month across all stocks and 1.55% after adjusting for the FF6 model. In addition, the raw return is 0.92% (0.48%) per month, and the FF6-adjusted return is statistically insignificant for the subsample excluding microcaps (credit rating downgrades). This analysis, based on *all* stocks, confirms our main findings from the extreme decile portfolios that machine learning signals weaken drastically in the subset of cheap-to-trade stocks.

In Panels A2 and B2, we report the returns for winner and loser portfolios from the intra-industry strategy, where winners (losers) consist of stocks that are expected to outperform (underperform) the industry average according to the NN3-predicted returns. We also implement the trading strategy by taking long positions in winner stocks and shorting loser stocks. The zero-investment trading profit is computed as the winner minus loser portfolio returns ("WML^{INTRA}"). In Panels A3 and B3, we report the returns for winner and loser portfolios from the inter-industry strategy, in which winners (losers) consist of industries that are expected to outperform (underperform) the market average according to the NN3-predicted returns. We implement the trading strategy by taking long positions in winner industries and shorting loser industries. The zero-investment trading profit is computed as the winner

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minus loser portfolio returns ("WML^{INTER}"). We also report the scaled results, i.e., $WML^{INTRA}\times H^{INTRA}/H$ and $WML^{INTER}\times H^{INTER}/H$, and they add up to WML, as shown in Equation (5).

Several findings are worth noting. First, the intra-industry strategy delivers substantially higher returns than the inter-industry strategy. As shown in Panels A2 and A3, the intra-industry strategy ($WML^{INTRA}\times H^{INTRA}/H$) accounts for 84% (i.e., 1.52% out of 1.81%) of the unconditional payoff in raw returns and 93% of that in risk-adjusted returns across all performance measures for the full sample. Meanwhile, the inter-industry strategy ($WML^{INTER}\times H^{INTER}/H$) accounts for the remaining 16% (i.e., 0.29% out of 1.81%) of the unconditional payoff in raw returns and 7% of that in risk-adjusted returns across all performance measures. All risk-adjusted returns are highly significant for the intra-industry strategy, while only one of five risk-adjusted returns is significant at the 5% level for the inter-industry strategy. A similar pattern also holds for the three subsamples with economic restrictions. This finding implies that the GKX method emphasizes stock selection more than industry rotation.

Second, the payoff of the intra-industry strategy (WML^{INTRA}) is higher than $WML^{INTRA}\times H^{INTRA}/H$ for the full sample as well as for the subsamples due to a lower cross-sectional dispersion in industry-adjusted returns, i.e., $H^{INTRA}/H < 1$. Improvements using an intra-industry strategy are particularly important for nonmicrocaps. As shown in Panel A2, the FF6-adjusted return regains significance by controlling for the industry benchmark, i.e., 0.34% per month (*t*-statistic = 3.32), as opposed to a statistically insignificant 0.18% in the unconditional strategy (Panel A1). Similarly, the monthly SY-adjusted return is statistically insignificant at 0.19% for the unconditional strategy and more than double for the intra-industry strategy, i.e., 0.40% (*t*-statistic = 2.57). The outperformance of the intra-industry strategy confirms that the GKX signal identifies mispricing in difficult-to-arbitrage stocks.

As a robustness check, we repeat the analysis using IPCA-predicted returns and CA2-predicted returns. For brevity, we present only the raw return and FF6-adjusted return for the full sample as well as three subsamples in Internet Appendix Table IA9. We confirm that the intra-industry strategy not only delivers substantially higher returns than the inter-industry strategy, but also outperforms the unconditional strategy on a risk-adjusted basis for most subsamples with economic restrictions. Hence, adjusting the portfolio weights by the industry average further controls for firm fundamentals and better predicts the subsequent correction due to market frictions. From a practitioner's perspective, the out-

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of-sample performance of machine learning portfolios can be further enhanced via a simple industry adjustment.

VI. Conclusion

This paper provides large-scale evidence on the economic significance of machine learning methods. The deep learning techniques that we analyze face the usual challenge of cross-sectional return predictability. In particular, the anomalous return patterns concentrate in difficult-to-value and difficult-to-arbitrage stocks. In addition, to the extent that deep learning signals predict cross-sectional stock returns for the full sample, the trading strategy is more profitable during periods of high market volatility and low market liquidity. Machine learning signals also involve remarkably high turnover and often require taking extreme long-short positions for predetermined portfolio volatility in the tangency portfolio implied by the pricing kernel. Beyond economic restrictions, machine learning-based trading strategies nonetheless display smaller downside risk, yield considerable profit in the long positions, and remain viable in the post-2001 period and the crisis period. Finally, black-box-like machine learning methods generate economically interpretable trading strategies and are more informative for stock selection than for industry rotation.

Our findings provide timely evidence to help understand machine learning applications and propose a list of back-testing protocols for academic research and asset management. When assessing machine learning methods, it is imperative to consider common economic restrictions in both the cross-section and the time series and incorporate the trading costs due to portfolio turnover. Similarly, it is essential to estimate the SDF based on admissible stock positions. It is also important to confirm the external validity of machine learning models before applying them to different universes of stocks, markets, asset classes, and sample periods.

Our paper also suggests an important avenue for future research. In particular, the optimization routines that mix various anomalies could inherently display a high turnover of stocks in extreme long-short portfolios and thus generate high trading costs. Similarly, pricing kernel estimates could rely on rather extreme long and short positions that are inadmissible in real time. Thus, it would be useful to extend machine learning methods to endogenously account for trading costs and further impose

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plausible portfolio constraints. Machine learning methods can also be used to evaluate practical transaction costs of strategic trading through relaxing the assumed functional forms of dependence between the trading costs and the invested capital and introducing a nonlinear time series model of trading costs of a patient trader. These and other topics are left for future research.

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Table 1: Performance of Portfolios Sorted by Neural Network Predicted Returns

At the end of each month t, stocks are sorted into deciles according to their one-month-ahead out-of-sample predicted returns ($\hat{R}$) using a neural network with three hidden layers (NN3) (Gu, Kelly, and Xiu 2020). This table reports the value-weighted returns for month $t+1$ for each decile portfolio as well as the strategy of going long (short) on the highest (lowest) expected return stocks ("HML") over the entire sample period from 1987 to 2017. Portfolio returns are further adjusted by the CAPM, Fama-French-Carhart 4-factor and Pástor-Stambaugh liquidity factor model (FFC+PS), Fama-French 5-factor model (FF5), Fama-French 6-factor model (FF6), and Stambaugh-Yuan 4-factor model (SY). Panel A reports the results for the full sample as well as the subsample that excludes microcaps. Panel B reports similar statistics for the subsamples that exclude nonrated firms and credit rating downgrades. Newey-West adjusted t-statistics are shown in parentheses. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Panel A: Value-Weighted Returns to Investment Strategies Sorted by NN3-Predicted Returns (Full Sample and Microcaps Excluded)

Rank of $\hat{R}$	Full Sample						Nonmicrocaps					
	Return	CAPM	FFC+PS	FF5	FF6	SY	Return	CAPM	FFC+PS	FF5	FF6	SY
Low	0.328	-0.923***	-0.508***	-0.360*	-0.146	-0.033	0.364	-0.893***	-0.467***	-0.290	-0.079	0.006
	(0.78)	(-4.35)	(-3.07)	(-1.95)	(-0.99)	(-0.19)	(0.85)	(-4.09)	(-2.74)	(-1.55)	(-0.53)	(0.03)
2	0.678**	-0.346***	-0.158*	-0.163*	-0.039	0.011	0.647**	-0.390***	-0.195**	-0.156	-0.041	0.016
	(2.37)	(-3.38)	(-1.67)	(-1.76)	(-0.48)	(0.12)	(2.24)	(-3.65)	(-2.01)	(-1.63)	(-0.49)	(0.16)
3	0.855***	-0.090	-0.059	-0.102	-0.053	-0.015	0.862***	-0.090	0.006	-0.048	0.032	0.065
	(3.21)	(-1.04)	(-0.72)	(-1.19)	(-0.63)	(-0.16)	(3.24)	(-1.01)	(0.06)	(-0.52)	(0.36)	(0.66)
4	0.890***	-0.006	-0.038	-0.145**	-0.128**	-0.096	0.841***	-0.073	-0.108	-0.189**	-0.183**	-0.153
	(3.88)	(-0.08)	(-0.58)	(-2.51)	(-2.09)	(-1.23)	(3.47)	(-0.96)	(-1.47)	(-2.29)	(-2.16)	(-1.61)
5	0.917***	0.042	-0.049	-0.084	-0.109	-0.096	0.979***	0.109	0.040	-0.014	-0.019	0.006
	(3.96)	(0.52)	(-0.70)	(-1.22)	(-1.58)	(-1.33)	(4.26)	(1.18)	(0.51)	(-0.21)	(-0.28)	(0.08)
6	1.019***	0.165**	0.057	0.009	-0.029	-0.016	0.899***	0.028	-0.064	-0.125*	-0.153**	-0.155**
	(4.83)	(2.19)	(0.78)	(0.13)	(-0.44)	(-0.20)	(4.13)	(0.38)	(-0.93)	(-1.93)	(-2.33)	(-2.17)
7	1.185***	0.307***	0.167*	0.129	0.073	0.097	1.058***	0.194**	0.084	0.008	-0.027	-0.008
	(5.14)	(3.10)	(1.84)	(1.42)	(0.82)	(0.98)	(4.84)	(2.17)	(0.94)	(0.10)	(-0.34)	(-0.09)
8	1.078***	0.218**	0.055	0.054	-0.029	0.008	1.210***	0.340***	0.167**	0.171**	0.092	0.080
	(4.51)	(2.31)	(0.69)	(0.60)	(-0.36)	(0.07)	(5.20)	(3.60)	(2.26)	(2.19)	(1.26)	(0.90)
9	1.350***	0.435***	0.240**	0.275**	0.174	0.152	1.112***	0.251***	0.094	0.073	-0.002	0.045
	(5.14)	(3.02)	(2.10)	(2.37)	(1.63)	(1.20)	(4.81)	(2.73)	(1.09)	(0.75)	(-0.02)	(0.36)
High	1.883***	0.971***	0.853***	0.846***	0.770***	0.735***	1.410***	0.496***	0.304**	0.337***	0.234**	0.185
	(6.45)	(4.91)	(5.37)	(5.53)	(5.01)	(4.38)	(5.39)	(3.46)	(2.56)	(2.80)	(2.02)	(1.46)
HML	1.556***	1.894***	1.361***	1.206***	0.916***	0.769***	1.047***	1.389***	0.771***	0.627**	0.312	0.179
	(4.53)	(5.64)	(5.31)	(4.66)	(4.08)	(3.03)	(3.24)	(4.43)	(3.24)	(2.41)	(1.51)	(0.73)

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Table 1—Continued

Panel B: Value-Weighted Returns to Investment Strategies Sorted by NN3-Predicted Returns (Credit Rating Sample and Downgrades Excluded)												
Rank of $\hat{R}$	Credit Rating Sample						Nondowngrades					
	Return	CAPM	FFC+PS	FF5	FF6	SY	Return	CAPM	FFC+PS	FF5	FF6	SY
Low	0.398	-0.807***	-0.419**	-0.426*	-0.178	0.019	0.783**	-0.364*	-0.027	-0.081	0.123	0.368*
	(0.94)	(-3.70)	(-2.48)	(-1.93)	(-1.10)	(0.10)	(2.07)	(-1.72)	(-0.15)	(-0.37)	(0.65)	(1.87)
2	0.759***	-0.265**	-0.108	-0.203*	-0.063	0.025	0.956***	-0.039	0.112	0.027	0.143	0.251**
	(2.61)	(-2.46)	(-1.05)	(-1.84)	(-0.62)	(0.20)	(3.58)	(-0.38)	(1.16)	(0.24)	(1.40)	(2.05)
3	0.923***	-0.017	0.067	-0.048	0.039	0.028	1.135***	0.231*	0.302**	0.206	0.268*	0.282*
	(3.35)	(-0.14)	(0.60)	(-0.40)	(0.34)	(0.21)	(4.46)	(1.84)	(2.42)	(1.50)	(1.97)	(1.93)
4	0.855***	-0.066	-0.142	-0.235**	-0.221**	-0.176	0.981***	0.095	0.021	-0.050	-0.048	0.014
	(3.30)	(-0.63)	(-1.55)	(-2.51)	(-2.23)	(-1.40)	(4.08)	(0.98)	(0.23)	(-0.57)	(-0.51)	(0.11)
5	1.096***	0.222*	0.160*	-0.018	0.007	0.058	1.186***	0.326**	0.269**	0.082	0.104	0.176
	(4.78)	(1.86)	(1.80)	(-0.25)	(0.09)	(0.62)	(5.30)	(2.56)	(2.40)	(0.88)	(1.09)	(1.48)
6	0.890***	0.024	-0.091	-0.126	-0.161**	-0.158*	1.031***	0.174*	0.049	0.017	-0.033	-0.032
	(3.96)	(0.27)	(-1.19)	(-1.56)	(-2.14)	(-1.86)	(4.78)	(1.81)	(0.56)	(0.18)	(-0.40)	(-0.36)
7	1.004***	0.171*	0.036	-0.052	-0.085	-0.054	1.127***	0.306***	0.188*	0.100	0.064	0.105
	(4.67)	(1.74)	(0.41)	(-0.66)	(-1.10)	(-0.61)	(5.43)	(3.07)	(1.86)	(1.18)	(0.74)	(1.08)
8	1.216***	0.352***	0.161**	0.134*	0.056	0.043	1.313***	0.461***	0.272***	0.255***	0.167**	0.123
	(5.38)	(3.39)	(2.00)	(1.67)	(0.75)	(0.46)	(6.06)	(4.54)	(3.42)	(2.95)	(2.21)	(1.33)
9	1.070***	0.223**	0.038	0.016	-0.055	0.037	1.139***	0.306***	0.109	0.079	0.000	0.081
	(4.61)	(2.27)	(0.42)	(0.16)	(-0.60)	(0.28)	(5.04)	(2.97)	(1.18)	(0.76)	(0.00)	(0.61)
High	1.422***	0.537***	0.323***	0.358***	0.255**	0.169	1.506***	0.632***	0.418***	0.436***	0.326***	0.238
	(5.62)	(3.83)	(2.59)	(2.88)	(2.18)	(1.21)	(6.05)	(4.35)	(3.20)	(3.40)	(2.68)	(1.64)
HML	1.024***	1.344***	0.743***	0.784***	0.433**	0.150	0.723**	0.996***	0.445**	0.517*	0.204	-0.129
	(3.18)	(4.40)	(3.23)	(2.70)	(2.05)	(0.59)	(2.49)	(3.40)	(2.00)	(1.87)	(0.92)	(-0.52)

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Table 2: Robustness Check: Neural Network Estimation Based on Subsamples

At the end of each month t, stocks are sorted into deciles according to their NN3-predicted returns ($\hat{R}$) (Gu, Kelly, and Xiu 2020). This table reports the value-weighted return for month $t+1$ for the strategy of going long (short) on the highest (lowest) expected return stocks. Portfolio returns are further adjusted by the CAPM, Fama-French-Carhart 4-factor and Pástor-Stambaugh liquidity factor model (FFC+PS), Fama-French 5-factor model (FF5), Fama-French 6-factor model (FF6), and Stambaugh-Yuan 4-factor model (SY). We employ the full sample as well as subsamples that exclude microcaps, nonrated firms, and credit rating downgrades for the NN3 estimations (in the "Training" column) and the portfolio sorts (in the "Testing" column). Newey-West adjusted t-statistics are shown in parentheses. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Value-Weighted Returns to Investment Strategies Sorted by NN3-Predicted Returns								
Training	Testing	Sample Period	Return	CAPM	FFC+PS	FF5	FF6	SY
Nonmicrocaps	Nonmicrocaps	1987–2017	1.191***	1.512***	0.838***	0.838***	0.486***	0.306
			(4.22)	(5.38)	(4.48)	(3.49)	(2.96)	(1.57)
Credit Rating Sample	Credit Rating Sample	1999–2017	1.021**	1.460***	1.044***	0.605	0.406	0.137
			(1.98)	(3.80)	(3.33)	(1.54)	(1.39)	(0.42)
Nondowngrades	Nondowngrades	1999–2017	0.797**	0.764**	0.263	0.147	0.035	-0.132
			(2.33)	(2.02)	(0.82)	(0.46)	(0.13)	(-0.36)
Full Sample	Full Sample	1987–2017	1.556***	1.894***	1.361***	1.206***	0.916***	0.769***
			(4.53)	(5.64)	(5.31)	(4.66)	(4.08)	(3.03)
Full Sample	Nonmicrocaps	1987–2017	1.047***	1.389***	0.771***	0.627**	0.312	0.179
			(3.24)	(4.43)	(3.24)	(2.41)	(1.51)	(0.73)
Full Sample	Full Sample	1999–2017	1.753***	2.144***	1.610***	1.307***	1.154***	0.827**
			(3.44)	(5.00)	(5.16)	(4.02)	(3.98)	(2.50)
Full Sample	Nonmicrocaps	1999–2017	1.232**	1.647***	1.102***	0.771**	0.609**	0.326
			(2.56)	(4.17)	(3.82)	(2.43)	(2.36)	(1.05)
Full Sample	Credit Rating Sample	1999–2017	1.267**	1.641***	1.087***	0.907**	0.724**	0.265
			(2.59)	(4.11)	(3.55)	(2.41)	(2.50)	(0.78)
Full Sample	Nondowngrades	1999–2017	0.858*	1.177***	0.667**	0.628*	0.460	-0.113
			(1.97)	(3.01)	(2.13)	(1.69)	(1.43)	(-0.33)

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Table 3: Performance of Portfolios Sorted by Risk Loadings on the Stochastic Discount Factor

At the end of each month t, stocks are sorted into deciles according to their risk loadings (β) on the stochastic discount factor estimated from a combination of deep neural networks (Chen, Pelger, and Zhu 2020). This table reports the value-weighted returns for month $t+1$ for each decile portfolio as well as the strategy of going long (short) on the highest (lowest) risk loading stocks ("HML") over the entire sample period from 1987 to 2016. Portfolio returns are further adjusted by the CAPM, Fama-French-Carhart 4-factor and Pástor-Stambaugh liquidity factor model (FFC+PS), Fama-French 5-factor model (FF5), Fama-French 6-factor model (FF6), and Stambaugh-Yuan 4-factor model (SY). Panel A reports the results for the full sample as well as the subsample that excludes microcaps. Panel B reports similar statistics for the subsamples that exclude nonrated firms and credit rating downgrades. Newey-West adjusted t-statistics are shown in parentheses. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Panel A: Value-Weighted Returns to Investment Strategies Sorted by Risk Loadings (Full Sample and Microcaps Excluded)

Rank of β	Full Sample						Nonmicrocaps					
	Return	CAPM	FFC+PS	FF5	FF6	SY	Return	CAPM	FFC+PS	FF5	FF6	SY
Low	0.078	-0.962***	-0.593***	-0.859***	-0.590***	-0.425*	0.342	-0.662***	-0.355**	-0.617***	-0.398**	-0.284
	(0.21)	(-4.35)	(-3.28)	(-3.58)	(-3.01)	(-1.82)	(0.99)	(-3.48)	(-2.17)	(-3.11)	(-2.25)	(-1.36)
2	0.735**	-0.205	-0.055	-0.206	-0.103	-0.077	0.673**	-0.264*	-0.093	-0.222	-0.118	-0.048
	(2.45)	(-1.35)	(-0.37)	(-1.35)	(-0.65)	(-0.43)	(2.42)	(-1.81)	(-0.65)	(-1.54)	(-0.79)	(-0.26)
3	0.812***	-0.123	0.033	-0.026	0.040	0.109	1.079***	0.163	0.330***	0.270**	0.335***	0.385**
	(2.95)	(-0.99)	(0.30)	(-0.23)	(0.33)	(0.78)	(4.10)	(1.32)	(2.77)	(2.23)	(2.63)	(2.58)
4	0.859***	-0.035	0.094	0.012	0.061	0.127	0.705**	-0.206	-0.129	-0.179*	-0.177*	-0.124
	(3.36)	(-0.33)	(0.89)	(0.12)	(0.53)	(0.93)	(2.47)	(-1.51)	(-1.21)	(-1.71)	(-1.70)	(-1.17)
5	0.965***	0.055	0.064	0.042	0.025	0.018	0.987***	0.090	0.120	0.084	0.102	0.060
	(3.82)	(0.53)	(0.68)	(0.43)	(0.26)	(0.19)	(4.03)	(0.88)	(1.20)	(0.78)	(0.94)	(0.53)
6	1.131***	0.231*	0.173	0.220**	0.167	0.156	1.227***	0.345***	0.273***	0.291***	0.251***	0.250**
	(4.40)	(1.97)	(1.61)	(2.06)	(1.63)	(1.33)	(5.16)	(3.65)	(2.95)	(3.01)	(2.66)	(2.58)
7	1.255***	0.366***	0.283***	0.241**	0.202*	0.249**	1.054***	0.171	0.107	0.113	0.065	0.061
	(5.24)	(3.83)	(2.94)	(2.54)	(1.91)	(2.33)	(4.02)	(1.25)	(0.85)	(0.93)	(0.53)	(0.46)
8	1.224***	0.355***	0.233**	0.150	0.126	0.204*	1.138***	0.235**	0.127	0.076	0.024	0.059
	(4.95)	(3.11)	(2.24)	(1.28)	(1.09)	(1.73)	(4.46)	(2.30)	(1.22)	(0.73)	(0.21)	(0.49)
9	1.476***	0.533***	0.419***	0.238	0.277*	0.395**	1.289***	0.407***	0.287**	0.192	0.191	0.315**
	(4.92)	(2.74)	(2.71)	(1.49)	(1.78)	(2.51)	(5.05)	(2.85)	(2.13)	(1.30)	(1.27)	(2.20)
High	2.261***	1.094***	1.325***	1.010***	1.277***	1.558***	1.425***	0.420**	0.300*	0.102	0.150	0.264
	(5.05)	(3.58)	(5.04)	(3.01)	(4.39)	(4.95)	(4.55)	(2.30)	(1.92)	(0.65)	(0.96)	(1.64)
HML	2.183***	2.056***	1.918***	1.869***	1.867***	1.983***	1.083***	1.083***	0.655***	0.720***	0.548**	0.548**
	(6.37)	(5.68)	(5.66)	(5.29)	(4.86)	(5.30)	(4.28)	(4.06)	(2.79)	(2.93)	(2.23)	(2.27)

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Table 3—Continued

Panel B: Value-Weighted Returns to Investment Strategies Sorted by Risk Loadings (Credit Rating Sample and Downgrades Excluded)

Rank of β	Credit Rating Sample						Nondowngrades					
	Return	CAPM	FFC+PS	FF5	FF6	SY	Return	CAPM	FFC+PS	FF5	FF6	SY
Low	0.504	-0.516**	-0.257	-0.579**	-0.371*	-0.205	0.711**	-0.280	-0.030	-0.324	-0.150	0.025
	(1.38)	(-2.39)	(-1.31)	(-2.48)	(-1.79)	(-0.83)	(1.98)	(-1.25)	(-0.14)	(-1.35)	(-0.68)	(0.10)
2	0.853***	-0.063	0.100	-0.052	0.056	0.161	0.965***	0.074	0.222	0.088	0.174	0.289
	(2.85)	(-0.37)	(0.61)	(-0.27)	(0.28)	(0.78)	(3.14)	(0.39)	(1.30)	(0.41)	(0.83)	(1.42)
3	0.928***	0.024	0.149	0.072	0.121	0.209	1.041***	0.145	0.249*	0.235*	0.252*	0.291*
	(3.76)	(0.21)	(1.18)	(0.57)	(0.89)	(1.28)	(4.13)	(1.10)	(1.87)	(1.74)	(1.74)	(1.83)
4	0.757***	-0.091	-0.038	-0.238**	-0.202*	-0.104	0.841***	0.004	0.067	-0.141	-0.107	-0.004
	(3.04)	(-0.79)	(-0.34)	(-2.20)	(-1.90)	(-0.84)	(3.54)	(0.03)	(0.59)	(-1.29)	(-0.99)	(-0.03)
5	0.937***	0.059	0.071	0.012	0.029	0.038	1.034***	0.160	0.166	0.133	0.139	0.128
	(3.77)	(0.49)	(0.57)	(0.10)	(0.22)	(0.29)	(4.09)	(1.22)	(1.20)	(0.95)	(0.97)	(0.89)
6	1.120***	0.249**	0.193	0.219*	0.188	0.104	1.225***	0.361***	0.289**	0.336**	0.283**	0.214
	(4.73)	(2.02)	(1.57)	(1.74)	(1.47)	(0.78)	(5.09)	(2.63)	(2.16)	(2.43)	(2.09)	(1.47)
7	1.144***	0.292**	0.197	0.158	0.108	0.154	1.278***	0.424***	0.337**	0.306**	0.249*	0.287**
	(4.44)	(2.46)	(1.60)	(1.18)	(0.82)	(1.22)	(4.94)	(3.41)	(2.58)	(2.19)	(1.82)	(2.11)
8	1.164***	0.273**	0.158	0.105	0.058	0.123	1.269***	0.393***	0.270*	0.153	0.107	0.157
	(4.63)	(2.27)	(1.28)	(0.85)	(0.43)	(0.90)	(5.12)	(2.87)	(1.90)	(1.05)	(0.69)	(1.05)
9	1.253***	0.377**	0.245*	0.111	0.111	0.196	1.342***	0.483***	0.337**	0.212	0.204	0.291**
	(4.78)	(2.35)	(1.70)	(0.77)	(0.76)	(1.38)	(5.30)	(2.99)	(2.32)	(1.45)	(1.37)	(2.00)
High	1.316***	0.290	0.164	-0.020	0.052	0.134	1.626***	0.689***	0.567***	0.372*	0.423**	0.536***
	(3.81)	(1.35)	(0.88)	(-0.10)	(0.27)	(0.69)	(5.01)	(3.02)	(2.93)	(1.89)	(2.13)	(2.85)
HML	0.812***	0.806***	0.420	0.559*	0.423	0.339	0.915***	0.969***	0.597**	0.696**	0.574*	0.511
	(2.83)	(2.68)	(1.51)	(1.88)	(1.46)	(1.18)	(2.91)	(3.01)	(2.00)	(2.14)	(1.82)	(1.64)

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Table 4: Performance of Portfolios Sorted by IPCA and Conditional Autoencoder Predicted Returns

In Panel A, at the end of each month t, stocks are sorted into deciles according to their one-month-ahead out-of-sample predicted returns ($\hat{R}$) using instrumented principal component analysis (IPCA) (Kelly, Pruitt, and Su 2019). We report the value-weighted returns for month $t+1$ for each decile portfolio as well as the strategy of going long (short) on the highest (lowest) expected return stocks ("HML") over the entire sample period from 1987 to 2017. Portfolio returns are further adjusted by the Fama-French 6-factor model (FF6). We report results for the full sample as well as for subsamples that exclude microcaps, nonrated firms, and credit rating downgrades. Panel B reports similar statistics when decile portfolios are sorted by the one-month-ahead out-of-sample predicted returns ($\hat{R}$) using a conditional autoencoder model with two hidden layers and five latent factors (CA2) (Gu, Kelly, and Xiu 2021). Both IPCA and CA2 impose a zero-alpha restriction. Newey-West adjusted t-statistics are shown in parentheses. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Panel A: Value-Weighted Returns to Investment Strategies Sorted by IPCA-Predicted Returns

Rank of $\hat{R}$	Full Sample		Nonmicrocaps		Credit Rating Sample		Nondowngrades	
	Return	FF6	Return	FF6	Return	FF6	Return	FF6
Low	0.454	-0.331***	0.353	-0.457***	0.357	-0.475***	0.645**	-0.217*
	(1.59)	(-3.54)	(1.17)	(-4.19)	(1.17)	(-3.76)	(2.40)	(-1.72)
2	0.763***	-0.095	0.756***	-0.075	0.782***	-0.131	0.989***	0.082
	(2.79)	(-1.14)	(2.80)	(-0.86)	(2.85)	(-1.35)	(3.87)	(0.84)
3	0.881***	-0.020	0.877***	-0.007	0.888***	-0.044	1.024***	0.094
	(3.71)	(-0.29)	(3.44)	(-0.10)	(3.41)	(-0.47)	(4.30)	(0.95)
4	1.021***	0.109*	0.871***	0.011	0.963***	0.001	1.115***	0.166*
	(4.18)	(1.84)	(3.64)	(0.14)	(4.11)	(0.02)	(5.13)	(1.74)
5	0.949***	0.009	1.057***	0.093	0.964***	0.033	1.071***	0.153*
	(4.25)	(0.12)	(4.42)	(1.34)	(3.98)	(0.44)	(4.72)	(1.91)
6	1.120***	0.124	0.993***	0.050	1.071***	0.097	1.182***	0.209**
	(4.78)	(1.62)	(4.46)	(0.67)	(4.82)	(1.19)	(5.35)	(2.31)
7	1.005***	-0.002	1.112***	0.125	1.093***	0.036	1.221***	0.169**
	(4.17)	(-0.03)	(4.67)	(1.55)	(4.72)	(0.43)	(5.73)	(1.99)
8	1.055***	0.005	1.020***	0.007	0.967***	-0.080	1.037***	-0.019
	(4.57)	(0.07)	(4.22)	(0.11)	(4.13)	(-0.98)	(4.63)	(-0.19)
9	1.298***	0.188*	1.099***	0.030	1.106***	-0.047	1.212***	0.084
	(5.05)	(1.91)	(4.79)	(0.35)	(4.64)	(-0.46)	(5.29)	(0.76)
High	1.399***	0.293**	1.254***	0.157	1.249***	0.132	1.378***	0.214*
	(5.01)	(2.15)	(4.77)	(1.49)	(4.80)	(1.17)	(5.27)	(1.72)
HML	0.945***	0.624***	0.901***	0.613***	0.893***	0.607***	0.733***	0.430**
	(5.62)	(3.31)	(5.57)	(3.71)	(5.08)	(3.38)	(4.08)	(2.37)

Panel B: Value-Weighted Returns to Investment Strategies Sorted by CA2-Predicted Returns

Rank of $\hat{R}$	Full Sample		Nonmicrocaps		Credit Rating Sample		Nondowngrades	
	Return	FF6	Return	FF6	Return	FF6	Return	FF6
Low	-0.045	-0.506***	0.210	-0.254*	0.309	-0.241	0.683*	0.038
	(-0.11)	(-3.13)	(0.53)	(-1.75)	(0.78)	(-1.42)	(1.79)	(0.18)
2	0.588*	-0.163	0.622**	-0.110	0.673**	-0.177	0.918***	0.100
	(1.94)	(-1.49)	(2.08)	(-0.97)	(2.21)	(-1.33)	(3.40)	(0.81)
3	0.740***	-0.126	0.674***	-0.141	0.820***	-0.047	1.027***	0.182
	(2.99)	(-1.45)	(2.64)	(-1.55)	(3.19)	(-0.43)	(4.30)	(1.52)
4	0.857***	-0.046	0.809***	-0.072	0.795***	-0.136*	0.929***	0.009
	(3.75)	(-0.60)	(3.56)	(-1.11)	(3.36)	(-1.69)	(4.19)	(0.10)
5	1.041***	0.093	0.952***	0.017	0.896***	-0.075	1.010***	0.043
	(4.69)	(1.51)	(4.33)	(0.23)	(4.00)	(-1.19)	(4.74)	(0.62)
6	1.053***	0.085	1.034***	0.092	1.085***	0.069	1.154***	0.155**
	(4.27)	(1.18)	(4.49)	(1.58)	(4.95)	(1.07)	(5.49)	(2.13)
7	1.160***	0.115	1.102***	0.126	1.031***	0.020	1.184***	0.180*
	(4.45)	(1.21)	(4.35)	(1.34)	(4.09)	(0.22)	(4.89)	(1.79)
8	1.207***	0.176*	1.107***	0.066	1.187***	0.117	1.339***	0.276**
	(4.39)	(1.78)	(4.26)	(0.67)	(4.66)	(1.04)	(5.47)	(2.27)
9	1.220***	0.057	1.248***	0.180*	1.217***	0.112	1.335***	0.238*
	(3.89)	(0.50)	(4.44)	(1.68)	(4.32)	(0.88)	(5.04)	(1.81)
High	1.114***	0.241	1.316***	0.133	1.183***	-0.054	1.348***	0.086
	(2.89)	(1.37)	(4.28)	(1.23)	(3.53)	(-0.35)	(4.27)	(0.56)
HML	1.159***	0.746***	1.105***	0.387**	0.874***	0.187	0.665**	0.048
	(4.17)	(3.01)	(4.22)	(2.03)	(2.97)	(0.79)	(2.22)	(0.20)

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Table 5: Nonnormality and Turnover of Machine Learning Portfolios

In Panel A, at the end of each month t, stocks are sorted into deciles according to their NN3-predicted returns (Gu, Kelly, and Xiu 2020). We compute the value-weighted returns for month $t + 1$ for the strategy of going long (short) on the highest (lowest) expected return stocks over the entire sample period from 1987 to 2017. For the long-short strategy, we report the annualized Sharpe ratio, the skewness and excess kurtosis of the monthly returns, the maximum drawdown, the average monthly return during the crisis period, and the monthly turnover. We report the results for the full sample as well as subsamples that exclude microcaps, nonrated firms, and credit rating downgrades. Panels B to D report similar statistics when decile portfolios are sorted by the risk loadings on the stochastic discount factor (Chen, Pelger, and Zhu 2020), IPCA-predicted returns (Kelly, Pruitt, and Su 2019), and CA2-predicted returns (Gu, Kelly, and Xiu 2021), respectively. Panel E reports similar statistics on the value-weighted market portfolio in excess of the 1-month T-bill rate. All returns on the long-short strategy and market index are scaled to 10% volatility per year.

Characteristics of Value-weighted Machine Learning Portfolios						
	Sharpe Ratio	Skewness	Excess Kurtosis	Maximum Drawdown	Return in Crisis	Turnover
Panel A: Sorted by NN3-Predicted Returns						
Full Sample	0.944	0.631	5.222	0.350	4.100	0.976
Nonmicrocaps	0.644	0.361	7.062	0.349	3.563	0.869
Credit Rating Sample	0.639	0.064	7.875	0.420	3.435	0.889
Nondowngrades	0.449	0.146	8.550	0.333	2.931	0.920
Panel B: Sorted by Risk Loadings						
Full Sample	1.225	1.063	5.932	0.209	0.472	1.664
Nonmicrocaps	0.839	0.326	1.582	0.246	0.677	1.625
Credit Rating Sample	0.566	0.267	1.440	0.407	-0.023	1.652
Nondowngrades	0.602	0.344	1.675	0.447	0.903	1.678
Panel C: Sorted by IPCA-Predicted Returns						
Full Sample	0.967	-0.449	4.805	0.203	0.574	1.186
Nonmicrocaps	0.978	-0.267	5.369	0.234	1.493	1.130
Credit Rating Sample	0.880	-0.219	4.221	0.315	0.474	1.164
Nondowngrades	0.697	-0.069	3.158	0.349	-0.640	1.184
Panel D: Sorted by CA2-Predicted Returns						
Full Sample	0.784	-0.077	2.418	0.202	-0.047	1.565
Nonmicrocaps	0.748	0.291	4.684	0.207	-0.529	1.478
Credit Rating Sample	0.522	-0.471	4.119	0.252	-1.796	1.542
Nondowngrades	0.387	-0.616	4.930	0.345	-0.167	1.571
Panel E: Market Portfolio						
Full Sample	0.527	-0.978	3.323	0.486	-6.954	0.089
Nonmicrocaps	0.530	-0.959	3.222	0.485	-6.907	0.086
Credit Rating Sample	0.543	-0.932	3.423	0.498	-6.747	0.080
Nondowngrades	0.682	-0.856	3.311	0.408	-6.615	0.084

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Table 6: Performance and Weights of SDF-Implied Mean-Variance Efficient Portfolios

We follow Kozak, Nagel, and Santosh (2020) to construct the SDF-implied mean-variance efficient (MVE) portfolio based on ridge regression in the pre-2005 sample and estimate the out-of-sample performance in the 2005 to 2017 period. We report the monthly abnormal returns adjusted by CAPM and the Fama-French 6-factor model (FF6). CAPM-adjusted returns equal the SDF-implied MVE portfolio returns because the portfolio is orthogonal to the market. FF6-adjusted returns are estimated by regressing the SDF-implied MVE portfolio returns on benchmark portfolio returns, where the benchmark portfolio returns are estimated from unregularized MVE portfolio weights and five nonmarket factors in the pre-2005 period. We also report the annualized Sharpe ratio and the quantile distribution of the SDF-implied MVE portfolio weights. We report the results for the full sample as well as subsamples that exclude microcaps, nonrated firms, and credit rating downgrades. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Characteristics of SDF-Implied MVE Portfolios														
	CAPM	FF6	Sharpe Ratio	SDF-Implied MVE Portfolio Weights										
				Mean	Std.Dev.	Min	5%	10%	25%	Median	75%	90%	95%	Max
Full Sample	3.662***	3.338***	2.318	0.083	1.338	-2.994	-2.343	-1.994	-0.912	0.341	0.964	1.687	1.895	3.182
	(6.01)	(5.90)												
Nonmicrocaps	1.543***	0.895***	0.977	0.084	0.447	-0.863	-0.666	-0.592	-0.238	0.072	0.407	0.647	0.741	1.431
	(3.88)	(2.87)												
Credit Rating Sample	1.418***	0.717*	0.898	-0.006	0.254	-0.678	-0.473	-0.382	-0.137	-0.003	0.187	0.326	0.387	0.419
	(2.97)	(1.93)												
Nondowngrades	1.308***	0.545	0.828	-0.022	0.248	-0.543	-0.448	-0.370	-0.217	0.004	0.135	0.293	0.376	0.595
	(2.92)	(1.59)												

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Table 7: Performance of Machine Learning Portfolios by Market State

In Panel A, at the end of each month t, stocks are sorted into deciles according to their NN3-predicted returns (Gu, Kelly, and Xiu 2020). We report the value-weighted Fama-French 6-factor-adjusted returns for month $t+1$ for the strategy of going long (short) on the highest (lowest) expected return stocks in various subperiods, including when investor sentiment (SENT), aligned investor sentiment (PLS SENT), market volatility (MKTVOL), implied market volatility (VIX) and market illiquidity (MKTILLIQ) are high (above median) and low (below median) in month t. We obtain the median breakpoint for market illiquidity in the pre- and post-2001 periods separately. We report the results for the full sample as well as for subsamples that exclude microcaps, nonrated firms, and credit rating downgrades. Panels B to D report similar statistics when decile portfolios are sorted by the risk loadings on the stochastic discount factor (Chen, Pelger, and Zhu 2020), IPCA-predicted returns (Kelly, Pruitt, and Su 2019), and CA2-predicted returns (Gu, Kelly, and Xiu 2021), respectively. Internet Appendix A provides detailed definitions for each variable. Newey-West adjusted t-statistics are shown in parentheses. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Value-Weighted FF6-Adjusted Returns of Machine Learning Portfolios										
	SENT		PLS SENT		MKTVOL		VIX		MKTILLIQ	
	Low	High	Low	High	Low	High	Low	High	Low	High
Panel A: Sorted by NN3-Predicted Returns										
Full Sample	0.732***	0.879**	0.684**	1.041***	0.641**	1.283***	0.218	1.662***	0.736***	1.075***
	(3.00)	(2.40)	(2.27)	(2.82)	(2.34)	(3.96)	(0.95)	(4.16)	(2.70)	(3.36)
Nonmicrocaps	0.334	0.095	0.250	0.265	0.062	0.814***	-0.050	0.747**	0.222	0.461
	(1.45)	(0.29)	(0.89)	(0.82)	(0.26)	(2.87)	(-0.20)	(2.03)	(0.84)	(1.51)
Credit Rating Sample	0.389	0.274	0.335	0.452	0.095	0.896***	0.025	0.808*	0.512**	0.351
	(1.49)	(0.91)	(1.19)	(1.45)	(0.44)	(2.93)	(0.11)	(1.96)	(1.97)	(1.15)
Nondowngrades	0.355	-0.161	0.225	0.111	0.054	0.597*	-0.060	0.582	0.294	0.120
	(1.43)	(-0.50)	(0.81)	(0.33)	(0.23)	(1.70)	(-0.24)	(1.25)	(1.21)	(0.36)
Panel B: Sorted by Risk Loadings										
Full Sample	1.372***	2.453***	1.383***	2.198***	1.352***	2.364***	0.898**	2.451***	1.150***	2.128***
	(3.83)	(3.78)	(3.50)	(3.87)	(3.64)	(4.09)	(2.30)	(3.80)	(2.73)	(4.53)
Nonmicrocaps	0.512**	0.681	0.151	0.824**	0.423	0.665	0.321	0.386	0.527	0.432*
	(2.14)	(1.60)	(0.54)	(2.44)	(1.41)	(1.62)	(1.14)	(1.01)	(1.45)	(1.76)
Credit Rating Sample	0.269	0.658	0.244	0.500	0.448	0.527	0.097	0.337	0.691	0.073
	(0.84)	(1.35)	(0.61)	(1.29)	(1.12)	(1.12)	(0.31)	(0.69)	(1.63)	(0.20)
Nondowngrades	0.329	0.945*	0.411	0.685*	0.453	0.699	0.031	0.651	0.847*	0.155
	(0.93)	(1.79)	(0.87)	(1.66)	(0.96)	(1.45)	(0.09)	(1.19)	(1.86)	(0.39)

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Table 7—Continued

Value-Weighted FF6-Adjusted Returns of Machine Learning Portfolios										
	SENT		PLS SENT		MKTVOL		VIX		MKTILLIQ	
	Low	High	Low	High	Low	High	Low	High	Low	High
Panel C: Sorted by IPCA-Predicted Returns										
Full Sample	0.558**	0.667**	0.363*	0.753***	0.590***	0.694**	0.440**	0.763**	0.603**	0.645***
	(2.46)	(2.03)	(1.68)	(2.64)	(3.57)	(2.56)	(2.60)	(2.23)	(2.39)	(2.78)
Nonmicrocaps	0.606***	0.615**	0.387*	0.742***	0.566***	0.636**	0.478***	0.647**	0.759***	0.463**
	(3.06)	(2.15)	(1.96)	(2.94)	(3.28)	(2.59)	(2.98)	(2.25)	(3.34)	(2.23)
Credit Rating Sample	0.647***	0.512*	0.466**	0.667**	0.533***	0.653**	0.498**	0.649**	0.782***	0.447*
	(2.82)	(1.66)	(2.08)	(2.39)	(2.97)	(2.51)	(2.56)	(2.14)	(3.18)	(1.96)
Nondowngrades	0.535**	0.326	0.386	0.429	0.378**	0.451	0.478**	0.325	0.583**	0.292
	(2.37)	(1.10)	(1.64)	(1.55)	(2.04)	(1.60)	(2.23)	(0.96)	(2.22)	(1.26)
Panel D: Sorted by CA2-Predicted Returns										
Full Sample	0.748***	0.617	0.673**	0.869**	0.634**	0.606	0.453*	0.910**	0.359	0.787**
	(2.70)	(1.41)	(2.41)	(2.17)	(2.34)	(1.61)	(1.75)	(2.04)	(1.51)	(2.14)
Nonmicrocaps	0.584**	0.064	0.224	0.471	0.537**	0.254	0.524**	0.122	0.292	0.275
	(2.55)	(0.21)	(0.89)	(1.50)	(2.13)	(0.79)	(2.28)	(0.33)	(1.13)	(1.03)
Credit Rating Sample	0.304	-0.055	0.016	0.287	0.531	-0.074	0.508*	-0.300	-0.102	0.263
	(1.15)	(-0.15)	(0.05)	(0.78)	(1.49)	(-0.19)	(1.69)	(-0.67)	(-0.30)	(0.81)
Nondowngrades	0.204	-0.218	-0.173	0.153	0.376	-0.170	0.193	-0.248	-0.331	0.245
	(0.73)	(-0.54)	(-0.55)	(0.40)	(1.30)	(-0.45)	(0.73)	(-0.54)	(-0.89)	(0.84)

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Table 8: Performance of Machine Learning Portfolios in Recent Years

In Panel A, at the end of each month t, stocks are sorted into deciles according to their NN3-predicted returns (Gu, Kelly, and Xiu 2020). We report the value-weighted returns for month $t+1$ for the strategy of going long (short) on the highest (lowest) expected return stocks in the post-2001 period. Portfolio returns are further adjusted by the CAPM, Fama-French-Carhart 4-factor and Pástor-Stambaugh liquidity factor model (FFC+PS), Fama-French 5-factor model (FF5), Fama-French 6-factor model (FF6), and Stambaugh-Yuan 4-factor model (SY). We report the results for the full sample as well as for subsamples that exclude microcaps, nonrated firms, and credit rating downgrades. Panels B to D report similar statistics when decile portfolios are sorted by the risk loadings on the stochastic discount factor (Chen, Pelger, and Zhu 2020), IPCA-predicted returns (Kelly, Pruitt, and Su 2019), and CA2-predicted returns (Gu, Kelly, and Xiu 2021), respectively. Newey-West adjusted t-statistics are shown in parentheses. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Panel A: Value-Weighted Returns to Investment Strategies Sorted by NN3-Predicted Returns						
	Return	CAPM	FFC+PS	FF5	FF6	SY
Full Sample	1.568***	2.012***	1.555***	1.096***	1.181***	0.699**
	(3.07)	(4.63)	(5.20)	(3.95)	(4.32)	(2.27)
Nonmicrocaps	1.106**	1.583***	1.116***	0.592**	0.687***	0.276
	(2.27)	(3.97)	(4.28)	(2.19)	(2.88)	(0.93)
Credit Rating Sample	1.182**	1.635***	1.165***	0.700**	0.803***	0.288
	(2.37)	(4.22)	(4.04)	(2.36)	(3.14)	(0.87)
Nondowngrades	0.796*	1.188***	0.768***	0.454	0.545*	-0.037
	(1.82)	(3.15)	(2.67)	(1.58)	(1.90)	(-0.11)
Panel B: Value-Weighted Returns to Investment Strategies Sorted by Risk Loadings						
	Return	CAPM	FFC+PS	FF5	FF6	SY
Full Sample	1.864***	1.759***	1.428***	1.060**	1.021**	1.407***
	(3.36)	(2.98)	(3.33)	(2.27)	(2.40)	(2.78)
Nonmicrocaps	0.954**	0.968**	0.625**	0.157	0.216	0.338
	(2.40)	(2.35)	(2.08)	(0.54)	(0.75)	(1.06)
Credit Rating Sample	0.759*	0.725*	0.400	0.114	0.165	0.237
	(1.87)	(1.67)	(1.13)	(0.33)	(0.48)	(0.63)
Nondowngrades	0.765*	0.786*	0.453	0.168	0.203	0.347
	(1.87)	(1.82)	(1.30)	(0.48)	(0.59)	(0.90)
Panel C: Value-Weighted Returns to Investment Strategies Sorted by IPCA-Predicted Returns						
	Return	CAPM	FFC+PS	FF5	FF6	SY
Full Sample	0.881***	0.967***	0.775***	0.643**	0.691***	0.321
	(3.95)	(4.32)	(3.81)	(2.58)	(3.31)	(1.19)
Nonmicrocaps	0.732***	0.862***	0.682***	0.532**	0.578***	0.216
	(3.47)	(4.19)	(3.76)	(2.38)	(3.06)	(0.92)
Credit Rating Sample	0.830***	0.953***	0.763***	0.661***	0.714***	0.279
	(3.68)	(4.31)	(3.65)	(2.61)	(3.28)	(1.03)
Nondowngrades	0.643***	0.724***	0.505**	0.475*	0.531**	0.088
	(2.86)	(3.05)	(2.34)	(1.76)	(2.35)	(0.31)
Panel D: Value-Weighted Returns to Investment Strategies Sorted by CA2-Predicted Returns						
	Return	CAPM	FFC+PS	FF5	FF6	SY
Full Sample	1.651***	1.753***	1.301***	1.065***	1.122***	0.840**
	(3.96)	(4.08)	(4.22)	(3.11)	(3.38)	(2.46)
Nonmicrocaps	1.388***	1.591***	1.083***	0.647**	0.760***	0.419
	(3.30)	(3.67)	(3.95)	(2.01)	(2.84)	(1.40)
Credit Rating Sample	1.245***	1.358***	0.784***	0.436	0.566*	0.179
	(2.85)	(3.01)	(2.64)	(1.12)	(1.80)	(0.50)
Nondowngrades	0.964**	1.072**	0.566**	0.265	0.363	0.095
	(2.35)	(2.54)	(1.99)	(0.92)	(1.40)	(0.30)

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Table 9: Stock Characteristics of Machine Learning Portfolios

In Panel A, at the end of each month t, stocks are sorted into deciles according to their NN3-predicted returns (Gu, Kelly, and Xiu 2020). We report, for the bottom and top decile portfolios, the contemporaneous equal-weighted average Log(Price), Log(Size), Book-to-market, Log(Illiquidity), Beta, 1M Return, 12M Momentum, IdioVol, Absolute Accruals, Log(Age), Assets Growth, ΔShares Outstanding, Corporate Investment, Dividend-to-price, Gross Profitability, Leverage, ROA, ROE, %Rated, Credit Rating, Analyst Coverage, Analyst Dispersion and SUE, as well as the difference in values between high and low decile portfolios ("HML"). We also report similar statistics when decile portfolios are sorted by the risk loadings on the stochastic discount factor (Chen, Pelger, and Zhu 2020). Panel B reports similar statistics when decile portfolios are sorted by the IPCA-predicted returns (Kelly, Pruitt, and Su 2019) and CA2-predicted returns (Gu, Kelly, and Xiu 2021). Internet Appendix A provides detailed definitions for each variable. Newey-West adjusted t-statistics are shown in parentheses. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Panel A: Stock Characteristics of Machine Learning Portfolios

Stock Characteristics	Sorted by NN3-Predicted Returns				Sorted by Risk Loadings			
	Low	High	HML	*t*-stat	Low	High	HML	*t*-stat
Log (Price)	2.092	1.386	-0.706***	(-14.63)	1.670	1.304	-0.366***	(-12.89)
Log (Size)	5.483	3.613	-1.870***	(-25.33)	4.646	3.524	-1.123***	(-29.82)
Book-to-Market	0.780	1.391	0.612***	(10.62)	0.914	1.353	0.440***	(13.15)
Log (Illiquidity)	1.220	3.791	2.571***	(22.70)	2.760	3.911	1.150***	(18.94)
Beta	1.330	1.068	-0.263***	(-6.87)	1.307	1.096	-0.211***	(-11.74)
1M Return	0.027	-0.012	-0.039***	(-10.22)	0.145	-0.118	-0.263***	(-37.18)
12M Momentum	-0.124	0.185	0.309***	(15.64)	-0.246	0.083	0.329***	(20.49)
IdioVol	0.076	0.084	0.008***	(4.19)	0.081	0.082	0.001	(1.55)
Absolute Accruals	0.103	0.107	0.003	(1.08)	0.108	0.097	-0.011***	(-9.83)
Log (Age)	2.171	2.463	0.292***	(9.93)	2.700	2.716	0.016**	(2.48)
Assets Growth	0.586	0.014	-0.572***	(-11.76)	0.164	0.048	-0.116***	(-12.76)
ΔShares Outstanding	0.389	0.070	-0.319***	(-10.92)	0.158	0.064	-0.094***	(-15.58)
Corporate Investment	-0.107	0.022	0.129***	(9.03)	-0.026	0.001	0.027***	(4.72)
Dividend-to-Price	0.012	0.009	-0.003***	(-3.61)	0.009	0.011	0.002***	(6.68)
Gross Profitability	0.329	0.354	0.025	(1.57)	0.387	0.420	0.032***	(8.11)
Leverage	1.308	1.832	0.524***	(4.18)	1.340	1.388	0.048	(1.16)
ROA	-0.024	-0.011	0.013***	(5.41)	-0.019	-0.009	0.010***	(12.04)
ROE	-0.045	-0.017	0.028***	(5.87)	-0.039	-0.017	0.022***	(12.77)
%Rated	0.236	0.093	-0.142***	(-13.48)	0.196	0.102	-0.094***	(-17.53)
Credit Rating	11.350	12.047	0.697**	(2.44)	12.547	13.036	0.490***	(4.38)
Analyst Coverage	4.142	1.432	-2.710***	(-14.99)	3.022	1.441	-1.581***	(-17.34)
Analyst Dispersion	0.049	0.057	0.008	(0.68)	0.050	0.083	0.033***	(2.61)
SUE	-0.019	-0.009	0.010***	(3.17)	-0.023	-0.016	0.006***	(2.79)

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Table 9—Continued

Panel B: Stock Characteristics of Machine Learning Portfolios								
Stock Characteristics	Sorted by IPCA-Predicted Returns				Sorted by CA2-Predicted Returns			
	Low	High	HML	*t*-stat	Low	High	HML	*t*-stat
Log (Price)	2.301	1.853	-0.449***	(-21.81)	1.636	0.951	-0.685***	(-21.54)
Log (Size)	5.748	4.067	-1.681***	(-67.77)	4.690	3.255	-1.435***	(-31.52)
Book-to-Market	0.871	1.348	0.477***	(15.70)	0.888	1.240	0.352***	(7.01)
Log (Illiquidity)	1.108	2.880	1.772***	(50.20)	2.347	4.447	2.100***	(25.37)
Beta	1.142	1.056	-0.086***	(-4.89)	1.423	1.150	-0.273***	(-11.84)
1M Return	0.030	-0.028	-0.058***	(-24.57)	-0.012	0.015	0.027***	(10.50)
12M Momentum	-0.085	0.285	0.370***	(21.26)	-0.132	-0.056	0.077***	(4.62)
IdioVol	0.068	0.074	0.006***	(8.48)	0.088	0.093	0.004***	(4.40)
Absolute Accruals	0.107	0.082	-0.025***	(-19.71)	0.109	0.111	0.003*	(1.72)
Log (Age)	2.279	2.476	0.197***	(21.21)	2.151	2.284	0.133***	(11.50)
Assets Growth	0.542	0.016	-0.526***	(-19.13)	0.423	0.114	-0.310***	(-9.60)
ΔShares Outstanding	0.343	0.050	-0.294***	(-18.38)	0.291	0.139	-0.151***	(-9.96)
Corporate Investment	-0.068	0.006	0.074***	(9.50)	-0.069	-0.012	0.058***	(7.01)
Dividend-to-Price	0.014	0.010	-0.004***	(-12.93)	0.011	0.007	-0.004***	(-5.67)
Gross Profitability	0.316	0.383	0.066***	(9.95)	0.330	0.329	-0.001	(-0.15)
Leverage	1.054	1.830	0.777***	(11.37)	1.565	1.808	0.243**	(2.38)
ROA	-0.019	0.005	0.024***	(22.44)	-0.025	-0.026	-0.001	(-0.59)
ROE	-0.038	0.017	0.055***	(24.02)	-0.047	-0.048	-0.001	(-0.62)
%Rated	0.290	0.119	-0.171***	(-35.38)	0.170	0.075	-0.095***	(-12.73)
Credit Rating	10.573	11.009	0.436***	(4.46)	12.929	13.920	0.991***	(6.40)
Analyst Coverage	4.071	1.930	-2.141***	(-24.91)	3.248	1.136	-2.112***	(-13.47)
Analyst Dispersion	0.085	0.070	-0.014	(-1.24)	0.057	0.010	-0.047*	(-1.97)
SUE	-0.012	-0.001	0.012***	(12.45)	-0.025	-0.022	0.002	(0.98)

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Table 10: Performance of Machine Learning Portfolios and their Attribution

This table reports the monthly returns from the unconditional strategy (WML) and its decomposition into intra-industry (WML^{INTRA}) and inter-industry (WML^{INTER}) returns components, as depicted in Equation (5). In Panel A1, we report the returns for WML, when the strategy takes long (short) positions in the winner (loser) stocks that are expected to outperform (underperform) the market according to the NN3-predicted returns (Gu, Kelly, and Xiu 2020). In Panel A2, we report the returns for WML^{INTRA} when the strategy takes long (short) positions in the stocks that are expected to outperform (underperform) the industry portfolio. H^{INTRA} and H are the scaling factors for the unconditional and intra-industry strategies. Panel A3 reports the returns for WML^{INTER} when the strategy takes long (short) positions in the industry portfolios that are expected to outperform (underperform) the market portfolio. H^{INTER} is the scaling factor for the inter-industry strategy. Portfolio returns are further adjusted by the CAPM, Fama-French-Carhart 4-factor and Pástor-Stambaugh liquidity factor model (FFC+PS), Fama-French 5-factor model (FF5), Fama-French 6-factor model (FF6), and Stambaugh-Yuan 4-factor model (SY). Panel A reports the results for the full sample as well as the subsample that excludes microcaps. Panel B reports similar statistics for the subsamples that exclude nonrated firms and credit rating downgrades. Newey-West adjusted t-statistics are shown in parentheses. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Panel A: Returns to Investment Strategies Sorted by NN3-Predicted Returns (Full Sample and Microcaps Excluded)

Rank	Full Sample						Nonmicrocaps					
	Return	CAPM	FFC+PS	FF5	FF6	SY	Return	CAPM	FFC+PS	FF5	FF6	SY
Panel A1: Unconditional Payoff												
Loser	0.239	-0.907***	-0.510***	-0.458***	-0.204*	-0.103	0.450	-0.787***	-0.369***	-0.272*	-0.029	0.058
	(0.60)	(-4.61)	(-3.99)	(-2.64)	(-1.71)	(-0.58)	(1.12)	(-4.08)	(-2.91)	(-1.67)	(-0.30)	(0.35)
Winner	2.050***	1.093***	1.268***	1.207***	1.351***	1.496***	1.368***	0.433***	0.300***	0.183**	0.153**	0.251***
	(5.23)	(3.87)	(5.35)	(4.70)	(5.03)	(5.04)	(5.07)	(2.87)	(3.66)	(2.53)	(2.08)	(2.70)
WML	1.812***	2.000***	1.778***	1.665***	1.554***	1.599***	0.918***	1.220***	0.669***	0.455**	0.182	0.194
	(8.12)	(8.55)	(8.24)	(8.43)	(7.24)	(7.07)	(3.50)	(4.76)	(3.70)	(2.42)	(1.38)	(0.92)
Panel A2: Intra-Industry Payoff												
Loser	0.310	-0.842***	-0.533***	-0.466***	-0.267**	-0.219	0.541	-0.691***	-0.389***	-0.325***	-0.145*	-0.094
	(0.77)	(-4.45)	(-4.18)	(-3.28)	(-2.38)	(-1.34)	(1.38)	(-4.35)	(-3.63)	(-2.70)	(-1.80)	(-0.68)
Winner	1.996***	1.025***	1.286***	1.212***	1.407***	1.584***	1.293***	0.329**	0.288***	0.174***	0.194***	0.310***
	(5.05)	(3.66)	(5.19)	(4.31)	(4.91)	(4.91)	(4.79)	(2.54)	(4.13)	(2.64)	(3.01)	(3.93)
WML^{INTRA}	1.686***	1.867***	1.819***	1.678***	1.675***	1.802***	0.752***	1.021***	0.678***	0.499***	0.339***	0.404**
	(8.75)	(9.90)	(8.62)	(8.73)	(7.58)	(7.68)	(3.69)	(5.81)	(4.70)	(4.23)	(3.32)	(2.57)
$WML^{INTRA} \times H^{INTRA}/H$	1.519***	1.681***	1.642***	1.516***	1.516***	1.637***	0.643***	0.870***	0.588***	0.429***	0.300***	0.372***
	(8.78)	(9.98)	(8.57)	(8.69)	(7.53)	(7.61)	(3.69)	(5.82)	(4.64)	(4.29)	(3.33)	(2.71)
Panel A3: Inter-Industry Payoff												
Loser	0.670**	-0.330*	0.042	-0.015	0.256	0.472**	0.678**	-0.394**	-0.086	-0.076	0.133	0.282*
	(1.99)	(-1.80)	(0.31)	(-0.06)	(1.50)	(2.12)	(2.08)	(-2.37)	(-0.76)	(-0.47)	(1.26)	(1.95)
Winner	1.353***	0.403*	0.358**	0.362**	0.372**	0.436**	1.184***	0.230	0.068	-0.015	-0.059	0.011
	(3.93)	(1.86)	(2.47)	(2.55)	(2.44)	(2.53)	(4.24)	(1.49)	(0.81)	(-0.19)	(-0.75)	(0.11)
WML^{INTER}	0.683***	0.733***	0.316*	0.376	0.116	-0.036	0.506**	0.625***	0.154	0.061	-0.192	-0.271
	(3.46)	(3.49)	(1.91)	(1.61)	(0.67)	(-0.16)	(2.42)	(2.70)	(0.95)	(0.31)	(-1.37)	(-1.41)
$WML^{INTER} \times H^{INTER}/H$	0.293***	0.319***	0.136*	0.149	0.038	-0.037	0.275**	0.350***	0.080	0.026	-0.119	-0.179
	(3.27)	(3.27)	(1.76)	(1.52)	(0.49)	(-0.37)	(2.46)	(2.77)	(0.88)	(0.24)	(-1.43)	(-1.59)

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Table 10—Continued

Panel B: Returns to Investment Strategies Sorted by NN3-Predicted Returns (Credit Rating Sample and Downgrades Excluded)

Rank	Credit Rating Sample						Nondowngrades					
	Return	CAPM	FFC+PS	FF5	FF6	SY	Return	CAPM	FFC+PS	FF5	FF6	SY
Panel B1: Unconditional Payoff												
Loser	0.337	-0.911***	-0.493***	-0.653***	-0.308**	-0.055	1.161***	0.040	0.305***	0.154	0.377***	0.642***
	(0.78)	(-4.22)	(-3.78)	(-2.67)	(-2.30)	(-0.25)	(3.44)	(0.25)	(2.91)	(0.91)	(3.28)	(3.63)
Winner	1.332***	0.386**	0.278**	0.057	0.090	0.271**	1.639***	0.738***	0.573***	0.380***	0.365***	0.505***
	(4.39)	(2.14)	(2.46)	(0.53)	(0.89)	(2.04)	(6.21)	(4.62)	(5.98)	(4.81)	(4.54)	(4.47)
WML	0.995***	1.296***	0.770***	0.710***	0.398**	0.326	0.478**	0.698***	0.269**	0.226	-0.012	-0.137
	(3.76)	(5.40)	(4.20)	(3.14)	(2.48)	(1.43)	(2.42)	(3.81)	(2.13)	(1.35)	(-0.10)	(-0.72)
Panel B2: Intra-Industry Payoff												
Loser	0.509	-0.713***	-0.414***	-0.601***	-0.320**	-0.103	1.228***	0.127	0.286**	0.131	0.291**	0.513***
	(1.22)	(-3.68)	(-3.25)	(-2.88)	(-2.47)	(-0.48)	(3.77)	(0.88)	(2.57)	(0.93)	(2.56)	(3.04)
Winner	1.227***	0.241	0.230**	0.007	0.097	0.308**	1.581***	0.649***	0.555***	0.367***	0.396***	0.569***
	(3.92)	(1.42)	(2.25)	(0.05)	(0.99)	(2.48)	(5.83)	(4.37)	(6.28)	(4.29)	(4.77)	(5.34)
WML^{INTRA}	0.718***	0.954***	0.644***	0.607***	0.417***	0.411**	0.353**	0.522***	0.269**	0.236**	0.104	0.056
	(3.56)	(5.57)	(4.20)	(3.89)	(3.08)	(2.13)	(2.41)	(3.92)	(2.45)	(2.00)	(0.94)	(0.36)
$WML^{INTRA} \times H^{INTRA}/H$	0.608***	0.805***	0.550***	0.511***	0.356***	0.363**	0.304**	0.443***	0.238***	0.204**	0.099	0.068
	(3.65)	(5.69)	(4.35)	(3.92)	(3.19)	(2.32)	(2.57)	(4.10)	(2.72)	(2.15)	(1.13)	(0.56)
Panel B3: Inter-Industry Payoff												
Loser	0.499	-0.631***	-0.310***	-0.491**	-0.200	0.075	1.179***	0.135	0.349***	0.156	0.369***	0.642***
	(1.34)	(-3.30)	(-2.64)	(-2.30)	(-1.59)	(0.42)	(3.74)	(0.83)	(3.07)	(0.90)	(2.89)	(3.95)
Winner	1.195***	0.244	0.093	-0.126	-0.110	0.053	1.486***	0.574***	0.400***	0.196**	0.179**	0.311**
	(4.02)	(1.37)	(0.86)	(-1.30)	(-1.15)	(0.38)	(5.71)	(3.55)	(3.74)	(2.40)	(2.14)	(2.51)
WML^{INTER}	0.695***	0.876***	0.403**	0.365*	0.090	-0.022	0.307*	0.439***	0.051	0.040	-0.189	-0.332**
	(3.30)	(4.17)	(2.48)	(1.73)	(0.60)	(-0.12)	(1.85)	(2.71)	(0.39)	(0.23)	(-1.40)	(-2.05)
$WML^{INTER} \times H^{INTER}/H$	0.387***	0.491***	0.221**	0.199*	0.042	-0.037	0.174*	0.255***	0.031	0.022	-0.112	-0.205**
	(3.27)	(4.11)	(2.36)	(1.68)	(0.47)	(-0.33)	(1.82)	(2.72)	(0.41)	(0.23)	(-1.47)	(-2.15)

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Internet Appendix to

Machine Learning versus Economic Restrictions:

Evidence from Stock Return Predictability

This Internet Appendix presents additional empirical results. Most of the results presented here are discussed in the paper.

Section 1. Data Description

- Internet Appendix A: Variable Definitions
- Table IA1: Number of Stocks Over Time

Section 2. Robustness of Performance of Machine Learning Portfolios

- Table IA2: Portfolios Sorted by Neural Network Predicted Returns: Equal-Weighted Returns (Robustness of Table 1)
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Internet Appendix A: Variable Definitions

Variables	Definitions
A. Stock Characteristics	
Log(Price)	The logarithm of stock price as reported in CRSP.
Log(Size)	The logarithm of stock market capitalization, computed as the number of common shares outstanding times share price as reported in CRSP.
Book-to-Market	Book value of equity divided by the market capitalization at fiscal year-end, following Fama and French (2015).
Log(Illiquidity)	The logarithm of the stock illiquidity, computed in a given month t as follows: $ILLIQ_{i,t} = \left(\sum_{d\in t}\left|R_{i,d,t}\right|/VOLD_{i,d,t}\right)/D_{i,t} \times 10^6$, where $R_{i,d,t}$ refers to the percentage return of stock i in day d of month t, $VOLD_{i,d,t}$ refers to the dollar trading volume at the same time, and $D_{i,t}$ is the number of trading days for stock i in month t, following Amihud (2002).
Beta	The beta in a given month t is computed as follows: $R_{i,w,t} = \alpha_{i,t} + \beta_{i,t}MKT_{w,t} + e_{i,w,t}$, where $R_{i,w,t}$ refers to the accumulated return of stock i in week w of month t, and $MKT_{w,t}$ refers to the contemporaneous equal-weighted market returns. The $\beta_{i,t}$ is estimated in a three-year rolling window with at least 52 weeks of returns prior to the formation period, following Fama and MacBeth (1973).
1M Return	The monthly stock return as reported in CRSP.
12M Momentum	Formation period return in a given month t, computed as the cumulative 11-month return from month $t-11$ to month $t-1$, following Jegadeesh (1990).
IdioVol	The idiosyncratic volatility for stock i in month t, computed as the standard deviation of the residual $e_{i,w,t}$, where $e_{i,w,t}$ is defined as in *Beta*, following Ali, Hwang, and Trombley (2003).
Absolute Accruals	The absolute value of accruals, computed in a given year t as follows: $Accruals_{i,t} = \left(IB_{i,t} - OANCF_{i,t}\right)/\overline{ASSET}_{i,t}$, where $IB_{i,t}$ refers to the income before extraordinary items (COMPUSTAT annual item IB) of stock i in year t, $OANCF_{i,t}$ refers to the operating cash flows (item OANCF), and $\overline{ASSET}_{i,t}$ refers to the average total assets (item AT) of the beginning and end of year t. If $OANCF_{i,t}$ is missing, then set to $\left(\Delta CA_{i,t} - \Delta Cash_{i,t}\right) - \left(\Delta CL_{i,t} - \Delta STD_{i,t} - \Delta TP_{i,t}\right) - Dep_{i,t}$, where $\Delta CA_{i,t}$ refers to the change in current assets (COMPUSTAT annual item ACT) of stock i in year t, $\Delta Cash_{i,t}$ refers to the change in cash and short-term investments (item CHE), $\Delta CL_{i,t}$ refers to the change in current liabilities (item LCT), $\Delta STD_{i,t}$ refers to the change in debt included in current liabilities (item DLC), $\Delta TP_{i,t}$ refers to the change in income taxes payable (item TXP), and $Dep_{i,t}$ refers to the depreciation and amortization expense (item DP), following Sloan (1996) and Bandyopadhyay, Huang, and Wirjanto (2010).
Log(Age)	The logarithm of the number of years since the first COMPUSTAT coverage, following Jiang, Lee, and Zhang (2005).
Asset Growth	Asset growth in a given year t, computed as follows: $ASSETG_{i,t} = \left(ASSET_{i,t} - ASSET_{i,t-1}\right)/ASSET_{i,t-1}$, where $ASSET_{i,t}$ refers to the total assets (COMPUSTAT annual item AT) of stock i in year t, following Cooper, Gulen, and Schill (2008).
ΔShares Outstanding	The annual percent change in shares outstanding, following Pontiff and Woodgate (2008).
Corporate Investment	Corporate investment in a given quarter t is computed as follows: $CI_{i,t} = PPE_{i,t} - \left(PPE_{i,t-1} + PPE_{i,t-2} + PPE_{i,t-3}\right)/3$, where $PPE_{i,t}$ refers to the ratio of change in net property, plant, and equipment (COMPUSTAT quarterly item PPENTQ) divided by sales (item SALEQ) of stock i in quarter t. If SALEQ is 0 or negative, then replace SALEQ with 0.01, following Titman, Wei, and Xie (2004).
Dividend-to-Price	Total dividends (COMPUSTAT annual item DVT) divided by market capitalization at fiscal year-end, following Litzenberger and Ramaswamy (1982).
Gross Profitability	Gross profitability in a given year t is computed as follows: $GP_{i,t} = \left(REVT_{i,t} - COGS_{i,t}\right)/ASSET_{i,t}$, where $REVT_{i,t}$ refers to the total revenue (COMPUSTAT annual item REVT) of stock i in year t, $COGS_{i,t}$ refers to the cost of goods sold (item COGS), and $ASSET_{i,t}$ is the total assets (item AT), following Novy-Marx (2013).
Leverage	Total liabilities (COMPUSTAT annual item LT) divided by market capitalization at fiscal year-end, following Bhandari (1988).
ROA	Return on assets in a given quarter t, computed as follows: $ROA_{i,t} = INCOME_{i,t}/ASSET_{i,t-1}$, where $INCOME_{i,t}$ refers to the income before extraordinary items (COMPUSTAT quarterly item IBQ) of stock i in quarter t, and $ASSET_{i,t-1}$ refers to the total assets (item ATQ), following Balakrishnan, Bartov, and Faurel (2010).
ROE	Return on equity in a given quarter t is computed as follows: $ROE_{i,t} = INCOME_{i,t}/EQUITY_{i,t-1}$, where $INCOME_{i,t}$ refers to the income before extraordinary items (COMPUSTAT quarterly item

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	IBQ) of stock i in quarter t, and $EQUITY_{i,t-1}$ refers to the shareholders' equity. Depending on availability, we use stockholders' equity (item SEQQ), or common equity (item CEQQ) plus redemption value (item PSTKRQ), or common equity (item CEQQ) plus the carrying value of preferred stock (item PSTKQ), or total assets (item ATQ) minus total liabilities (item LTQ), in that order, as shareholders' equity, following Hou, Xue, and Zhang (2015).
%Rated Credit Rating	The number of firms with S&P long-term issuer credit rating divided by the total number of firms. The monthly S&P long-term issuer credit rating is as reported in COMPUSTAT. We transform the S&P ratings into ascending numerical scores, where AAA = 1, AA+ = 2, AA = 3, AA– = 4, A+ = 5, A = 6, A– = 7, BBB+ = 8, BBB = 9, BBB– = 10, BB+ = 11, BB = 12, BB– = 13, B+ = 14, B = 15, B– = 16, CCC+ = 17, CCC = 18, CCC– = 19, CC = 20, C = 21, and D = 22, following Avramov, Chordia, Jostova, and Philipov (2009).
Analyst Coverage	The number of analysts following the firm as reported in I/B/E/S in each quarter.
Analyst Dispersion	The standard deviation of analysts' earnings (earnings per share, EPS) forecasts divided by the absolute value of the median earnings forecast as reported in I/B/E/S in each quarter.
SUE	The standardized unexpected earnings (SUE) in a given quarter t is computed as follows: $SUE_{i,t} = (EPS_{i,t} - F_{i,t})/P_{i,t}$, where $EPS_{i,t}$ refers to the actual earnings per share for stock i in quarter t, $F_{i,t}$ refers to the median earnings forecast for the current fiscal quarter, and $P_{i,t}$ refers to the stock price. If $EPS_{i,t}$ or $F_{i,t}$ is missing then $SUE_{i,t} = \frac{INCOME_{i,t} - INCOME_{i,t-4}}{MV_{i,t}}$, where $INCOME_{i,t}$ refers to the income before extraordinary items (COMPUSTAT quarterly item IBQ) of stock i in quarter t, and $MV_{i,t}$ refers to the stock market capitalization.
B. Market State Measures	
SENT	The monthly Baker and Wurgler (2007) investor sentiment. We use the raw version of the Baker-Wurgler sentiment index that excludes the NYSE turnover variable.
MKTVOL	The standard deviation of daily CRSP value-weighted index returns in a month.
VIX	The average daily VIX index of implied volatilities of S&P 500 index options in a month.
MKTILLIQ	The value-weighted average of stock-level Amihud (2002) illiquidity for all NYSE/AMEX stocks in a month, following Avramov, Cheng, and Hameed (2016). The stock-level Amihud illiquidity is defined as in *Log(Illiquidity)*.
DOWN	A dummy variable that takes a value of 1 if CRSP value-weighted index return is negative in a month and 0 otherwise.
TERM	The difference between the average yield of 10-year Treasury bonds and three-month T-bills, following Welch and Goyal (2008).
DEF	The difference between the average yield of bonds rated BAA and AAA by Moody's, following Welch and Goyal (2008).

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Table IA1: Number of Stocks Over Time

This table reports the average number of stocks per month on a year-by-year basis for the full sample as well as for subsamples for which we exclude the microcaps, require firms to have an S&P long-term issuer credit rating, and exclude observations from 12 months before to 12 months after an issuer credit rating downgrade.

Average Number of Stocks Over Time								
Year	GKX Sample				CPZ Sample			
	Full Sample	Nonmicrocaps	Credit Rating Sample	Nondowngrades	Full Sample	Nonmicrocaps	Credit Rating Sample	Nondowngrades
1987	5,320	1,928	1,042	764	2,080	939	554	406
1988	5,711	2,055	1,030	787	2,113	977	534	420
1989	5,799	2,110	985	731	2,191	994	509	400
1990	5,783	2,174	926	638	2,164	974	461	349
1991	5,827	2,337	901	633	2,266	1,056	450	337
1992	5,869	2,469	930	722	2,433	1,135	478	378
1993	6,209	2,696	1,005	809	2,504	1,171	501	405
1994	6,784	2,955	1,095	895	2,541	1,202	521	425
1995	7,270	3,327	1,165	955	2,551	1,295	542	439
1996	7,519	3,555	1,282	1,072	2,553	1,326	578	478
1997	7,830	3,667	1,405	1,150	2,635	1,372	640	512
1998	7,877	3,709	1,541	1,190	2,667	1,401	701	533
1999	7,523	3,733	1,638	1,188	2,689	1,449	735	519
2000	7,308	3,881	1,623	1,133	2,602	1,476	739	504
2001	7,068	3,576	1,601	1,048	2,563	1,430	739	491
2002	6,761	3,424	1,578	1,029	2,620	1,472	768	519
2003	6,402	3,387	1,550	1,108	2,649	1,541	795	567
2004	6,208	3,465	1,575	1,243	2,675	1,653	841	647
2005	6,154	3,447	1,559	1,238	2,755	1,679	850	648
2006	6,160	3,470	1,535	1,189	2,706	1,669	847	631
2007	6,148	3,424	1,483	1,109	2,533	1,607	822	612
2008	6,228	3,371	1,415	914	2,368	1,519	779	514
2009	5,910	3,272	1,360	904	2,296	1,497	753	512
2010	5,404	3,059	1,351	1,090	2,263	1,494	757	619
2011	5,279	3,036	1,354	1,129	2,234	1,508	773	657
2012	5,227	3,057	1,362	1,128	2,212	1,505	781	651
2013	5,117	3,054	1,376	1,176	2,169	1,490	788	666
2014	5,143	3,146	1,414	1,182	2,081	1,460	786	642
2015	5,270	3,270	1,464	1,109	2,024	1,413	788	564
2016	5,292	3,236	1,427	1,109	1,933	1,344	763	563
2017	5,255	3,213	1,385	1,274				
Total	21,882	13,119	4,715	4,499	7,904	5,083	2,436	2,294

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Table IA2: Performance of Portfolios Sorted by Neural Network Predicted Returns: Equal-Weighted Returns

At the end of each month t, stocks are sorted into deciles according to their one-month-ahead out-of-sample predicted returns ($\hat{R}$) using a neural network with three hidden layers (NN3) (Gu, Kelly, and Xiu 2020). This table reports the equal-weighted returns for month $t+1$ for each decile portfolio as well as the strategy of going long (short) on the highest (lowest) expected return stocks ("HML") over the entire sample period from 1987 to 2017. Portfolio returns are further adjusted by the CAPM, Fama-French-Carhart 4-factor and Pástor-Stambaugh liquidity factor model (FFC+PS), Fama-French 5-factor model (FF5), Fama-French 6-factor model (FF6), and Stambaugh-Yuan 4-factor model (SY). Panel A reports the results for the full sample as well as the subsample that excludes microcaps. Panel B reports similar statistics for the subsamples that exclude nonrated firms and credit rating downgrades. Newey-West adjusted t-statistics are shown in parentheses. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Panel A: Equal-Weighted Returns to Investment Strategies Sorted by NN3-Predicted Returns (Full Sample and Microcaps Excluded)

Rank of $\hat{R}$	Full Sample						Nonmicrocaps					
	Return	CAPM	FFC+PS	FF5	FF6	SY	Return	CAPM	FFC+PS	FF5	FF6	SY
Low	-0.121	-1.383***	-0.879***	-0.763***	-0.466***	-0.381*	0.274	-1.067***	-0.545***	-0.391**	-0.112	-0.024
	(-0.26)	(-5.55)	(-5.23)	(-3.61)	(-3.09)	(-1.73)	(0.60)	(-4.38)	(-3.32)	(-2.05)	(-0.90)	(-0.12)
2	0.569	-0.498***	-0.189*	-0.203	0.011	0.135	0.666**	-0.447***	-0.182**	-0.172	0.002	0.088
	(1.59)	(-2.98)	(-1.70)	(-1.35)	(0.11)	(0.88)	(1.98)	(-3.31)	(-2.06)	(-1.32)	(0.03)	(0.72)
3	0.713**	-0.235	-0.086	-0.146	-0.005	0.098	0.791***	-0.201*	-0.094	-0.149	-0.042	0.074
	(2.32)	(-1.62)	(-0.88)	(-1.21)	(-0.05)	(0.80)	(2.72)	(-1.73)	(-1.21)	(-1.59)	(-0.55)	(0.79)
4	0.931***	0.042	0.103	-0.009	0.088	0.195	0.862***	-0.069	-0.053	-0.140*	-0.083	0.010
	(3.29)	(0.30)	(1.09)	(-0.08)	(0.90)	(1.63)	(3.38)	(-0.66)	(-0.87)	(-1.88)	(-1.31)	(0.12)
5	0.962***	0.113	0.106	0.012	0.063	0.144	1.005***	0.106	0.058	-0.077	-0.049	0.042
	(3.61)	(0.77)	(1.04)	(0.11)	(0.59)	(1.15)	(4.03)	(0.92)	(0.74)	(-1.03)	(-0.70)	(0.50)
6	1.115***	0.260*	0.228**	0.121	0.155	0.238*	1.073***	0.186	0.105	-0.002	0.004	0.082
	(4.23)	(1.73)	(2.15)	(1.13)	(1.41)	(1.88)	(4.43)	(1.46)	(1.19)	(-0.03)	(0.04)	(0.82)
7	1.205***	0.338**	0.280**	0.184	0.203*	0.290**	1.163***	0.256*	0.140*	0.041	0.026	0.075
	(4.36)	(2.04)	(2.37)	(1.56)	(1.69)	(2.14)	(4.60)	(1.96)	(1.67)	(0.60)	(0.36)	(0.88)
8	1.344***	0.460**	0.403***	0.302***	0.324***	0.427***	1.201***	0.289**	0.161**	0.036	0.016	0.111
	(4.49)	(2.47)	(3.27)	(2.63)	(2.69)	(3.03)	(4.73)	(2.15)	(2.00)	(0.53)	(0.23)	(1.25)
9	1.563***	0.664***	0.614***	0.547***	0.570***	0.685***	1.333***	0.419***	0.272***	0.170**	0.133*	0.225**
	(4.95)	(3.15)	(4.68)	(4.14)	(4.12)	(4.53)	(5.09)	(2.86)	(3.31)	(2.41)	(1.89)	(2.36)
High	2.349***	1.360***	1.618***	1.586***	1.770***	1.931***	1.517***	0.559***	0.396***	0.290***	0.241**	0.333***
	(5.35)	(4.17)	(5.52)	(4.96)	(5.22)	(5.12)	(5.21)	(3.13)	(3.95)	(3.02)	(2.50)	(2.87)
HML	2.470***	2.742***	2.497***	2.349***	2.236***	2.312***	1.243***	1.625***	0.940***	0.680***	0.353**	0.357
	(9.03)	(9.78)	(9.27)	(9.35)	(8.06)	(7.91)	(3.75)	(5.04)	(4.19)	(2.89)	(2.02)	(1.37)

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Table IA2—Continued

Panel B: Equal-Weighted Returns to Investment Strategies Sorted by NN3-Predicted Returns (Credit Rating Sample and Downgrades Excluded)												
Rank of $\hat{R}$	Credit Rating Sample						Nondowngrades					
	Return	CAPM	FFC+PS	FF5	FF6	SY	Return	CAPM	FFC+PS	FF5	FF6	SY
Low	0.056	-1.289***	-0.741***	-0.904***	-0.488***	-0.213	1.103***	-0.147	0.297*	0.148	0.468***	0.762***
	(0.12)	(-4.94)	(-4.47)	(-3.02)	(-2.99)	(-0.84)	(2.63)	(-0.65)	(1.85)	(0.64)	(2.87)	(3.13)
2	0.785**	-0.329*	-0.140	-0.358**	-0.118	0.111	1.290***	0.225	0.388***	0.165	0.357***	0.617***
	(2.12)	(-1.84)	(-1.23)	(-2.11)	(-1.03)	(0.55)	(4.01)	(1.43)	(3.24)	(1.07)	(3.15)	(3.26)
3	0.863**	-0.158	-0.108	-0.357**	-0.195	-0.006	1.216***	0.251	0.282**	0.049	0.175	0.365**
	(2.51)	(-0.85)	(-0.84)	(-2.50)	(-1.56)	(-0.03)	(4.16)	(1.54)	(2.40)	(0.40)	(1.53)	(2.21)
4	0.972***	-0.035	-0.021	-0.251**	-0.128	0.063	1.229***	0.271*	0.248**	0.066	0.148	0.324**
	(3.11)	(-0.21)	(-0.20)	(-2.38)	(-1.43)	(0.41)	(4.51)	(1.86)	(2.31)	(0.67)	(1.50)	(2.24)
5	1.188***	0.256*	0.192**	-0.023	0.058	0.210	1.415***	0.517***	0.434***	0.230**	0.283***	0.433***
	(4.30)	(1.69)	(2.07)	(-0.24)	(0.68)	(1.52)	(5.65)	(3.73)	(4.14)	(2.45)	(3.06)	(3.22)
6	0.986***	0.065	-0.045	-0.192*	-0.155	-0.049	1.302***	0.413***	0.282***	0.121	0.135	0.245**
	(3.61)	(0.43)	(-0.46)	(-1.88)	(-1.55)	(-0.39)	(5.29)	(2.92)	(2.74)	(1.35)	(1.44)	(2.06)
7	1.124***	0.211	0.083	-0.115	-0.090	0.034	1.377***	0.493***	0.357***	0.161*	0.169*	0.301**
	(4.16)	(1.36)	(0.78)	(-1.21)	(-0.93)	(0.27)	(5.53)	(3.24)	(3.07)	(1.68)	(1.68)	(2.34)
8	1.186***	0.273*	0.151	-0.074	-0.044	0.130	1.452***	0.568***	0.421***	0.216**	0.215**	0.367***
	(4.42)	(1.78)	(1.37)	(-0.73)	(-0.45)	(0.93)	(5.93)	(3.92)	(3.93)	(2.42)	(2.33)	(2.86)
9	1.275***	0.361**	0.210**	0.030	0.030	0.206*	1.512***	0.626***	0.454***	0.296***	0.277***	0.439***
	(4.77)	(2.33)	(2.32)	(0.34)	(0.35)	(1.78)	(6.08)	(4.15)	(5.06)	(3.46)	(3.31)	(3.83)
High	1.431***	0.470**	0.342**	0.143	0.163	0.334**	1.770***	0.850***	0.659***	0.471***	0.442***	0.558***
	(4.42)	(2.31)	(2.53)	(1.08)	(1.27)	(2.26)	(6.18)	(4.52)	(5.49)	(4.70)	(4.21)	(4.16)
HML	1.374***	1.759***	1.083***	1.048***	0.652***	0.548*	0.667**	0.997***	0.362*	0.323	-0.026	-0.205
	(4.08)	(5.75)	(4.59)	(3.63)	(3.16)	(1.93)	(2.11)	(3.40)	(1.70)	(1.32)	(-0.13)	(-0.73)

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Table IA3: Performance of Portfolios Sorted by OLS Predicted Returns and Individual Anomalies

At the end of each month t, stocks are sorted into deciles according to their one-month-ahead out-of-sample predicted returns using ordinary least squares (OLS). Panel A reports the equal- and value-weighted return for month $t+1$ for the strategy of going long (short) on the highest (lowest) expected return stocks ("HML") over the entire sample period from 1987 to 2017. Portfolio returns are further adjusted by the CAPM, Fama-French-Carhart 4-factor and Pástor-Stambaugh liquidity factor model (FFC+PS), Fama-French 5-factor model (FF5), Fama-French 6-factor model (FF6), and Stambaugh-Yuan 4-factor model (SY). We report results for the full sample as well as subsamples that exclude microcaps, nonrated firms, and credit rating downgrades. Panel B reports similar statistics, while at the end of each month t, stocks are sorted into deciles according to each of the 94 firm characteristics. We adjust the sign of those characteristics so that a higher value predicts higher future performance according to the original studies. For each characteristic, we compute the equal- and value-weighted return for month $t+1$ for the strategy of going long (short) on the highest (lowest) expected return stocks ("HML"). Then, we compute an equal-weighted average of portfolio returns across 94 characteristics. Newey-West adjusted t-statistics are shown in parentheses. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Panel A: Returns to Investment Strategies Sorted by OLS-Predicted Returns

	Equal-Weighted						Value-Weighted					
	Return	CAPM	FFC+PS	FF5	FF6	SY	Return	CAPM	FFC+PS	FF5	FF6	SY
Full Sample	1.215***	1.162***	1.272***	1.521***	1.545***	1.475***	0.084	0.118	0.164	0.308	0.268	0.127
	(5.25)	(4.78)	(5.28)	(5.95)	(5.85)	(4.69)	(0.39)	(0.50)	(0.64)	(1.09)	(0.92)	(0.37)
Nonmicrocaps	0.472**	0.410*	0.367*	0.565**	0.514**	0.409	0.013	0.060	0.128	0.237	0.223	0.074
	(2.37)	(1.84)	(1.66)	(2.42)	(2.19)	(1.47)	(0.07)	(0.30)	(0.61)	(1.04)	(0.95)	(0.25)
Credit Rating Sample	0.437*	0.361	0.488**	0.458**	0.563**	0.632**	0.024	0.061	0.135	0.103	0.115	0.133
	(1.78)	(1.50)	(2.02)	(2.00)	(2.33)	(2.02)	(0.12)	(0.27)	(0.61)	(0.46)	(0.49)	(0.51)
Nondowngrades	0.506**	0.480**	0.506**	0.412*	0.470**	0.550**	-0.008	0.035	0.078	0.082	0.065	0.032
	(2.39)	(2.13)	(2.29)	(1.96)	(2.08)	(1.97)	(-0.04)	(0.15)	(0.34)	(0.34)	(0.26)	(0.12)

Panel B: Returns to Investment Strategies Sorted by Individual Anomalies

	Equal-Weighted						Value-Weighted					
	Return	CAPM	FFC+PS	FF5	FF6	SY	Return	CAPM	FFC+PS	FF5	FF6	SY
Full Sample	0.366***	0.472***	0.346***	0.308***	0.250***	0.221***	0.192***	0.306***	0.189***	0.139***	0.092***	0.043
	(5.92)	(8.18)	(7.91)	(6.03)	(6.07)	(3.95)	(2.91)	(5.41)	(4.88)	(3.27)	(2.89)	(1.19)
Nonmicrocaps	0.233***	0.344***	0.218***	0.167***	0.115***	0.079**	0.170***	0.272***	0.151***	0.095**	0.049*	-0.003
	(3.71)	(6.11)	(6.21)	(4.40)	(4.60)	(2.16)	(2.74)	(4.87)	(4.02)	(2.57)	(1.87)	(-0.09)
Credit Rating Sample	0.302***	0.387***	0.263***	0.244***	0.182***	0.121**	0.131**	0.215***	0.111***	0.092**	0.045	-0.017
	(5.11)	(6.78)	(6.24)	(4.59)	(4.56)	(2.55)	(2.47)	(4.75)	(3.11)	(2.20)	(1.42)	(-0.52)
Nondowngrades	0.124***	0.197***	0.103***	0.083**	0.039	-0.017	0.071	0.142***	0.046	0.038	-0.004	-0.071*
	(2.60)	(4.45)	(3.31)	(2.11)	(1.22)	(-0.44)	(1.48)	(3.28)	(1.44)	(0.93)	(-0.13)	(-1.93)

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Table IA4: Neural Network Estimation Based on Subsamples: Equal-Weighted Returns

At the end of each month t, stocks are sorted into deciles according to their NN3-predicted returns ($\hat{R}$) (Gu, Kelly, and Xiu 2020). This table reports the equal-weighted return for month $t+1$ for the strategy of going long (short) on the highest (lowest) expected return stocks. Portfolio returns are further adjusted by the CAPM, Fama-French-Carhart 4-factor and Pástor-Stambaugh liquidity factor model (FFC+PS), Fama-French 5-factor model (FF5), Fama-French 6-factor model (FF6), and Stambaugh-Yuan 4-factor model (SY). We employ the full sample as well as subsamples that exclude microcaps, nonrated firms, and credit rating downgrades for the NN3 estimations (in the "Training" column) and the portfolio sorts (in the "Testing" column). Newey-West adjusted t-statistics are shown in parentheses. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Equal-Weighted Returns to Investment Strategies Sorted by NN3-Predicted Returns								
Training	Testing	Sample Period	Return	CAPM	FFC+PS	FF5	FF6	SY
Nonmicrocaps	Nonmicrocaps	1987–2017	1.505***	1.891***	1.147***	1.015***	0.655***	0.571**
			(4.79)	(6.31)	(5.89)	(4.23)	(4.25)	(2.49)
Credit Rating Sample	Credit Rating Sample	1999–2017	1.552**	2.024***	1.482***	1.087**	0.834***	0.442
			(2.48)	(4.01)	(4.40)	(2.32)	(2.96)	(1.05)
Nondowngrades	Nondowngrades	1999–2017	0.935***	0.904**	0.460	0.203	0.113	0.079
			(2.81)	(2.44)	(1.59)	(0.69)	(0.42)	(0.23)
Full Sample	Full Sample	1987–2017	2.470***	2.742***	2.497***	2.349***	2.236***	2.312***
			(9.03)	(9.78)	(9.27)	(9.35)	(8.06)	(7.91)
Full Sample	Nonmicrocaps	1987–2017	1.243***	1.625***	0.940***	0.680***	0.353**	0.357
			(3.75)	(5.04)	(4.19)	(2.89)	(2.02)	(1.37)
Full Sample	Full Sample	1999–2017	2.417***	2.692***	2.483***	2.297***	2.249***	2.300***
			(6.05)	(7.03)	(7.32)	(6.64)	(6.15)	(5.67)
Full Sample	Nonmicrocaps	1999–2017	1.174**	1.618***	1.111***	0.638**	0.465**	0.379
			(2.28)	(3.64)	(4.30)	(2.18)	(2.30)	(1.14)
Full Sample	Credit Rating Sample	1999–2017	1.258**	1.689***	1.136***	0.878**	0.663***	0.376
			(2.49)	(3.99)	(4.08)	(2.29)	(2.66)	(1.01)
Full Sample	Nondowngrades	1999–2017	0.354	0.733*	0.194	-0.027	-0.215	-0.579
			(0.73)	(1.73)	(0.74)	(-0.08)	(-0.86)	(-1.50)

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Table IA5: NN3-Predicted Alpha and Value-Weighted Loss Function

At the end of each month t, stocks are sorted into deciles according to their NN3-predicted alphas ($\hat{\alpha}$). We employ NN3 to predict FF6-adjusted returns ($\hat{\alpha}$) and adopt a value-weighted loss function in the NN3 estimation. Panel A reports the month $t+1$ value-weighted return for each decile portfolio as well as the strategy of going long (short) on the highest (lowest) expected alpha stocks ("HML") over the entire sample period from 1987 to 2017. Portfolio returns are further adjusted by the Fama-French 6-factor model (FF6). We report the results for the full sample as well as the subsamples that exclude microcaps, nonrated firms, and credit rating downgrades. Panel B reports similar statistics when decile portfolios are sorted by the NN3-predicted returns ($\hat{R}$) (Gu, Kelly, and Xiu 2020) using the same universe of stocks. Newey-West adjusted t-statistics are shown in parentheses. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Panel A: Value-Weighted Returns to Investment Strategies Sorted by NN3-Predicted Alphas

Rank of $\hat{\alpha}$	Full Sample		Nonmicrocaps		Credit Rating Sample		Nondowngrades	
	Return	FF6	Return	FF6	Return	FF6	Return	FF6
Low	0.168	-0.403**	0.538	-0.192	0.515	-0.216	1.012***	0.152
	(0.36)	(-2.05)	(1.43)	(-1.30)	(1.28)	(-1.10)	(2.83)	(0.77)
2	0.648*	-0.155	0.692**	-0.170*	0.736**	-0.147	1.099***	0.240
	(1.90)	(-1.11)	(2.21)	(-1.81)	(2.20)	(-1.14)	(3.66)	(1.54)
3	0.781***	-0.113	0.941***	0.021	0.941***	-0.028	1.197***	0.261**
	(2.63)	(-0.98)	(3.72)	(0.25)	(3.25)	(-0.25)	(4.60)	(2.04)
4	0.943***	-0.059	0.950***	-0.038	0.895***	-0.023	1.047***	0.162
	(3.84)	(-0.80)	(4.15)	(-0.46)	(3.51)	(-0.23)	(4.51)	(1.49)
5	0.854***	-0.121	0.909***	-0.084	1.008***	-0.064	1.149***	0.082
	(3.73)	(-1.57)	(3.80)	(-1.10)	(4.35)	(-0.71)	(5.09)	(0.82)
6	1.079***	0.053	1.083***	0.072	1.042***	0.009	1.158***	0.112
	(4.75)	(0.64)	(5.14)	(0.99)	(4.58)	(0.11)	(5.21)	(1.24)
7	0.979***	-0.053	0.888***	-0.126	1.002***	-0.092	1.080***	-0.006
	(4.08)	(-0.69)	(3.74)	(-1.46)	(4.47)	(-1.13)	(4.77)	(-0.06)
8	0.994***	-0.062	1.023***	-0.093	0.909***	-0.163*	1.035***	-0.016
	(4.64)	(-0.77)	(4.59)	(-1.20)	(3.76)	(-1.78)	(4.52)	(-0.17)
9	1.076***	0.104	1.148***	0.161*	1.079***	-0.048	1.163***	0.047
	(4.32)	(1.16)	(4.83)	(1.96)	(4.92)	(-0.57)	(5.34)	(0.55)
High	1.224***	0.203	1.238***	0.184	1.215***	0.124	1.366***	0.282**
	(4.18)	(1.40)	(4.73)	(1.64)	(4.74)	(1.06)	(5.29)	(2.42)
HML	1.056***	0.606**	0.700***	0.376*	0.701**	0.340	0.354	0.130
	(2.83)	(2.20)	(2.61)	(1.86)	(2.20)	(1.37)	(1.22)	(0.53)

Panel B: Value-Weighted Returns to Investment Strategies Sorted by NN3-Predicted Returns

Rank of $\hat{R}$	Full Sample		Nonmicrocaps		Credit Rating Sample		Nondowngrades	
	Return	FF6	Return	FF6	Return	FF6	Return	FF6
Low	0.398	-0.148	0.418	-0.100	0.446	-0.173	0.797**	0.099
	(1.01)	(-1.10)	(1.03)	(-0.71)	(1.08)	(-1.05)	(2.12)	(0.49)
2	0.695**	-0.095	0.701**	-0.038	0.774***	-0.085	0.970***	0.113
	(2.48)	(-1.12)	(2.45)	(-0.41)	(2.66)	(-0.82)	(3.67)	(1.06)
3	0.923***	-0.026	0.866***	-0.057	0.867***	-0.030	1.101***	0.228*
	(3.58)	(-0.31)	(3.34)	(-0.69)	(3.17)	(-0.27)	(4.31)	(1.67)
4	0.965***	-0.036	0.959***	-0.031	0.897***	-0.147	0.998***	-0.006
	(4.27)	(-0.59)	(4.12)	(-0.37)	(3.57)	(-1.56)	(4.27)	(-0.07)
5	0.945***	-0.086	0.931***	-0.095	1.072***	-0.022	1.165***	0.075
	(4.22)	(-1.21)	(4.00)	(-1.24)	(4.77)	(-0.30)	(5.31)	(0.82)
6	1.006***	-0.085	0.952***	-0.095	0.935***	-0.128*	1.055***	-0.014
	(4.65)	(-0.94)	(4.47)	(-1.43)	(4.24)	(-1.67)	(4.97)	(-0.17)
7	1.112***	0.009	1.063***	-0.019	0.989***	-0.119	1.122***	0.042
	(4.72)	(0.11)	(4.80)	(-0.23)	(4.58)	(-1.44)	(5.43)	(0.46)
8	1.181***	0.023	1.180***	0.040	1.247***	0.087	1.349***	0.205***
	(5.02)	(0.25)	(5.12)	(0.55)	(5.48)	(1.21)	(6.17)	(2.77)
9	1.320***	0.148	1.122***	0.027	1.029***	-0.095	1.097***	-0.038
	(4.91)	(1.21)	(4.83)	(0.33)	(4.44)	(-1.03)	(4.85)	(-0.40)
High	1.973***	0.846***	1.454***	0.248**	1.502***	0.333***	1.584***	0.403***
	(6.62)	(5.46)	(5.54)	(2.04)	(5.91)	(2.79)	(6.31)	(3.20)
HML	1.575***	0.994***	1.036***	0.349*	1.056***	0.507**	0.787***	0.303
	(4.82)	(4.70)	(3.31)	(1.71)	(3.29)	(2.30)	(2.63)	(1.26)

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Table IA6: Performance of Portfolios Sorted by Risk Loadings on the Stochastic Discount Factor: Equal-Weighted Returns

At the end of each month t, stocks are sorted into deciles according to their risk loadings (β) on the stochastic discount factor estimated from a combination of deep neural networks (Chen, Pelger, and Zhu 2020). This table reports the equal-weighted returns for month $t+1$ for each decile portfolio as well as the strategy of going long (short) on the highest (lowest) risk loading stocks ("HML") over the entire sample period from 1987 to 2016. Portfolio returns are further adjusted by the CAPM, Fama-French-Carhart 4-factor and Pástor-Stambaugh liquidity factor model (FFC+PS), Fama-French 5-factor model (FF5), Fama-French 6-factor model (FF6), and Stambaugh-Yuan 4-factor model (SY). Panel A reports the results for the full sample as well as the subsample that excludes microcaps. Panel B reports similar statistics for the subsamples that exclude nonrated firms and credit rating downgrades. Newey-West adjusted t-statistics are shown in parentheses. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Panel A: Equal-Weighted Returns to Investment Strategies Sorted by Risk Loadings (Full Sample and Microcaps Excluded)

Rank of β	Full Sample						Nonmicrocaps					
	Return	CAPM	FFC+PS	FF5	FF6	SY	Return	CAPM	FFC+PS	FF5	FF6	SY
Low	-0.110	-1.141***	-0.889***	-1.087***	-0.826***	-0.644**	0.315	-0.808***	-0.512***	-0.769***	-0.505***	-0.370*
	(-0.24)	(-4.10)	(-4.18)	(-4.93)	(-4.10)	(-2.31)	(0.78)	(-3.96)	(-3.47)	(-4.12)	(-3.50)	(-1.70)
2	0.607	-0.390**	-0.267**	-0.412***	-0.263**	-0.175	0.634**	-0.387***	-0.255***	-0.403***	-0.265**	-0.162
	(1.62)	(-2.09)	(-2.02)	(-3.01)	(-1.98)	(-1.01)	(1.98)	(-2.72)	(-2.62)	(-3.36)	(-2.52)	(-1.19)
3	0.876**	-0.092	-0.032	-0.133	-0.035	0.050	1.018***	0.009	0.066	-0.057	0.031	0.103
	(2.56)	(-0.58)	(-0.29)	(-1.24)	(-0.32)	(0.37)	(3.26)	(0.07)	(0.72)	(-0.60)	(0.34)	(0.98)
4	1.007***	0.053	0.071	-0.056	0.015	0.120	0.945***	-0.021	-0.012	-0.157*	-0.098	0.006
	(3.13)	(0.35)	(0.71)	(-0.56)	(0.15)	(0.94)	(3.22)	(-0.17)	(-0.13)	(-1.71)	(-1.13)	(0.06)
5	1.219***	0.281*	0.263***	0.167*	0.220**	0.308***	1.094***	0.131	0.074	-0.017	0.001	0.055
	(4.05)	(1.86)	(3.06)	(1.70)	(2.30)	(2.79)	(3.86)	(1.02)	(0.98)	(-0.22)	(0.01)	(0.64)
6	1.249***	0.310**	0.242***	0.157*	0.186**	0.287***	1.279***	0.318**	0.246***	0.140*	0.156**	0.237***
	(4.15)	(2.04)	(2.82)	(1.68)	(2.05)	(2.80)	(4.46)	(2.39)	(2.96)	(1.77)	(2.04)	(2.78)
7	1.436***	0.492***	0.415***	0.292***	0.322***	0.446***	1.220***	0.256*	0.183*	0.039	0.065	0.172
	(4.71)	(3.00)	(4.47)	(3.00)	(3.45)	(3.97)	(4.35)	(1.73)	(1.96)	(0.36)	(0.62)	(1.56)
8	1.687***	0.745***	0.706***	0.528***	0.581***	0.716***	1.412***	0.444***	0.346***	0.184**	0.192**	0.329***
	(5.61)	(4.22)	(6.31)	(4.49)	(5.03)	(5.83)	(4.81)	(2.96)	(3.83)	(2.11)	(2.31)	(3.49)
9	1.998***	1.056***	1.073***	0.907***	1.017***	1.170***	1.567***	0.600***	0.494***	0.268***	0.294***	0.429***
	(5.93)	(4.70)	(6.26)	(4.54)	(5.05)	(5.44)	(5.49)	(3.65)	(4.87)	(2.75)	(3.28)	(4.05)
High	3.336***	2.274***	2.583***	2.379***	2.667***	2.923***	1.901***	0.837***	0.820***	0.537***	0.656***	0.848***
	(7.01)	(6.42)	(7.99)	(5.95)	(6.83)	(6.86)	(5.53)	(3.85)	(5.84)	(3.21)	(4.76)	(5.23)
HML	3.446***	3.415***	3.473***	3.465***	3.492***	3.566***	1.586***	1.645***	1.332***	1.306***	1.161***	1.218***
	(11.25)	(11.17)	(9.28)	(8.94)	(8.04)	(8.01)	(6.80)	(6.98)	(6.09)	(6.42)	(5.13)	(5.06)

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Table IA6—Continued

Panel B: Equal-Weighted Returns to Investment Strategies Sorted by Risk Loadings (Credit Rating Sample and Downgrades Excluded)												
Rank of β	Credit Rating Sample						Nondowngrades					
	Return	CAPM	FFC+PS	FF5	FF6	SY	Return	CAPM	FFC+PS	FF5	FF6	SY
Low	0.489	-0.640**	-0.372*	-0.762***	-0.430*	-0.137	1.013**	-0.033	0.228	-0.127	0.147	0.401
	(1.02)	(-2.24)	(-1.68)	(-2.73)	(-1.87)	(-0.41)	(2.48)	(-0.13)	(0.98)	(-0.52)	(0.64)	(1.23)
2	0.720**	-0.315*	-0.220	-0.514***	-0.329**	-0.094	1.001***	0.027	0.090	-0.146	-0.017	0.178
	(2.00)	(-1.73)	(-1.56)	(-3.07)	(-2.31)	(-0.45)	(3.11)	(0.16)	(0.65)	(-1.00)	(-0.12)	(0.98)
3	1.015***	0.038	0.087	-0.151	-0.010	0.175	1.280***	0.339**	0.378***	0.176	0.281**	0.475***
	(3.20)	(0.23)	(0.73)	(-1.05)	(-0.08)	(1.04)	(4.47)	(2.28)	(2.80)	(1.29)	(2.17)	(2.89)
4	0.947***	-0.022	-0.042	-0.267**	-0.183	-0.051	1.205***	0.278*	0.254**	0.025	0.084	0.222
	(3.16)	(-0.14)	(-0.39)	(-2.08)	(-1.57)	(-0.37)	(4.48)	(1.83)	(2.08)	(0.19)	(0.69)	(1.57)
5	1.057***	0.116	0.041	-0.129	-0.072	0.057	1.280***	0.367***	0.266**	0.150	0.170	0.267**
	(3.74)	(0.85)	(0.42)	(-1.24)	(-0.73)	(0.46)	(4.83)	(2.82)	(2.36)	(1.41)	(1.54)	(2.08)
6	1.207***	0.262*	0.165	0.001	0.033	0.175	1.385***	0.465***	0.348***	0.189*	0.194*	0.315**
	(4.21)	(1.89)	(1.45)	(0.01)	(0.32)	(1.34)	(5.20)	(3.31)	(2.75)	(1.74)	(1.74)	(2.35)
7	1.203***	0.250	0.125	-0.025	0.003	0.150	1.435***	0.505***	0.355***	0.217*	0.216*	0.361***
	(4.03)	(1.54)	(1.11)	(-0.21)	(0.02)	(1.11)	(5.17)	(3.12)	(3.14)	(1.84)	(1.86)	(2.83)
8	1.238***	0.296*	0.173	-0.021	0.006	0.178	1.459***	0.542***	0.395***	0.192	0.191	0.355***
	(4.22)	(1.70)	(1.49)	(-0.18)	(0.05)	(1.44)	(5.32)	(3.11)	(3.13)	(1.54)	(1.60)	(2.61)
9	1.415***	0.441**	0.331***	0.075	0.133	0.278**	1.635***	0.684***	0.553***	0.288**	0.331***	0.468***
	(4.59)	(2.23)	(2.67)	(0.59)	(1.15)	(2.11)	(5.56)	(3.45)	(4.43)	(2.33)	(2.84)	(3.55)
High	2.007***	0.881***	0.968***	0.613**	0.840***	1.120***	2.426***	1.412***	1.369***	1.087***	1.189***	1.395***
	(4.69)	(2.93)	(4.17)	(2.15)	(3.53)	(4.17)	(7.04)	(5.59)	(6.91)	(5.44)	(6.21)	(7.01)
HML	1.519***	1.522***	1.340***	1.376***	1.270***	1.257***	1.413***	1.444***	1.141***	1.214***	1.041***	0.994***
	(5.74)	(5.83)	(4.44)	(4.89)	(3.93)	(3.77)	(4.85)	(5.15)	(3.88)	(4.53)	(3.50)	(2.88)

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Table IA7: IPCA Estimation Based on Subsamples and the Unrestricted IPCA Model

At the end of each month t, stocks are sorted into deciles according to their one-month-ahead out-of-sample predicted returns ($\hat{R}$) using instrumented principal component analysis (IPCA) (Kelly, Pruitt, and Su 2019). Panel A reports the month $t+1$ value-weighted returns for the strategy of going long (short) on the highest (lowest) expected return stocks. Portfolio returns are further adjusted by the Fama-French 6-factor model (FF6). We employ the full sample as well as subsamples that exclude microcaps, nonrated firms, and credit rating downgrades for the IPCA estimations (in the "Estimation" column) and the portfolio sorts (in the "Testing" column). Panel B employs the unrestricted IPCA model (without zero-alpha restriction) and reports the month $t+1$ value-weighted returns and FF6-adjusted returns for each decile portfolio as well as the strategy of going long (short) on the highest (lowest) expected return stocks ("HML"). We report the results for the full sample as well as subsamples that exclude microcaps, nonrated firms, and credit rating downgrades over the entire sample period from 1987 to 2017. Panel C reports similar statistics for the post-2001 period. Newey-West adjusted t-statistics are shown in parentheses. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Panel A: Value-Weighted Returns to Investment Strategies Sorted by IPCA-Predicted Returns

Estimation	Testing	Sample Period	Return	FF6
Nonmicrocaps	Nonmicrocaps	1987–2017	1.059***	0.757***
			(5.87)	(5.30)
Credit Rating Sample	Credit Rating Sample	1996–2017	0.722*	0.434*
			(1.87)	(1.79)
Nondowngrades	Nondowngrades	1996–2017	0.586	0.118
			(1.63)	(0.44)
Full Sample	Full Sample	1987–2017	0.945***	0.624***
			(5.62)	(3.31)
Full Sample	Nonmicrocaps	1987–2017	0.901***	0.613***
			(5.57)	(3.71)
Full Sample	Full Sample	1996–2017	1.000***	0.684***
			(4.61)	(3.02)
Full Sample	Nonmicrocaps	1996–2017	0.918***	0.637***
			(4.52)	(3.32)
Full Sample	Credit Rating Sample	1996–2017	1.012***	0.708***
			(4.70)	(3.33)
Full Sample	Nondowngrades	1996–2017	0.797***	0.506**
			(3.54)	(2.21)

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Panel B: Value-Weighted Returns to Investment Strategies Sorted by IPCA-Predicted Returns (1987–2017)

Rank of $\hat{R}$	Full Sample		Nonmicrocaps		Credit Rating Sample		Nondowngrades	
	Return	FF6	Return	FF6	Return	FF6	Return	FF6
Low	0.374	-0.241***	0.320	-0.267***	0.359	-0.318***	0.672**	-0.039
	(1.22)	(-2.97)	(1.01)	(-3.06)	(1.12)	(-2.91)	(2.42)	(-0.30)
2	0.821***	-0.018	0.732***	-0.080	0.747***	-0.124	0.916***	0.043
	(3.14)	(-0.28)	(2.72)	(-1.04)	(2.65)	(-1.65)	(3.51)	(0.50)
3	0.906***	0.025	0.882***	0.045	0.929***	0.093	1.066***	0.267***
	(3.64)	(0.36)	(3.60)	(0.76)	(3.57)	(1.12)	(4.52)	(2.66)
4	0.907***	-0.057	0.941***	0.004	0.980***	-0.006	1.121***	0.162*
	(4.04)	(-1.06)	(3.88)	(0.06)	(4.07)	(-0.08)	(4.90)	(1.93)
5	0.995***	-0.048	0.947***	-0.051	0.972***	-0.100	1.113***	0.030
	(4.29)	(-0.64)	(4.16)	(-0.89)	(4.22)	(-1.34)	(5.21)	(0.38)
6	1.129***	0.008	0.995***	-0.041	0.958***	-0.123	1.045***	-0.036
	(4.82)	(0.12)	(4.36)	(-0.55)	(4.13)	(-1.65)	(4.77)	(-0.38)
7	1.092***	0.054	1.130***	0.015	1.044***	-0.026	1.095***	0.047
	(4.56)	(0.67)	(4.95)	(0.25)	(4.63)	(-0.34)	(4.91)	(0.58)
8	1.202***	0.124	1.082***	0.052	1.098***	0.007	1.256***	0.166*
	(4.93)	(1.26)	(4.53)	(0.64)	(4.73)	(0.08)	(5.46)	(1.73)
9	1.453***	0.279***	1.240***	0.110	1.206***	0.055	1.284***	0.140
	(5.40)	(2.73)	(5.18)	(1.17)	(5.07)	(0.52)	(5.38)	(1.25)
High	1.480***	0.244*	1.409***	0.219**	1.263***	0.063	1.380***	0.157
	(4.88)	(1.71)	(5.01)	(1.97)	(4.74)	(0.53)	(5.05)	(1.26)
HML	1.106***	0.485***	1.089***	0.486***	0.904***	0.381**	0.708***	0.196
	(4.61)	(2.88)	(4.85)	(3.58)	(3.89)	(2.42)	(3.18)	(1.19)

Panel C: Value-Weighted Returns to Investment Strategies Sorted by IPCA-Predicted Returns (2001–2017)

Rank of $\hat{R}$	Full Sample		Nonmicrocaps		Credit Rating Sample		Nondowngrades	
	Return	FF6	Return	FF6	Return	FF6	Return	FF6
Low	0.103	-0.421***	0.070	-0.434***	0.050	-0.528***	0.356	-0.271*
	(0.22)	(-3.78)	(0.15)	(-3.48)	(0.10)	(-3.37)	(0.91)	(-1.67)
2	0.532	-0.155*	0.433	-0.251***	0.429	-0.305***	0.587	-0.126
	(1.35)	(-1.88)	(1.06)	(-2.63)	(1.01)	(-3.15)	(1.53)	(-1.12)
3	0.658*	0.009	0.613*	0.017	0.534	-0.063	0.706**	0.151
	(1.77)	(0.08)	(1.69)	(0.21)	(1.35)	(-0.54)	(2.01)	(1.08)
4	0.678**	-0.043	0.659*	-0.061	0.748**	0.023	0.867***	0.183
	(2.12)	(-0.67)	(1.84)	(-0.57)	(2.21)	(0.20)	(2.83)	(1.63)
5	0.716**	0.008	0.764**	0.028	0.619*	-0.140	0.769**	-0.009
	(2.21)	(0.09)	(2.35)	(0.40)	(1.80)	(-1.44)	(2.50)	(-0.10)
6	0.848***	0.073	0.712**	-0.017	0.721**	0.002	0.817***	0.118
	(2.73)	(0.89)	(2.26)	(-0.18)	(2.25)	(0.02)	(2.76)	(0.90)
7	0.779**	0.067	0.813***	0.033	0.814***	0.069	0.842***	0.090
	(2.48)	(0.66)	(2.61)	(0.42)	(2.65)	(0.71)	(2.86)	(0.89)
8	0.825***	0.096	0.772**	0.097	0.759**	0.072	0.910***	0.227**
	(2.69)	(0.96)	(2.52)	(1.04)	(2.52)	(0.74)	(3.16)	(2.39)
9	1.164***	0.364***	0.942***	0.222**	0.906***	0.193*	0.957***	0.240**
	(3.37)	(3.00)	(3.15)	(2.44)	(3.00)	(1.68)	(3.25)	(2.05)
High	1.168***	0.278	1.070***	0.182	1.047***	0.168	1.146***	0.238
	(3.18)	(1.64)	(2.97)	(1.43)	(3.01)	(1.17)	(3.17)	(1.50)
HML	1.065***	0.699***	1.000***	0.616***	0.997***	0.696***	0.790***	0.509**
	(3.26)	(3.34)	(3.08)	(3.68)	(3.03)	(3.63)	(2.67)	(2.52)

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Table IA8: Performance of Machine Learning Portfolios in Predictive Regressions

Panel A presents the results of the following monthly time series regressions, as well as their corresponding Newey-West adjusted t-statistics:

$$HML_t = \alpha_0 + \beta_1 High\ SENT_{t-1} + \beta_2 High\ MKTVOL_{t-1} + \beta_3 High\ MKTILLIQ_{t-1} + \beta_4 M_{t-1} + c_1' F_t + c_2' F_t \times High\ SENT_{t-1} + c_3' F_t \times High\ MKTVOL_{t-1} + c_4' F_t \times High\ MKTILLIQ_{t-1} + e_t,$$

where HML_t refers to the value-weighted returns on the high minus low decile portfolios sorted by the NN3-predicted returns (Gu, Kelly, and Xiu 2020) across all stocks in month t; $High\ SENT_{t-1}$ refers to a dummy variable that takes a value of 1 if the Baker and Wurgler (2007) investor sentiment is above the median over the entire sample period and 0 otherwise; $High\ MKTVOL_{t-1}$ refers to a dummy variable that takes a value of 1 if the market volatility is above the median over the entire sample period and 0 otherwise, and $High\ MKTILLIQ_{t-1}$ refers to a dummy variable that takes a value of 1 if market illiquidity is above the median and 0 otherwise. We obtain the median breakpoint for market illiquidity in the pre- and post-2001 periods separately. $High\ SENT_{t-1}$ can be replaced with $High\ PLS\ SENT_{t-1}$, defined as a dummy variable that takes a value of 1 if the aligned investor sentiment from Huang, Jiang, Tu, and Zhou (2015) is above the median over the entire sample period and 0 otherwise. $High\ MKTVOL_{t-1}$ can also be replaced with $High\ VIX_{t-1}$, defined as a dummy variable that takes a value of 1 if the implied market volatility is above the median over the entire sample period and 0 otherwise. M_{t-1} refers to a set of other proxies for market conditions, including down market state (DOWN), defined as a dummy variable that takes a value of 1 if the CRSP value-weighted index return is negative and 0 otherwise; term spread (TERM), defined as the difference between the average yield of 10-year Treasury bonds and three-month T-bills, and default spread (DEF), defined as the difference between the average yield of bonds rated BAA and AAA by Moody's. The vector F stacks the Fama-French six factors, including the market factor (MKT), the size factor (SMB), the book-to-market factor (HML), the profitability factor (RMW), the investment factor (CMA), and the momentum factor (MOM). Panels B to D report similar statistics when decile portfolios are sorted by the risk loadings on the stochastic discount factor (Chen, Pelger, and Zhu 2020), IPCA-predicted returns (Kelly, Pruitt, and Su 2019), and CA2-predicted returns (Gu, Kelly, and Xiu 2021). Internet Appendix A provides detailed definitions for each variable. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

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Panel A: Value-Weighted Returns to Investment Strategies Sorted by NN3-Predicted Returns Regressed on Lagged Market States								
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Constant	0.867	0.941	1.255	1.181	1.500**	1.103	1.679*	1.079
	(1.14)	(1.39)	(1.43)	(1.64)	(2.02)	(1.59)	(1.94)	(1.44)
High SENT	0.227		-0.006		-0.184		-0.263	
	(0.58)		(-0.01)		(-0.38)		(-0.46)	
High PLS SENT		0.459		0.236		0.463		0.449
		(0.95)		(0.48)		(0.88)		(0.80)
High MKTVOL	0.959*	0.922*			0.661	0.672		
	(1.93)	(1.89)			(1.23)	(1.21)		
High VIX			1.648***	1.599***			1.418**	1.591***
			(3.29)	(3.28)			(2.56)	(2.87)
High MKTILLIQ	0.591	0.693	0.694	0.727	0.669	0.839*	0.391	0.442
	(1.37)	(1.52)	(1.41)	(1.46)	(1.43)	(1.78)	(0.77)	(0.87)
DOWN	-0.691*	-0.712*	-0.803*	-0.827*	-0.310	-0.422	-0.295	-0.437
	(-1.71)	(-1.70)	(-1.74)	(-1.76)	(-0.76)	(-1.05)	(-0.64)	(-0.97)
TERM	-0.145	-0.221	-0.193	-0.226	-0.161	-0.260	-0.154	-0.214
	(-0.80)	(-1.06)	(-0.95)	(-1.02)	(-0.97)	(-1.29)	(-0.80)	(-1.00)
DEF	-0.248	-0.282	-0.735	-0.687	-0.804	-0.475	-1.232*	-0.851
	(-0.33)	(-0.39)	(-0.94)	(-0.93)	(-1.28)	(-0.71)	(-1.80)	(-1.20)
MKT	-0.196***	-0.193***	-0.218***	-0.216***	0.292**	0.245**	0.171	0.174
	(-3.35)	(-3.31)	(-3.01)	(-2.99)	(2.27)	(2.01)	(1.50)	(1.30)
MKT × High SENT					-0.119		-0.150	
					(-1.00)		(-1.15)	
MKT × High PLS SENT						-0.110		-0.186
						(-0.87)		(-1.26)
MKT × High MKTVOL					-0.364***	-0.343***		
					(-3.20)	(-2.85)		
MKT × High VIX							-0.260**	-0.268**
							(-1.99)	(-2.07)
MKT × High MKTILLIQ					-0.241**	-0.220**	-0.230**	-0.185
					(-2.34)	(-2.23)	(-1.99)	(-1.50)
SMB	0.532***	0.528***	0.538***	0.534***	0.268	0.346**	0.158	0.301
	(4.56)	(4.57)	(4.21)	(4.20)	(1.57)	(1.97)	(0.81)	(1.57)
SMB × High SENT					0.291		0.316	
					(1.64)		(1.62)	
SMB × High PLS SENT						0.102		0.027
						(0.52)		(0.13)
SMB × High MKTVOL					-0.091	-0.130		
					(-0.54)	(-0.81)		
SMB × High VIX							0.060	-0.005
							(0.34)	(-0.03)
SMB × High MKTILLIQ					0.500**	0.539***	0.543**	0.597***
					(2.48)	(2.63)	(2.43)	(2.60)
HML	0.125	0.125	0.139	0.135	-0.131	0.113	-0.209	0.087
	(1.02)	(1.02)	(1.09)	(1.07)	(-0.51)	(0.44)	(-0.82)	(0.34)
HML × High SENT					0.640***		0.685***	
					(3.21)		(2.99)	
HML × High PLS SENT						0.220		0.193
						(0.78)		(0.68)
HML × High MKTVOL					-0.227	-0.248		
					(-0.88)	(-0.93)		
HML × High VIX							-0.086	-0.145
							(-0.31)	(-0.51)
HML × High MKTILLIQ					0.116	0.034	0.087	0.020
					(0.53)	(0.15)	(0.34)	(0.07)
RMW	0.582***	0.581***	0.585***	0.584***	0.420**	0.332*	0.372**	0.377**
	(6.84)	(6.82)	(6.72)	(6.70)	(2.25)	(1.73)	(2.10)	(2.11)
RMW × High SENT					0.016		0.091	
					(0.10)		(0.52)	
RMW × High PLS SENT						0.179		0.178
						(1.13)		(1.06)
RMW × High MKTVOL					-0.018	-0.031		
					(-0.10)	(-0.17)		
RMW × High VIX							0.005	-0.079
							(0.03)	(-0.41)
RMW × High MKTILLIQ					0.172	0.166	0.201	0.182
					(1.17)	(1.10)	(1.24)	(1.15)
CMA	0.733***	0.731***	0.715***	0.712***	-0.307	-0.002	-0.050	0.282
	(6.15)	(6.28)	(5.57)	(5.65)	(-1.15)	(-0.01)	(-0.20)	(0.96)
CMA × High SENT					0.182		0.186	
					(0.83)		(0.75)	
CMA × High PLS SENT						-0.237		-0.336
						(-0.85)		(-1.09)
CMA × High MKTVOL					0.878***	0.930***		
					(3.28)	(3.83)		
CMA × High VIX							0.671**	0.713**
							(2.00)	(2.52)
CMA × High MKTILLIQ					0.092	-0.026	0.009	-0.038
					(0.32)	(-0.10)	(0.03)	(-0.13)
MOM	0.520***	0.517***	0.497***	0.498***	0.261	0.124	0.603**	0.367
	(3.20)	(3.16)	(2.94)	(2.94)	(0.80)	(0.36)	(2.05)	(1.28)
MOM × High SENT					-0.304		-0.428	
					(-0.93)		(-1.09)	
MOM × High PLS SENT						-0.161		-0.195
						(-0.47)		(-0.50)
MOM × High MKTVOL					0.455	0.477		
					(1.26)	(1.29)		
MOM × High VIX							-0.191	-0.050
							(-0.56)	(-0.14)
MOM × High MKTILLIQ					-0.144	0.107	0.035	0.318
					(-0.51)	(0.37)	(0.11)	(1.01)
Obs	372	372	335	335	372	372	335	335
R-squared	0.522	0.523	0.543	0.544	0.614	0.603	0.612	0.602

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Panel B: Value-Weighted Returns to Investment Strategies Sorted by Risk Loadings Regressed on Lagged Market States

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Constant	0.435	1.257	0.530	1.554	0.962	1.759	0.857	2.125*
	(0.32)	(1.03)	(0.36)	(1.20)	(0.80)	(1.54)	(0.64)	(1.73)
High SENT	1.160*		1.124*		0.865		1.010	
	(1.91)		(1.75)		(1.26)		(1.42)	
High PLS SENT		0.863		0.623		0.559		0.614
		(1.25)		(0.89)		(0.87)		(0.94)
High MKTVOL	1.066	1.022			0.921	0.901		
	(1.56)	(1.50)			(1.38)	(1.28)		
High VIX			1.564**	1.382**			1.755***	1.326**
			(2.28)	(2.06)			(2.60)	(1.98)
High MKTILLIQ	1.608**	1.830**	1.849**	2.145**	1.203*	1.370*	1.160	1.379*
	(2.20)	(2.27)	(2.31)	(2.47)	(1.79)	(1.84)	(1.55)	(1.71)
DOWN	-0.874	-0.844	-1.152	-1.092	-0.602	-0.595	-0.977	-0.801
	(-1.28)	(-1.22)	(-1.60)	(-1.50)	(-0.92)	(-0.90)	(-1.39)	(-1.14)
TERM	0.037	-0.137	-0.001	-0.207	-0.021	-0.097	-0.103	-0.227
	(0.13)	(-0.41)	(-0.00)	(-0.59)	(-0.08)	(-0.36)	(-0.36)	(-0.76)
DEF	-0.291	-0.664	-0.568	-0.943	-0.552	-1.243	-0.709	-1.619
	(-0.30)	(-0.70)	(-0.55)	(-0.98)	(-0.55)	(-1.07)	(-0.68)	(-1.37)
MKT	0.163	0.170	0.203	0.212*	0.054	0.292	-0.057	0.202
	(1.51)	(1.59)	(1.59)	(1.68)	(0.31)	(1.58)	(-0.36)	(1.16)
MKT × High SENT					0.106		0.129	
					(0.55)		(0.59)	
MKT × High PLS SENT						-0.185		-0.267
						(-0.99)		(-1.36)
MKT × High MKTVOL					-0.026	-0.093		
					(-0.16)	(-0.54)		
MKT × High VIX							0.084	0.031
							(0.47)	(0.18)
MKT × High MKTILLIQ					0.229	0.142	0.305	0.238
					(1.35)	(0.75)	(1.51)	(1.15)
SMB	0.626***	0.632***	0.634***	0.642***	0.447*	0.185	0.275	0.098
	(3.82)	(3.91)	(3.65)	(3.73)	(1.82)	(0.74)	(1.00)	(0.39)
SMB × High SENT					-0.019		0.101	
					(-0.07)		(0.32)	
SMB × High PLS SENT						0.398		0.425
						(1.62)		(1.47)
SMB × High MKTVOL					-0.463*	-0.550**		
					(-1.86)	(-2.30)		
SMB × High VIX							-0.217	-0.337
							(-0.75)	(-1.16)
SMB × High MKTILLIQ					1.024***	1.149***	0.930***	1.072***
					(3.58)	(3.89)	(2.77)	(3.22)
HML	0.479***	0.489***	0.462***	0.465***	0.252	0.054	0.742**	0.279
	(2.92)	(2.95)	(2.79)	(2.77)	(0.76)	(0.17)	(2.45)	(0.97)
HML × High SENT					-0.105		-0.303	
					(-0.30)		(-0.80)	
HML × High PLS SENT						0.347		0.623*
						(0.93)		(1.70)
HML × High MKTVOL					0.305	0.254		
					(0.78)	(0.65)		
HML × High VIX							-0.305	-0.352
							(-0.86)	(-0.93)
HML × High MKTILLIQ					0.034	-0.045	0.066	-0.076
					(0.09)	(-0.12)	(0.18)	(-0.21)
RMW	0.013	0.011	0.028	0.030	0.496	0.234	0.858***	0.488*
	(0.09)	(0.08)	(0.19)	(0.20)	(1.65)	(0.95)	(2.76)	(1.96)
RMW × High SENT					-0.380		-0.509*	
					(-1.42)		(-1.84)	
RMW × High PLS SENT						-0.045		-0.094
						(-0.21)		(-0.44)
RMW × High MKTVOL					-0.133	-0.040		
					(-0.52)	(-0.18)		
RMW × High VIX							-0.571**	-0.402*
							(-2.04)	(-1.70)
RMW × High MKTILLIQ					-0.416	-0.353	-0.322	-0.228
					(-1.33)	(-1.24)	(-1.02)	(-0.77)
CMA	0.134	0.149	0.179	0.199	-0.475	-0.146	-0.469	-0.067
	(0.57)	(0.63)	(0.72)	(0.78)	(-1.24)	(-0.35)	(-1.23)	(-0.16)
CMA × High SENT					-0.218		-0.054	
					(-0.63)		(-0.14)	
CMA × High PLS SENT						-0.519		-0.616
						(-1.35)		(-1.58)
CMA × High MKTVOL					0.168	0.232		
					(0.50)	(0.70)		
CMA × High VIX							0.153	0.254
							(0.46)	(0.81)
CMA × High MKTILLIQ					1.379**	1.157**	1.499**	1.332**
					(2.53)	(2.41)	(2.51)	(2.53)
MOM	-0.192	-0.201	-0.216	-0.212	0.212	0.626	0.071	0.523
	(-0.62)	(-0.64)	(-0.65)	(-0.63)	(0.48)	(1.51)	(0.16)	(1.39)
MOM × High SENT					0.367		0.568	
					(0.59)		(0.86)	
MOM × High PLS SENT						-0.354		-0.513
						(-0.71)		(-1.02)
MOM × High MKTVOL					-0.789*	-0.970**		
					(-1.74)	(-2.13)		
MOM × High VIX							-0.596	-0.624
							(-1.34)	(-1.41)
MOM × High MKTILLIQ					0.520	0.558	0.544	0.556
					(1.13)	(1.18)	(1.18)	(1.20)
Obs	360	360	323	323	360	360	323	323
R-squared	0.168	0.164	0.184	0.180	0.273	0.269	0.296	0.289

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Panel C: Value-Weighted Returns to Investment Strategies Sorted by IPCA-Predicted Returns Regressed on Lagged Market States

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Constant	0.176	0.274	0.084	0.302	0.394	0.239	0.233	0.312
	(0.36)	(0.57)	(0.16)	(0.59)	(0.77)	(0.51)	(0.42)	(0.62)
High SENT	0.158		0.276		0.175		0.186	
	(0.49)		(0.82)		(0.41)		(0.43)	
High PLS SENT		0.152		0.226		0.453		0.501
		(0.38)		(0.58)		(1.10)		(1.20)
High MKTVOL	0.142	0.134			-0.005	-0.090		
	(0.52)	(0.49)			(-0.02)	(-0.28)		
High VIX			0.302	0.248			0.312	0.191
			(0.83)	(0.70)			(0.74)	(0.44)
High MKTILLIQ	0.130	0.168	0.258	0.345	0.118	0.123	0.203	0.289
	(0.41)	(0.47)	(0.70)	(0.86)	(0.30)	(0.31)	(0.45)	(0.64)
DOWN	0.157	0.159	-0.055	-0.044	0.257	0.307	0.137	0.137
	(0.44)	(0.45)	(-0.15)	(-0.12)	(0.67)	(0.83)	(0.33)	(0.34)
TERM	-0.006	-0.035	-0.037	-0.097	-0.018	-0.064	-0.047	-0.156
	(-0.05)	(-0.23)	(-0.25)	(-0.56)	(-0.14)	(-0.42)	(-0.30)	(-0.90)
DEF	0.204	0.159	0.274	0.197	0.088	0.155	0.128	0.106
	(0.47)	(0.36)	(0.62)	(0.43)	(0.19)	(0.30)	(0.30)	(0.19)
MKT	0.048	0.050	0.085	0.087	0.033	0.170*	0.020	0.183*
	(0.98)	(1.01)	(1.57)	(1.64)	(0.37)	(1.78)	(0.22)	(1.83)
MKT × High SENT					-0.016		0.080	
					(-0.17)		(0.80)	
MKT × High PLS SENT						-0.261***		-0.225**
						(-2.94)		(-2.28)
MKT × High MKTVOL					0.071	0.122		
					(0.79)	(1.40)		
MKT × High VIX							0.086	0.103
							(0.78)	(0.97)
MKT × High MKTILLIQ					-0.049	-0.051	-0.081	-0.039
					(-0.53)	(-0.52)	(-0.75)	(-0.36)
SMB	0.164**	0.164**	0.173**	0.174**	0.166	0.074	0.084	0.027
	(2.05)	(2.06)	(2.02)	(2.03)	(1.05)	(0.49)	(0.58)	(0.21)
SMB × High SENT					-0.146		-0.092	
					(-0.87)		(-0.53)	
SMB × High PLS SENT						-0.036		-0.063
						(-0.21)		(-0.35)
SMB × High MKTVOL					0.013	0.016		
					(0.08)	(0.11)		
SMB × High VIX							0.105	0.123
							(0.66)	(0.77)
SMB × High MKTILLIQ					0.152	0.182	0.179	0.211
					(0.87)	(1.06)	(0.95)	(1.15)
HML	0.067	0.068	0.014	0.015	0.072	0.142	0.205	0.200
	(0.74)	(0.76)	(0.16)	(0.17)	(0.35)	(0.78)	(1.19)	(1.23)
HML × High SENT					0.021		-0.138	
					(0.12)		(-0.77)	
HML × High PLS SENT						-0.085		-0.026
						(-0.45)		(-0.12)
HML × High MKTVOL					0.034	0.087		
					(0.19)	(0.45)		
HML × High VIX							-0.056	-0.031
							(-0.28)	(-0.15)
HML × High MKTILLIQ					-0.139	-0.119	-0.206	-0.220
					(-0.72)	(-0.65)	(-0.98)	(-1.04)
RMW	0.368***	0.367***	0.374***	0.374***	0.311**	0.360***	0.300**	0.349***
	(5.53)	(5.48)	(5.31)	(5.29)	(2.53)	(3.44)	(2.35)	(3.30)
RMW × High SENT					0.099		0.104	
					(0.89)		(0.82)	
RMW × High PLS SENT						0.072		0.078
						(0.71)		(0.74)
RMW × High MKTVOL					-0.120	-0.121		
					(-1.30)	(-1.36)		
RMW × High VIX							-0.114	-0.124
							(-1.09)	(-1.29)
RMW × High MKTILLIQ					0.160	0.090	0.174	0.110
					(1.48)	(0.92)	(1.46)	(0.99)
CMA	0.004	0.005	0.030	0.033	-0.239	-0.060	-0.086	0.102
	(0.04)	(0.05)	(0.29)	(0.32)	(-1.00)	(-0.23)	(-0.43)	(0.44)
CMA × High SENT					-0.248		-0.111	
					(-1.15)		(-0.51)	
CMA × High PLS SENT						-0.443*		-0.478*
						(-1.76)		(-1.76)
CMA × High MKTVOL					0.251	0.313		
					(1.19)	(1.57)		
CMA × High VIX							0.085	0.203
							(0.35)	(0.89)
CMA × High MKTILLIQ					0.079	0.068	0.003	0.101
					(0.35)	(0.32)	(0.01)	(0.43)
MOM	0.262**	0.260*	0.305**	0.306**	-0.237	-0.054	-0.154	0.087
	(1.98)	(1.96)	(2.14)	(2.15)	(-0.96)	(-0.22)	(-0.65)	(0.37)
MOM × High SENT					0.208		0.394	
					(0.77)		(1.29)	
MOM × High PLS SENT						-0.081		-0.201
						(-0.33)		(-0.73)
MOM × High MKTVOL					0.499*	0.453		
					(1.87)	(1.61)		
MOM × High VIX							0.245	0.282
							(0.82)	(0.98)
MOM × High MKTILLIQ					0.034	0.107	0.150	0.258
					(0.12)	(0.38)	(0.49)	(0.84)
Obs	372	372	335	335	372	372	335	335
R-squared	0.296	0.296	0.317	0.316	0.340	0.351	0.349	0.356

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Panel D: Value-Weighted Returns to Investment Strategies Sorted by CA2-Predicted Returns Regressed on Lagged Market States

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Constant	-0.827	-0.511	-0.745	-0.276	0.192	0.192	0.300	0.407
	(-0.81)	(-0.58)	(-0.67)	(-0.30)	(0.21)	(0.24)	(0.29)	(0.47)
High SENT	0.463		0.488		-0.011		0.010	
	(1.03)		(1.00)		(-0.02)		(0.02)	
High PLS SENT		0.385		0.193		0.034		0.058
		(0.72)		(0.35)		(0.06)		(0.10)
High MKTVOL	0.077	0.058			-0.265	-0.238		
	(0.13)	(0.10)			(-0.49)	(-0.42)		
High VIX			0.190	0.126			0.307	0.364
			(0.34)	(0.23)			(0.55)	(0.61)
High MKTILLIQ	0.490	0.589	0.821	0.938	0.267	0.251	0.281	0.359
	(0.79)	(0.90)	(1.24)	(1.34)	(0.51)	(0.47)	(0.49)	(0.64)
DOWN	-0.963*	-0.955	-1.114*	-1.081*	-0.588	-0.699	-0.711	-0.848
	(-1.66)	(-1.63)	(-1.84)	(-1.78)	(-1.23)	(-1.38)	(-1.35)	(-1.52)
TERM	0.262	0.187	0.208	0.129	0.124	0.084	0.030	-0.036
	(1.25)	(0.78)	(0.90)	(0.52)	(0.73)	(0.43)	(0.16)	(-0.17)
DEF	0.777	0.632	0.784	0.607	0.152	0.266	0.075	0.082
	(0.81)	(0.68)	(0.79)	(0.64)	(0.17)	(0.27)	(0.08)	(0.08)
MKT	0.022	0.025	0.049	0.052	-0.107	-0.083	-0.070	-0.018
	(0.27)	(0.31)	(0.48)	(0.51)	(-0.59)	(-0.51)	(-0.38)	(-0.10)
MKT × High SENT					0.133		0.128	
					(0.91)		(0.74)	
MKT × High PLS SENT						0.057		-0.037
						(0.39)		(-0.22)
MKT × High MKTVOL					0.052	0.033		
					(0.34)	(0.22)		
MKT × High VIX							-0.112	-0.089
							(-0.60)	(-0.49)
MKT × High MKTILLIQ					0.001	0.027	0.142	0.145
					(0.01)	(0.18)	(0.82)	(0.88)
SMB	0.346**	0.348**	0.325**	0.329**	0.089	-0.062	0.078	0.001
	(2.46)	(2.46)	(2.23)	(2.25)	(0.45)	(-0.34)	(0.38)	(0.00)
SMB × High SENT					-0.233		-0.169	
					(-1.15)		(-0.77)	
SMB × High PLS SENT						-0.002		-0.113
						(-0.01)		(-0.55)
SMB × High MKTVOL					0.099	0.095		
					(0.46)	(0.44)		
SMB × High VIX							0.066	0.106
							(0.30)	(0.50)
SMB × High MKTILLIQ					0.821***	0.839***	0.805***	0.829***
					(2.94)	(2.98)	(2.85)	(3.01)
HML	0.201	0.206	0.172	0.175	0.306	0.430*	0.279	0.389
	(1.42)	(1.45)	(1.18)	(1.19)	(1.17)	(1.68)	(1.02)	(1.57)
HML × High SENT					0.270		0.303	
					(1.00)		(1.02)	
HML × High PLS SENT						0.151		0.312
						(0.59)		(1.13)
HML × High MKTVOL					-0.516*	-0.535*		
					(-1.82)	(-1.91)		
HML × High VIX							-0.362	-0.442
							(-1.24)	(-1.54)
HML × High MKTILLIQ					0.151	0.129	0.023	-0.057
					(0.57)	(0.47)	(0.09)	(-0.20)
RMW	0.333***	0.333***	0.344***	0.346***	0.610***	0.533***	0.815***	0.738***
	(4.04)	(4.05)	(3.99)	(3.99)	(3.16)	(3.23)	(3.54)	(4.14)
RMW × High SENT					-0.053		-0.101	
					(-0.32)		(-0.57)	
RMW × High PLS SENT						0.144		0.143
						(1.11)		(1.14)
RMW × High MKTVOL					-0.183	-0.189		
					(-1.02)	(-1.15)		
RMW × High VIX							-0.431**	-0.465***
							(-1.98)	(-2.69)
RMW × High MKTILLIQ					-0.287	-0.290*	-0.244	-0.246
					(-1.64)	(-1.86)	(-1.44)	(-1.61)
CMA	0.127	0.133	0.158	0.168	-0.458	-0.423	-0.175	-0.019
	(0.60)	(0.62)	(0.70)	(0.75)	(-1.55)	(-1.29)	(-0.57)	(-0.05)
CMA × High SENT					0.064		0.089	
					(0.21)		(0.28)	
CMA × High PLS SENT						0.137		-0.131
						(0.41)		(-0.40)
CMA × High MKTVOL					0.458	0.533*		
					(1.46)	(1.71)		
CMA × High VIX							0.005	0.158
							(0.01)	(0.46)
CMA × High MKTILLIQ					0.654	0.628	0.863**	0.836**
					(1.50)	(1.60)	(1.98)	(2.21)
MOM	0.364	0.361	0.358	0.359	-0.012	0.282	0.263	0.489
	(1.38)	(1.35)	(1.26)	(1.27)	(-0.03)	(0.73)	(0.55)	(1.20)
MOM × High SENT					-0.205		-0.325	
					(-0.43)		(-0.63)	
MOM × High PLS SENT						-0.753*		-1.102**
						(-1.75)		(-2.49)
MOM × High MKTVOL					0.280	0.190		
					(0.62)	(0.45)		
MOM × High VIX							-0.263	-0.086
							(-0.58)	(-0.21)
MOM × High MKTILLIQ					0.713*	0.703*	1.091**	1.143***
					(1.67)	(1.68)	(2.53)	(2.96)
Obs	360	360	324	324	360	360	324	324
R-squared	0.185	0.185	0.189	0.187	0.284	0.290	0.297	0.307

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Table IA9: Performance of Machine Learning Portfolios and their Attribution: IPCA and Conditional Autoencoder

This table reports the monthly returns from the unconditional strategy (WML) and its decomposition into intra-industry (WML^{INTRA}) and inter-industry (WML^{INTER}) return components, as depicted in Equation (5). In Panel A1, we report the returns for WML, where the strategy takes long (short) positions in the winner (loser) stocks that are expected to outperform (underperform) the market according to the IPCA-predicted returns (Kelly, Pruitt, and Su 2019). In Panel A2, we report the returns for WML^{INTRA}, where the strategy takes long (short) positions in the stocks that are expected to outperform (underperform) the industry portfolio. H^{INTRA} and H are the scaling factors for the unconditional and intra-industry strategies. Panel A3 reports the return for WML^{INTER}, where the strategy takes long (short) positions in the industry portfolios that are expected to outperform (underperform) the market portfolio. H^{INTER} is the scaling factor for the inter-industry strategy. Portfolio returns are further adjusted by the Fama-French 6-factor model (FF6). We report the results for the full sample as well as subsamples that exclude microcaps, nonrated firms, and credit rating downgrades. Panel B reports similar statistics when portfolios are sorted according to the CA2-predicted returns (Gu, Kelly, and Xiu 2021). Both IPCA and CA2 impose a zero-alpha restriction. Newey-West adjusted t-statistics are shown in parentheses. Numbers with "*", "**", and "***" are significant at the 10%, 5%, and 1% levels, respectively.

Panel A: Returns to Investment Strategies Sorted by IPCA-Predicted Returns

Rank	Full Sample		Nonmicrocaps		Credit Rating Sample		Nondowngrades	
	Return	FF6	Return	FF6	Return	FF6	Return	FF6
Panel A1: Unconditional Payoff								
Loser	0.072	-0.642***	0.383	-0.460***	0.400	-0.510***	1.011***	0.023
	(0.21)	(-4.83)	(1.23)	(-5.37)	(1.19)	(-4.25)	(3.56)	(0.21)
Winner	2.019***	1.216***	1.451***	0.365***	1.432***	0.216**	1.716***	0.466***
	(6.02)	(8.56)	(5.19)	(6.40)	(4.69)	(2.50)	(6.47)	(6.10)
WML	1.947***	1.858***	1.067***	0.825***	1.032***	0.726***	0.705***	0.443***
	(14.24)	(15.11)	(8.15)	(8.02)	(7.70)	(5.84)	(5.83)	(4.39)
Panel A2: Intra-Industry Payoff								
Loser	0.160	-0.611***	0.426	-0.489***	0.524	-0.530***	1.041***	-0.063
	(0.47)	(-5.04)	(1.38)	(-6.86)	(1.64)	(-5.04)	(3.74)	(-0.65)
Winner	1.951***	1.208***	1.408***	0.392***	1.363***	0.258***	1.721***	0.561***
	(5.93)	(8.03)	(5.11)	(6.47)	(4.25)	(2.80)	(6.37)	(6.68)
WML^{INTRA}	1.791***	1.819***	0.982***	0.881***	0.839***	0.788***	0.681***	0.623***
	(15.91)	(15.69)	(10.09)	(9.96)	(9.46)	(7.79)	(8.56)	(7.13)
$WML^{INTRA} \times H^{INTRA}/H$	1.609***	1.636***	0.846***	0.761***	0.691***	0.651***	0.554***	0.508***
	(15.90)	(15.65)	(10.11)	(9.97)	(9.45)	(7.77)	(8.59)	(7.15)
Panel A3: Inter-Industry Payoff								
Loser	0.608*	-0.033	0.721**	-0.108	0.602*	-0.233*	1.207***	0.263**
	(1.77)	(-0.20)	(2.36)	(-0.95)	(1.72)	(-1.80)	(4.20)	(2.16)
Winner	1.400***	0.462***	1.190***	0.031	1.204***	-0.090	1.463***	0.156**
	(4.02)	(3.58)	(4.06)	(0.47)	(4.08)	(-1.01)	(5.55)	(2.02)
WML^{INTER}	0.791***	0.495***	0.469***	0.138	0.601***	0.143	0.257*	-0.107
	(4.38)	(3.52)	(2.76)	(1.06)	(3.57)	(1.01)	(1.84)	(-0.86)
$WML^{INTER} \times H^{INTER}/H$	0.338***	0.222***	0.221***	0.064	0.341***	0.075	0.151*	-0.065
	(4.34)	(3.56)	(2.66)	(0.98)	(3.59)	(0.89)	(1.87)	(-0.90)

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Table IA9—Continued

Panel B: Returns to Investment Strategies Sorted by CA2-Predicted Returns								
Rank	Full Sample		Nonmicrocaps		Credit Rating Sample		Nondowngrades	
	Return	FF6	Return	FF6	Return	FF6	Return	FF6
Panel B1: Unconditional Payoff								
Loser	0.206	-0.322**	0.299	-0.297***	0.086	-0.663***	1.178***	0.299**
	(0.48)	(-2.18)	(0.74)	(-3.01)	(0.19)	(-4.25)	(3.31)	(2.24)
Winner	1.863***	1.348***	1.243***	0.138**	1.194***	0.040	1.616***	0.393***
	(4.22)	(4.51)	(4.32)	(2.01)	(3.34)	(0.30)	(5.48)	(3.81)
WML	1.657***	1.670***	0.944***	0.436***	1.107***	0.703***	0.438**	0.094
	(8.91)	(6.96)	(4.51)	(3.47)	(4.73)	(3.88)	(2.40)	(0.68)
Panel B2: Intra-Industry Payoff								
Loser	0.266	-0.320**	0.348	-0.363***	0.263	-0.603***	1.217***	0.255*
	(0.61)	(-2.28)	(0.88)	(-4.73)	(0.59)	(-4.06)	(3.49)	(1.89)
Winner	1.815***	1.363***	1.187***	0.172***	1.099***	0.046	1.588***	0.450***
	(4.06)	(4.51)	(4.09)	(2.77)	(2.94)	(0.33)	(5.34)	(4.24)
WML^{INTRA}	1.550***	1.684***	0.838***	0.535***	0.835***	0.649***	0.371**	0.194
	(9.45)	(7.41)	(5.17)	(5.75)	(4.48)	(4.04)	(2.57)	(1.47)
$WML^{INTRA} \times H^{INTRA}/H$	1.467***	1.598***	0.739***	0.478***	0.724***	0.569***	0.327***	0.184*
	(9.52)	(7.52)	(5.25)	(5.83)	(4.61)	(4.17)	(2.76)	(1.68)
Panel B3: Inter-Industry Payoff								
Loser	0.655*	0.128	0.638*	0.006	0.399	-0.378**	1.193***	0.307**
	(1.84)	(0.81)	(1.92)	(0.05)	(1.01)	(-2.34)	(3.80)	(2.46)
Winner	1.232***	0.372**	1.054***	-0.054	1.095***	-0.146	1.392***	0.139
	(3.63)	(2.36)	(3.79)	(-0.70)	(3.54)	(-1.44)	(5.09)	(1.47)
WML^{INTER}	0.578***	0.244	0.416**	-0.060	0.697***	0.232	0.198	-0.168
	(2.84)	(1.41)	(2.17)	(-0.39)	(3.19)	(1.25)	(1.22)	(-1.26)
$WML^{INTER} \times H^{INTER}/H$	0.190**	0.072	0.205**	-0.043	0.383***	0.134	0.111	-0.090
	(2.53)	(1.09)	(2.11)	(-0.53)	(3.17)	(1.30)	(1.23)	(-1.28)

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